

Transformers

10.1. INTRODUCTION

Electrical energy is generated at places where it is easier to get water head, oil or coal for hydro-electric, diesel or thermal power stations respectively. Then energy is to be transmitted at considerable distances for use in villages, towns and cities located at distant places. As already shown transmission of electrical energy at high voltages is economical, therefore, some means are required for stepping up the voltage at generating stations and stepping down the same at the places where it is to be used. Electric machine used for this purpose is "transformer". In our country the electrical energy is usually generated at 6.6 or 11 or 33 kV, stepped up to 132, 220, 400, or 765 kV with the help of step-up transformers for transmission and then stepped down to 66 kV or 33 kV at grid substations for feeding various substations, which further step down the voltage to 11 kV for feeding distributing transformers stepping down the voltage further to 415/240 volts for the consumer uses.

Transformers are extensively used like, the *instrument transformers* for metering purposes. One-to-one ratio transformers (*i.e.* the transformers having equal primary and secondary voltages) are used to electrically isolate the two parts of an electrical circuit. In high voltage laboratories the transformers are used to provide very high voltages for testing purposes, termed as *testing transformers*. In electric communication circuits transformers are employed for a variety of purposes *e.g.*, as an impedance transformation device to allow maximum transfer of power from the input circuit to the output device. In radio and television circuits input transformers, inter-stage transformers and output transformers are widely employed. Transformers are also employed in telephone circuits, instrumentation circuits and control circuits. Indeed the transformer is a device which plays a vital and essential role in many facets of electrical engineering.

Transformer is an ac machine that (i) transfers electrical energy from one electric circuit to another (ii) does so without a change of frequency (iii) does so by the principle of electro-magnetic induction and (iv) has electric circuits that are linked by a common magnetic circuit. The energy transfer usually takes place with a change of voltage, although this is not always necessary. When the transformer raises the voltage *i.e.* when the out-put voltage of a transformer is higher than its input voltage, it is called the *step-up transformer* and when it lowers the voltage it is called the *step down transformer*.

Since its basic construction requires no moving parts, so it is often called the '*static transformer*'. and it is very rugged machine requiring the minimum amount of repair and maintenance. Owing to the lack of rotating parts there are no friction or windage losses. Further, the other losses are relatively low, so that the efficiency of a transformer is high. Typical transformer efficiencies at full load lie between 96% and 97% and with extremely large capacity transformers the efficiencies are as high as 99%. The cost per kVA output of transformers is quite low as compared with other electrical machines. As there are no

teeth, slots or rotating parts, and the windings can be immersed in oil, it is not difficult to insulate transformers for very high voltages.

10.2. OPERATING PRINCIPLE

An elementary transformer consists of a soft iron or silicon steel core and two windings placed on it. The windings are insulated from both the core and each other. The core is built up of thin soft iron or silicon steel laminations to provide a path of low reluctance to the magnetic flux. The winding connected to the supply main is called the *primary* and the winding connected to the load circuit is called the *secondary*. The winding connected to higher voltage circuit is called the *high-voltage (hv) winding* while that connected to the lower voltage circuit is called the *low-voltage (lv) winding*. In case of a step-up transformer, low-voltage winding is the primary and high voltage winding is the secondary while in case of a step-down transformer the high-voltage winding is the primary and low-voltage winding is the secondary.

✓ The action of a transformer is based on the principle that energy may be efficiently transferred by induction from one set of coils to another by means of a varying magnetic flux, provided that both the sets of coils are on a common magnetic circuit. ✓ In a transformer, the coils and magnetic circuit are all stationary with respect to one another. The emfs are induced by the variation in the magnitude of flux with time, as illustrated in Fig. 10.1.

Although in the actual construction the two windings are usually wound one over the other, for the sake of simplicity, the figures for analysing transformer theory show the windings on opposite sides of the core as in Fig. 10.1.

When the primary winding is connected to an a.c. supply mains, a current flows through it. Since this winding links with an iron core, so current flowing through this winding produces an alternating flux ϕ in the core. Since this flux is alternating and links with the secondary winding also, so induces an emf in the secondary winding. The frequency of induced emf in secondary winding is the same as that of the flux or that of the supply voltage. The induced emf in the secondary winding enables it to deliver current to an external load connected across it. Thus the energy is transformed from primary winding to the secondary winding by means of electro-magnetic induction without any change in frequency. The flux ϕ of the iron core links not only with the secondary winding but also with the primary winding, so produces self induced emf in the primary winding. This induced emf in the primary winding opposes the applied voltage and, therefore, sometimes it is known as *back emf* of the primary. In fact the induced emf in the primary winding limits the primary current in much the same way that the back emf in a dc motor limits the armature current.

10.3. TRANSFORMER ON DC

A transformer cannot operate on dc supply and never be connected to a dc source. If a rated dc voltage is applied to the primary of a transformer, the flux produced in the transformer core will not vary but remain constant in magnitude and, therefore, no emf will be induced in the secondary winding except at the moment of switching on. Thus the transformer is not capable of raising or lowering the dc voltage. Also there will be no self induced emf in the primary winding, which is only possible with varying flux linkage, to oppose the applied voltage and since the resistance of primary winding is quite low, therefore, a heavy current will flow through the primary winding which may result in the burning out of the primary winding. This is reason that *dc is never applied to a transformer*.

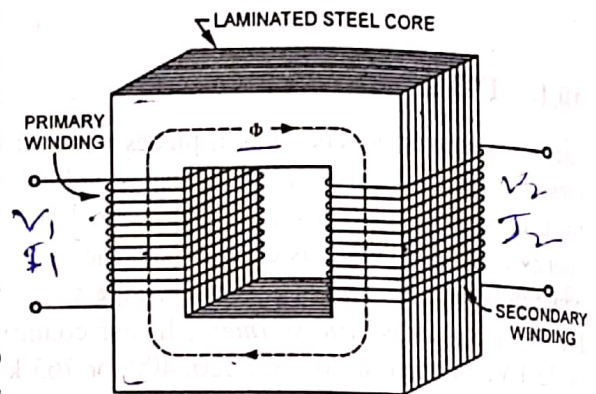


Fig. 10.1 Simple Transformer

10.4. IDEAL TRANSFORMER

For a better understanding and an easier explanation of a practical transformer, certain idealizing assumptions are made which are close approximations for a practical transformer. A transformer having these ideal properties is hypothetical (has no real existence) and is referred to as the ideal transformer. It possesses certain essential features of a real transformer but some details of minor significance are ignored which will be introduced step-by-step while analysing a transformer. The idealizing assumptions made are as follows:

- No winding resistance i.e. the primary and secondary windings have zero resistance. It means that there is no ohmic power loss and no resistive voltage drop in an ideal transformer.
- No magnetic leakage i.e. there is no leakage flux and all the flux set up is confined to the core and links both the windings.
- No iron loss i.e. hysteresis and eddy current losses in transformer core are zero.
- Zero-magnetizing current i.e. the core has infinite permeability and zero reluctance so that zero magnetizing current is required for establishing the requisite amount of flux in the core.

From the above discussion an ideal transformer is supposed to consist of two purely inductive coils wound on a loss-free core.

10.5. EMF EQUATION

When an alternating (sinusoidal) voltage is applied to the primary winding of a transformer, an alternating (sinusoidal) flux, as shown in Fig. 10.2, is set up in the iron core which links both the windings (primary and secondary windings).

Let $\phi_{\max}$ = Maximum value of flux in webers
and f = Supply frequency in hertz.

As illustrated in Fig. 10.2, the magnetic flux increases from zero to its maximum value $\phi_{\max}$ in one-fourth of a cycle i.e. in $\frac{1}{4f}$ second.

So average rate of change of flux, $\frac{d\phi}{dt} = \frac{\phi_{\max}}{1/4f} = 4f\phi_{\max}$

Since average emf induced per turn in volts is equal to the average rate of change of flux.

So average emf induced per turn = $4f\phi_{\max}$ volts

Since flux ϕ varies sinusoidally, so emf induced will be sinusoidal and form-factor-for-sinusoidal wave is 1.11 i.e. the rms or effective value is 1.11 times the average value.

$\therefore$ RMS value of emf induced per turn = $1.11 \times 4f\phi_{\max}$ volts ... (10.1)

If the number of turns on primary and secondary windings are N_1 and N_2 respectively, then

RMS value of emf induced in primary, E_1 = EMF induced per turn $\times$ number of primary turns
 $= 4.44 f \phi_{\max} \times N_1 = 4.44 f N_1 \phi_{\max}$ volts ... (10.2)

Similarly rms value of emf induced in secondary,

$E_2 = 4.44 f \phi_{\max} \times N_2$ volts ... (10.3)

The above relations for emf induced in primary and secondary windings can be derived alternatively as below

The instantaneous value of sinusoidally varying flux may be given as
 $\phi = \phi_{\max} \sin \omega t$

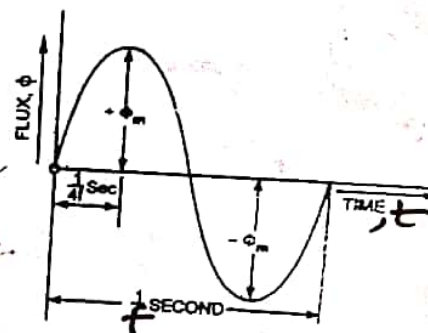


Fig. 10.2 Sinusoidal Variation of Flux With Time

$$\mathcal{E} = - \frac{d\phi}{dt}$$

$$\begin{aligned} \text{Instantaneous value of emf induced per turn} &= \frac{-d\phi}{dt} \text{ volts} = -\omega \phi_{\max} \cos \omega t \\ &= \omega \phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right) \text{ volts} \end{aligned}$$

It is clear from the above equation that the maximum value of emf induced per turn
 $= \omega \phi_{\max} = 2\pi f \phi_{\max} \text{ volts} \quad \because \omega = 2\pi f$

$$\text{and rms value of emf induced per turn} = \frac{1}{\sqrt{2}} \times 2\pi f \phi_{\max} = 4.44 f \phi_{\max} \text{ volts}$$

Hence rms value of emf induced in primary, $E_1 = 4.44 f N_1 \phi_{\max} \text{ volts}$ ✓

and rms value of emf induced in secondary, $E_2 = 4.44 f N_2 \phi_{\max} \text{ volts}$ ✓

In an ideal transformer the voltage drops in primary and secondary windings are negligible, so

EMF induced in primary winding, $E =$ Applied voltage to primary, V_1

and terminal voltage, $V_2 =$ EMF induced in secondary, E_2

Note: If $B_{\max}$ is the maximum allowable flux density in Wb/m^2 (or T) and a is the area of x-section of iron core in square metres, then in Eqs. (10.1), (10.2) and (10.3), $\phi_{\max}$ is given as

$$\phi_{\max} = B_{\max} a \text{ webers}$$

10.6. VOLTAGE AND CURRENT TRANSFORMATION RATIOS

Referring to Eq. (10.1), it is clear that the volts per turn is exactly the same for both the primary and secondary windings i.e. in any transformer, the secondary and primary induced emfs are related to each other by the ratio of the number of secondary and primary turns. Thus

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = K \quad \text{---(10.4)}$$

The same relationship can be derived by dividing Eq. (10.3) by Eq. (10.2).

The constant K in Eq. (10.4) is called the voltage transformation ratio. ✓

For step-up transformer, $V_2 > V_1$ or voltage transformation ratio, $K > 1$. ✓

For step down transformer, $V_2 < V_1$ or voltage transformation ratio, $K < 1$. ✓

In an ideal transformer, the losses are negligible, so the volt-ampere input to the primary and volt-ampere output from secondary can be approximately equated i.e.

$$\text{Output VA} = \text{Input VA}$$

$$\text{or } V_2 I_2 = V_1 I_1$$

$$\text{or } \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{1}{K} \quad \text{---(10.5)}$$

Primary and secondary currents are inversely proportional to their respective turns.

Note: For a transformer the true ratio of transformation (or turn-ratio) is constant, while the voltage ratio $\left(\frac{V_2}{V_1} \right)$ varies about 1 to 8 percent depending upon the load and its power factor.

Example 10.1. It is desired to have a 4.13 mWb maximum core flux in a transformer at 110 V and 50 Hz. Determine the required number of turns in the primary.

[Pb. Technical Univ. Electrical Engineering December, 2000]

Solution:

$$\text{EMF induced in primary, } E_1 = 110 \text{ V}$$

Supply frequency, $f = 50 \text{ Hz}$

Maximum core flux, $\phi_{\text{max}} = 4.13 \text{ m Wb} = 4.13 \times 10^{-3} \text{ Wb}$

$$\begin{aligned} \text{Required number of turns on primary, } N_1 &= \frac{E_1}{4.44 f \phi_{\text{max}}} \quad \dots \text{Refer Eq. (10.2)} \\ &= \frac{110}{4.44 \times 50 \times 4.13 \times 10^{-3}} = 120 \text{ Ans.} \end{aligned}$$

Example 10.2. The emf per turn of a single phase 10 kVA, 2200 / 220 V, 50 Hz transformer is 10 V. Calculate (i) the number of primary and secondary turns, (ii) the net cross-sectional area of core for a maximum flux density of 1.5 T.

Solution:

EMF per turn = 10 V

Primary induced emf, $E_1 = V_1 = 2200 \text{ V}$

Secondary induced emf, $E_2 = V_2 = 220 \text{ V}$

Supply frequency, $f = 50 \text{ Hz}$

Maximum flux density, $B_{\text{max}} = 1.5 \text{ T}$

$$(i) \text{ Number of primary turns, } N_1 = \frac{E_1}{\text{EMF per turn}} = \frac{2200}{10} = 220 \text{ Ans.}$$

$$\text{Number of secondary turns, } N_2 = \frac{E_2}{\text{EMF per turn}} = \frac{220}{10} = 22 \text{ Ans.}$$

$$\text{Maximum value of flux, } \phi_{\text{max}} = \frac{\text{EMF per turn}}{4.44 f} = \frac{10}{4.44 \times 50} = 0.045 \text{ Wb} \quad \dots \text{refer Eq. (10.1)}$$

$$(ii) \text{ Net cross-sectional area of core, } a = \frac{\phi_{\text{max}}}{B_{\text{max}}} = \frac{0.045}{1.5} = 0.03 \text{ m}^2 \text{ Ans.}$$

Example 10.3. A single phase transformer has 350 primary and 1,050 secondary turns. The net cross-sectional area of the core is 55 cm^2 . If the primary winding be connected to a 400 V, 50 Hz single phase supply, calculate (i) maximum value of the flux density in the core and (ii) the voltage induced in the secondary winding.

[R.G.P.V. Bhopal Basic Electrical Engineering 2001; G.G.S.I.P. Delhi Univ; Electrical Science May-June 2006]

Solution: Net cross-sectional area of core, $a = 55 \text{ cm}^2 = 0.0055 \text{ m}^2$

$$\text{Maximum value of flux, } \phi_{\text{max}} = \frac{E_1}{4.44 f N_1} \quad \dots \text{refer Eq. (10.2)}$$

$$= \frac{400}{4.44 \times 50 \times 350} = 5.148 \times 10^{-3} \text{ Wb}$$

$$(i) \text{ Peak value of flux density in the core, } B_{\text{max}} = \frac{\phi_{\text{max}}}{a} = \frac{5.148 \times 10^{-3}}{0.0055} = 0.936 \text{ T Ans.}$$

$$(ii) \text{ Voltage induced in the secondary winding, } E_2 = E_1 \times \frac{N_2}{N_1} \quad \dots \text{refer Eq. (10.4)}$$

$$= 400 \times \frac{1,050}{350} = 1,200 \text{ V Ans.}$$

Example 10.4. A 25 kVA, single phase transformer has 250 turns on the primary and 40 turns on the secondary winding. The primary is connected to 1,500 V, 50-Hz mains calculate (i) primary and secondary currents on full load (ii) secondary emf (iii) maximum flux in the core.

[Bharathidhasan Univ., April 1998]

Solution: Supply voltage, $V_1 = 1500$ V
Primary induced emf, $E_1 = V_1 = 1500$ V

$$(ii) \text{ Secondary emf, } E_2 = \frac{E_1 \times N_2}{N_1} = \frac{1,500 \times 40}{250} = 240 \text{ V Ans.}$$

Approximate value of primary current on full load,

$$I_1 = \frac{\text{Rated kVA} \times 1,000}{V_1} = \frac{25 \times 1,000}{1,500} = 16.667 \text{ A Ans.}$$

Approximate value of secondary current on full load,

$$I_2 = \frac{\text{Rated kVA} \times 1,000}{V_2 \text{ (or } E_2)} = \frac{25 \times 1,000}{240} = 104.167 \text{ A Ans. } \because E_2 = V_2$$

Maximum value of flux in the core,

$$\phi_{\max} = \frac{E_1}{4.44 f N_1} = \frac{1,500}{4.44 \times 50 \times 250} = 0.027 \text{ Wb or } 27 \text{ mWb Ans.}$$

Example 10.5. A 200/50 V, 50 Hz single phase transformer is connected to a 200 V, 50 Hz supply with secondary winding open. Primary winding has 400 turns.

- What is the value of maximum flux through the core, if the primary winding has 400 turns?
- What is the peak value of flux if the primary voltage is 200 V, 25 Hz?
- What happens to no-load current?

[R.G.P.V. Bhopal Basic Electrical Engineering December 2002]

Solution: (i) Maximum value of flux,

$$\phi_{\max} = \frac{E_1}{4.44 f N_1} = \frac{V_1}{4.44 f N_1} = \frac{200}{4.44 \times 50 \times 400} = 2.25 \times 10^{-3} \text{ Wb or } 2.25 \text{ mWb Ans.}$$

(ii) When primary voltage is 200 V and supply frequency is 25 Hz

$$\phi'_{\max} = \frac{200}{4.44 \times 25 \times 400} = 4.5 \text{ mWb Ans.}$$

(iii) The no-load primary current I_0 , called the exciting current, is very small in comparison to the full-load primary current (2–5% of full-load primary current). This current is made up of a relatively larger quadrature or magnetising component I_m and a comparatively small in-phase or energy component I_w .

When supply frequency is reduced, keeping supply voltage constant, maximum value of flux is increased in inverse ratio of supply frequencies, and therefore, the magnetising component of no-load current and thus the no-load current is increased.

Example 10.6. A single phase, 50 Hz, core type transformer has square cores of 20 cm side, permissible maximum flux density is 1 Wb/m². Calculate the number of turns per limb on the high and low-voltage sides for a 300/220 V ratio.

[Manamalan Sundaram Univ., April 1998]

Solution: Core area = $20 \times 20 = 400 \text{ cm}^2 = 0.04 \text{ m}^2$

Permissible flux density, $B_{\max} = 1.0 \text{ Wb/m}^2$

Maximum value of permissible flux, $\phi_{\max} = B_{\max} \times a = 1 \times 0.04 = 0.04 \text{ Wb}$

a heavy current will flow through the primary winding which may result in the burning out of the primary.

Number of turns on low-voltage (secondary) winding,

$$N_2 = \frac{E_2}{4.44 f \phi_{\max}} = \frac{220}{4.44 \times 50 \times 0.04} = 24.77; 26 \text{ (say)}$$

Number of turns on high-voltage (primary) winding,

$$N_1 = \frac{E_1}{E_2} \times N_2 = \frac{3,000}{220} \times 26 = 354$$

$$\text{Number of } hv \text{ turns on each limb} = \frac{354}{2} = 177 \text{ Ans.}$$

$$\text{Number of } lv \text{ turns on each limb} = \frac{26}{2} = 13 \text{ Ans.}$$

Note: LV turns are calculated first, rounded one figure off to the next higher even number in order that maximum flux density does not exceed the permissible maximum flux density. Then *hv* turns are calculated with corrected number of turns by transformation ratio. Further there are two limbs and each limb accommodates half of the *lv* and half of the *hv* turns from the view point of reducing leakage reactance.

Example 10.7. A single phase, 50 Hz core type transformer has square cores of 20 cm side. The permissible flux density in the core is 1.0 Wb/m². Calculate the number of turns per limb on the high and low voltage sides for a 3,000/220 V ratio. To allow for insulation of stampings, assume the net iron length to be 0.9 × gross iron length.

Solution: Iron-length on one side of the core
= 0.9 × 20 = 18 cm = 0.18 m

Iron-length on the other side of the core
= 20 cm or 0.20 m

because only one side of the core is affected by insulation and other side remains unaffected (Fig. 10.3).

Core area, $a = 0.18 \times 0.20 = 0.036 \text{ m}^2$

Permissible flux density,

$$B_{\max} = 1.0 \text{ Wb/m}^2$$

$$\text{Maximum value of flux, } \phi_{\max} = B_{\max} \times a = 1.0 \times 0.036 = 0.036 \text{ Wb}$$

$$\text{Number of turns on } lv \text{ (secondary) winding, } N_2 = \frac{E_2}{4.44 f \phi_{\max}} = \frac{220}{4.44 \times 50 \times 0.036} = 27.5$$

$$\text{Number of turns on } hv \text{ (primary) winding, } N_1 = \frac{E_1}{E_2} \times N_2 = \frac{3,000}{220} \times 28 = 382$$

$$\text{Number of } hv \text{ turns on each limb} = \frac{382}{2} = 191 \text{ Ans.}$$

$$\text{Number of } lv \text{ turns on each limb} = \frac{28}{2} = 14 \text{ Ans.}$$

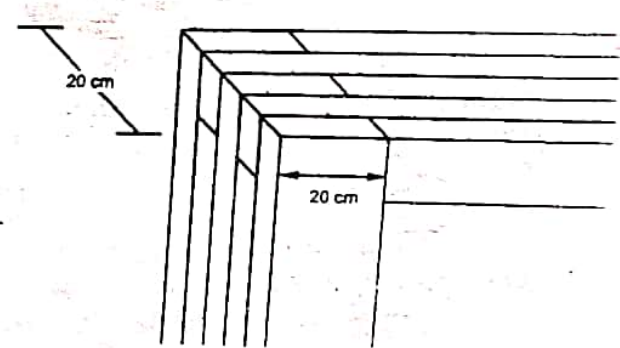


Fig. 10.3

Example 10.8. A single phase transformer has a core whose cross-sectional area is 150 cm²; operates at a maximum flux density of 1.1 Wb/m² from a 50 Hz supply. If the secondary winding has 66 turns, determine the output in KVA when connected to a load of 4 Ω impedance. Neglect any voltage drop in the transformer.

Solution:

Peak value of flux density, $B_{\max} = 1.1 \text{ Wb/m}^2$

Cross-sectional area of core, $a = 150 \text{ cm}^2 = 0.015 \text{ m}^2$

Maximum value of flux in the core, $\phi_{\max} = B_{\max} \times a = 1.1 \times 0.015 = 0.0165 \text{ Wb}$

Number of turns on secondary, $N_2 = 66$

Supply frequency, $f = 50 \text{ Hz}$

RMS value of induced emf in the secondary winding,

$$E_2 = 4.44 f \phi_{\max} N_2 = 4.44 \times 50 \times 0.0165 \times 66 = 240 \text{ volts}$$

Secondary voltage, $V_2 = E_2 = 240 \text{ V}$

neglecting voltage drop in secondary winding

$$\text{Secondary current, } I_2 = \frac{V_2}{\text{Load impedance}} = \frac{240}{4} = 60 \text{ A}$$

$$\text{Output in kVA} = \frac{V_2 I_2}{1,000} = \frac{240 \times 60}{1,000} = 14.4 \text{ Ans.}$$

Example 10.9. A single phase transformer with 10 : 1 turn-ratio and rated at 50 kVA, 2400/240 V, 50 Hz is used to step down the voltage of a distribution system. The low tension voltage is to be kept constant at 240 V. Find the value of load impedance of the *lt* side so that the transformer will be loaded fully. Find also the value of maximum flux inside the core if the *lt* side has 23 turns.

Solution : Low tension (secondary) voltage, $V_2 = 240 \text{ volts}$

$$\text{Full-load secondary current, } I_2 = \frac{\text{Rated kVA} \times 1,000}{V_2} = \frac{50 \times 1,000}{240} = 208.33 \text{ A}$$

$$\text{Load impedance connected on secondary side, } Z = \frac{V_2}{I_2} = \frac{240}{208.33} = 1.152 \Omega \text{ Ans.}$$

$$\text{Maximum value of flux in the core, } \phi_{\max} = \frac{E_2}{4.44 f N_2} = \frac{240}{4.44 \times 50 \times 23} = 0.047 \text{ Wb Ans.}$$

10.7. EFFECTS OF VOLTAGE AND FREQUENCY VARIATIONS

Power transformers are not ordinarily subjected to frequency variations and usually are subject to only modest voltage variations, but it is interesting to consider the effects thereof.

Variations in voltage and/or frequency affects the iron losses (hysteresis and eddy current losses) in a transformer. As long as the flux variations are sinusoidal with respect to line, hysteresis loss (P_h), and eddy current loss (P_e) varies according to the following relations

$$P_h \propto f (\phi_{\max})^x$$

where x lies between 1.5 and 2.5 depending on the grade of iron used in transformer core

$$\text{and } P_e \propto f^2 (\phi_{\max})^2$$

If the transformer is operated with the frequency and voltage changed in the same proportion, the flux density will remain unchanged as obvious from Eq. (10.2) and apparently the no-load current will also remain unaffected.

The transformer can be operated safely at frequency less than rated one with correspondingly reduced voltage. In this case iron losses will be reduced. But if the transformer is operated with increased voltage and frequency in the same proportion, the core losses may increase to an intolerable level. Increase in frequency with constant supply voltage will cause reduction in hysteresis loss and leave the eddy current

losses unaffected. Some increase in voltage could, therefore, be tolerated at higher frequencies, but exactly how much depends on the relative magnitude of the hysteresis and eddy current losses and the grade of iron used in the transformer core.

Example 10.10. A 1 kVA, 220/110 V, 400 Hz transformer is desired to be used at a frequency of 60 Hz. What will be the kVA rating of the transformer at reduced frequency ?

Solution: From Eq. (10.2)

$$V_1 = 4.44 \phi_{\max} N_1 f = 4.44 B_{\max} a N_1 f$$

Assuming flux density in the core remaining unchanged, we have

$$V_1 \propto f$$

$$\text{or } \frac{V_1'}{V_1} = \frac{f'}{f}$$

$$\text{or } V_1' = V_1 \times \frac{f'}{f} = 220 \times \frac{60}{400} = 33 \text{ volts}$$

KVA rating, being proportional to the voltage, will change to $1 \times \frac{33}{220} = 0.15$ or 0.0825 kVA or 82.5 VA Ans.

10.8. TRANSFORMER AS TWO PORT NETWORK

A box representing two-port network is illustrated in Fig. 10.4. The terminal behaviour of a two port device may be specified by two voltages and two currents (voltage V_1 and current I_1 at the input port and voltage V_2 and current I_2 at the output port). The conventional positive polarities* of voltage V_1 and V_2 and currents I_1 and I_2 are shown in the figure. Out of four quantities (V_1 , V_2 , I_1 and I_2) any two may be selected as independent variables and the remaining two can be expressed in terms of the selected independent variables. In general, we are not free to select the independent variables arbitrarily — for example in case of an ideal transformer.

Since the network is linear, the superposition principle applies. Each port voltage is the sum of two components, one arising from I_1 , the other from I_2 . Thus the voltages V_1 and V_2 can be written as

$$\left. \begin{aligned} V_1 &= z_{11}I_1 + z_{12}I_2 \\ \text{and } V_2 &= z_{21}I_1 + z_{22}I_2 \end{aligned} \right\} \dots(10.6)$$

The physical significance of z parameters can be realized by substituting one or the other of two ports on open circuit. If output port is open-circuited, I_2 is zero and the Eqs. (10.6) provide

$$z_{11} = \frac{V_1}{I_1} \text{ with } I_2 = 0 \text{ i.e. } z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \dots(10.7)$$

$$\text{and } z_{21} = \frac{V_2}{I_1} \text{ with } I_2 = 0 \text{ i.e. } z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} \dots(10.8)$$

Thus z_{11} is defined as the input impedance of the network when the output terminals are open-circuited. z_{21} is defined as the transfer impedance with output terminals open-circuited (the open-circuit output voltage resulting from the flow of unit current at the input).

* The currents are taken positive when they enter the device and negative when they leave it.



Fig. 10.4 Conventional Positive Polarities of Voltages and Currents in a Two-Port Network

Similarly with input port open-circuited, $I_1 = 0$ Eqs. (10.6) provide

$$\text{and } z_{22} = \frac{V_2}{I_2} \text{ with } I_1 = 0 \text{ i.e. } z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} \quad \dots(10.9)$$

$$\text{and } z_{12} = \frac{V_1}{I_2} \text{ with } I_1 = 0 \text{ i.e. } z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} \quad \dots(10.10)$$

Thus z_{22} is defined as the output impedance of the network when the input terminals are open-circuited. z_{12} is defined as the transfer impedance with input terminals open-circuited (the open-circuit voltage at the input terminals resulting from the flow of unit current at the output. All passive linear bilateral networks obey the principle of reciprocity and, therefore $z_{12} = z_{21}$.

Transformer is a two-port network but in a transformer two voltages V_2 and V_1 (or two currents I_1 and I_2) cannot be selected as the independent variables because their ratio is a constant and is equal to the turn-ratio of the transformer. Primary terminals of transformer are input terminals while secondary terminals are output terminals.

10.9. TRANSFORMER ON NO LOAD

When the primary of a transformer is connected to the source of ac supply and the secondary is open, the transformer is said to be at no-load (there is no load on secondary).

Consider an ideal transformer whose secondary side is open and the primary winding is connected to a sinusoidal alternating voltage V_1 . The alternating voltage applied to the primary winding will cause flow of alternating current in the primary winding. Since the primary coil is purely inductive and there is no output (secondary being open) the primary draws the magnetising current I_m only. (The function of this current is merely to magnetize the core.) If the transformer is truly ideal, the magnitude of I_m should be zero by virtue of assumption (iv) made in Art 10.4. Since the reluctance of the magnetic circuit is never zero, I_m has definite magnitude. The magnetising current, I_m is small in magnitude and lags behind supply voltage V_1 by 90° . This magnetising current I_m produces an alternating flux ϕ which is, at all times, proportional to the current (assuming permeability of the magnetic circuit to be constant) and, hence, in phase with it.

Let the instantaneous linking flux be given as

$$\phi = \phi_{\max} \sin \omega t$$

The varying flux is linked with both of the windings (primary and secondary) and so induces emfs in the primary and secondary windings. The instantaneous values of induced emfs in the primary and secondary windings will be

$$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt} (\phi_{\max} \sin \omega t)$$

$$= -N_1 \omega \phi_{\max} \cos \omega t = N_1 \omega \phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right)$$

...(10.12)

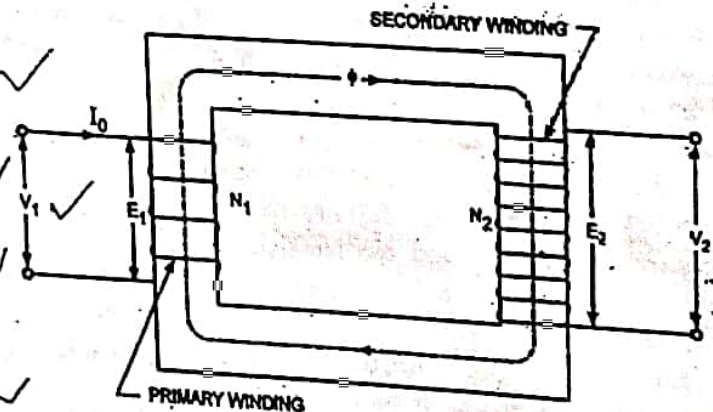


Fig. 10.5 Transformer on No Load

Similarly, $e_2 = N_2 \omega \phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right)$ (10.13)

Since primary winding has no ohmic resistance, (as assumed), therefore, applied voltage to primary winding is to only oppose the induced emf in the primary winding, hence instantaneous applied voltage to primary will be given by

$$v_1 = -e_1 = -N_1 \omega \phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right) \quad (10.14)$$

Comparing Eqs. (10.11), (10.12), (10.13), and (10.14) we conclude that

- Induced emfs in primary and secondary windings, E_1 and E_2 lag behind the main flux ϕ by $\frac{\pi}{2}$ so these emfs (E_1 and E_2) are in phase with each other, as shown in Fig. 10.6 vectorially.
- Applied voltage to the primary winding leads the main flux by $\frac{\pi}{2}$ and is in phase opposition to the induced emf in the primary winding, as shown in Fig. 10.6 vectorially.
- Secondary voltage $V_2 = E_2$ as there is no voltage drop in secondary.

The instantaneous value of applied voltage, induced emfs, flux and magnetising current, in case of an ideal transformer, are illustrated by sinusoidal waves in Fig. 10.7.

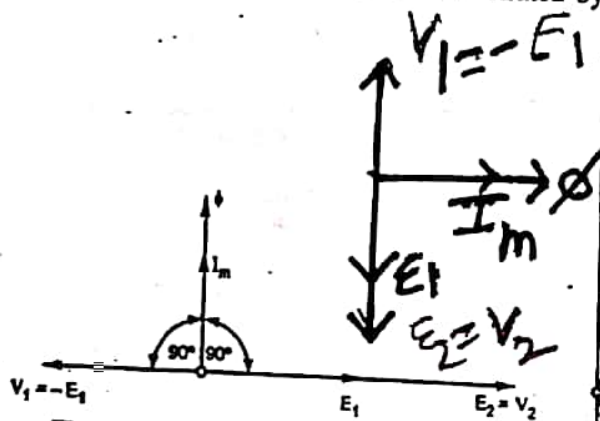


Fig. 10.6 No-Load Phasor Diagram For An Ideal Transformer

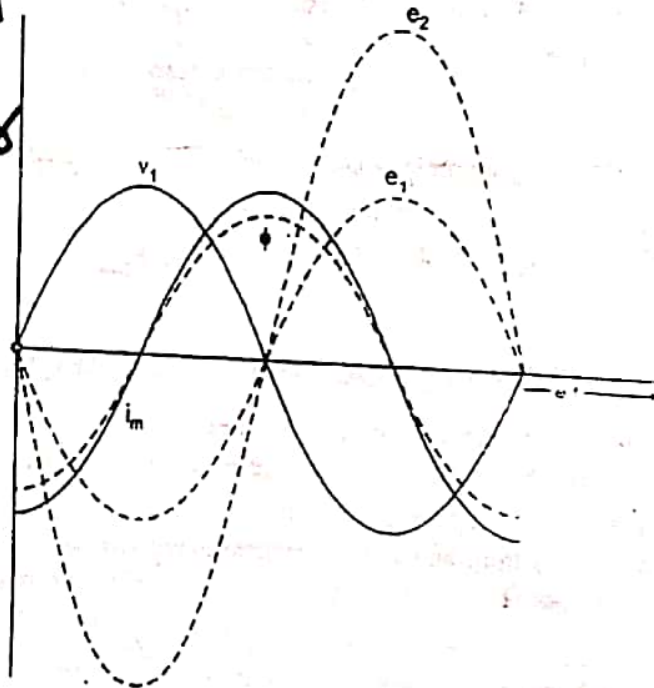
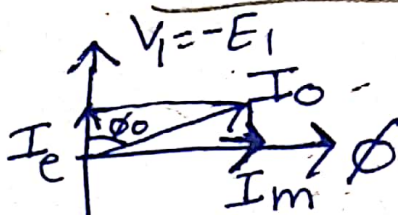


Fig. 10.7

However, when a varying flux is set up in magnetic material, there will be power loss, called the iron or core loss. So the input current to the primary under no-load condition has also to supply the hysteresis and eddy current losses (iron losses) occurring in the core in addition to small amount of copper loss occurring in primary winding (no copper loss occurs in secondary winding on open circuit or on no-load). Hence, the no-load primary current I_0 does not lag behind applied voltage V_1 by 90° but lags behind V_1 by angle $\phi_0 < 90^\circ$. Input power on no-load, $P_o = V_1 I_0 \cos \phi_0$ where $\cos \phi_0$ is the primary power factor under no-load conditions.

As seen from phasor diagram shown in Fig. 10.8, input current to the primary I_0 , called the exciting current, has two components (i) in-phase, active or energy component, I_e used to meet the iron loss in addition to small amount of copper loss occurring in the primary winding and (ii) quadrature component or watt less component called the magnetizing component, I_m used to create the alternating flux in the core.



Thus, $I_e = I_0 \cos \phi_0$
 and $I_m = I_0 \sin \phi_0$
 and $\sqrt{I_e^2 + I_m^2} = I_0$

Angle of lag, $\phi_0 = \tan^{-1} \frac{I_m}{I_e}$

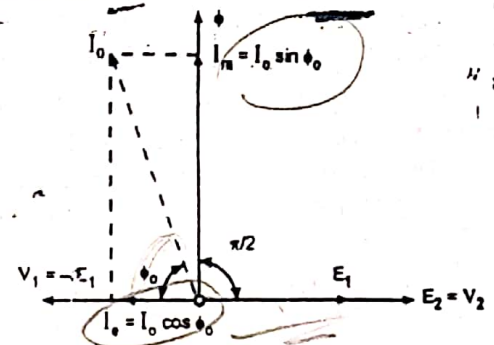


Fig. 10.8

The equivalent circuit of a transformer on no-load is illustrated in Fig. 10.9, where in two components of no-load current; I_e and I_m are represented by currents drawn by a non-inductive resistance R_0 and a pure inductive reactance X_0 respectively. Both these currents are drawn at induced emf $E_1 = V_1$ for resistance-less, no-leakage primary coil; even otherwise $E_1 = V_1$.

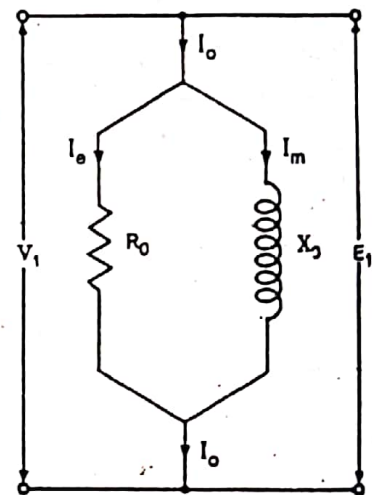


Fig. 10.9

The worthnoting points are given below:

1. The no-load primary current I_0 , called the exciting current, is very small in comparison to the full-load primary current. It ranges from 2 to 5 per cent of full-load primary current.
2. The exciting or no-load current I_0 is made up of a relatively large quadrature or magnetizing component I_m and a comparatively small in-phase or energy-component I_e so the power-factor of a transformer on no-load is very small (usually varies between 0.1 and 0.2 lag). The phase angle between I_0 and V_1 is about 78° to 87° .
3. No-load primary copper loss i.e. $I_0^2 R_1$ is very small and may be neglected. Thus the no-load primary input power is practically equal to the iron loss occurring in the core of the transformer.
4. Phase angle ϕ_0 is also known as the hysteresis angle of advance because it is principally the core-loss which is responsible for shift in the current phasor.
5. Since the permeability of the core material varies with the instantaneous value of the exciting current, so the waveform of the exciting or magnetizing current is not truly sinusoidal. As such, it should not be represented by a phasor because only sinusoidally varying quantities are represented by rotating phasors. But, in practice, it makes no appreciable difference.

Example 10.11. A voltage $v = 200 \sin 314 t$ is applied to the transformer winding in a no-load test. The resulting current is found to be

$$i = 3 \sin (314 t - 60^\circ)$$

Determine the core loss and the parameters of no-load approximate equivalent circuit.

[U.P. Technical Univ. Electrical Machines 2002-03]

Solution : No-load current, $I_0 = \frac{3}{\sqrt{2}} = 2.12 \text{ A}$

Core loss = Input at no-load

$$= V I_0 \cos \phi_0 = \frac{200}{\sqrt{2}} \times \frac{3}{\sqrt{2}} \cos 60^\circ = 150 \text{ W Ans.}$$

Energy component of no-load current, $I_e = I_0 \cos \phi_0 = \frac{3}{\sqrt{2}} \cos 60^\circ = 1.06 \text{ A}$

Magnetising component of no-load current, $I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{(2.12)^2 - (1.06)^2} = 1.836 \text{ A}$

No-load resistance, $R_0 = \frac{V}{I_e} = \frac{200/\sqrt{2}}{1.06} = 133.42 \Omega \text{ Ans.}$

No-load reactance, $X_0 = \frac{V}{I_m} = \frac{200/\sqrt{2}}{1.836} = 77 \Omega \text{ Ans.}$

Example 10.12. A 50 Hz transformer takes 75 W input at 1.5 A and 120 V. If the primary winding resistance is 0.4Ω , find the no-load power factor.

[Pb. Technical Univ. Electrical Engineering June 2000; May 2001]

Solution: No-load power input to transformer, $P_0 = 75 \text{ W}$

No-load primary current, $I_0 = 1.5 \text{ A}$

Primary voltage, $V_1 = 120 \text{ V}$

Primary resistance, $R_1 = 0.4 \Omega$

Copper loss in primary winding, $P_C = I_0^2 R_1 = (1.5)^2 \times 0.4 = 0.9 \text{ W}$

No-load iron loss, $P_i = P_0 - P_C = 75 - 0.9 = 74.1 \text{ W}$

No-load power factor, $\cos \phi_0 = \frac{P_0}{V_1 \times I_0} = \frac{74.1}{120 \times 1.5} = 0.412 \text{ (lag) Ans.}$

Example 10.13. A 100 kVA, 2400/240 V, 50 Hz single phase transformer has an exciting current of 0.64 A and a core loss of 700 W, when its high voltage side is energised at rated voltage and frequency. Calculate the components of no-load current and no-load branch parameters.

[Pb. Technical Univ. Basic Electrical Engineering December 2002]

Solution: Primary voltage, $V_1 = 2,400 \text{ V}$

Exciting or no-load current, $I_0 = 0.64 \text{ A}$

Iron loss = 700 W

Iron-loss current, $I_e = I_0 \cos \phi_0 = \frac{\text{Iron loss}}{V_1} = \frac{700}{2,400} = 0.292 \text{ A Ans.}$

Magnetising component of no-load current,

$I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{(0.64)^2 - (0.292)^2} = 0.57 \text{ A Ans.}$

No-load branch parameters, $R_0 = \frac{V_1}{I_e} = \frac{2,400}{0.292} = 8,220 \Omega \text{ or } 8.22 \text{ k } \Omega \text{ Ans.}$

$X_0 = \frac{V_1}{I_m} = \frac{2,400}{0.57} = 4,210 \Omega \text{ or } 4.21 \text{ k } \Omega \text{ Ans.}$

Example 10.14. Derive the EMF equation of single phase transformer.

The no-load current of a transformer is 4.0 A at 0.25 pf when supplied at 250 V, 50 Hz. The number of turns on the primary winding is 200. Calculate

(i) flux in the core (ii) the core loss, and (iii) magnetising current.

[U.P. Technical Univ. Electrical Machines 2003-04]

Solution:

No-load current, $I_0 = 4 \text{ A}$

No-load power factor, $\cos \phi_0 = 0.25$

Primary induced emf, E_1 = Supply voltage, $V_1 = 250$ V

Supply frequency, $f = 50$ Hz

Number of primary turns, $N_1 = 200$

$$\begin{aligned} \text{(i) Maximum value of flux in the core, } \phi_{\max} &= \frac{E_1}{4.44 f N_1} \\ &= \frac{250}{4.44 \times 50 \times 200} = 0.00563 \text{ Wb or } 5.63 \text{ mWb Ans.} \end{aligned}$$

$$\begin{aligned} \text{(ii) Core loss} &= \text{Input on no-load} \\ &= V_1 I_0 \cos \phi = 250 \times 4 \times 0.25 = 250 \text{ W Ans.} \end{aligned}$$

$$\begin{aligned} \text{(iii) Magnetising current, } I_m &= I_0 \sin \phi_0 \\ &= I_0 \sqrt{1 - (\cos \phi_0)^2} = 4 \sqrt{1 - (0.25)^2} = 3.873 \text{ A Ans.} \end{aligned}$$

10.10. TRANSFORMER ON LOAD

When the secondary circuit of a transformer is completed through an impedance, or load, the transformer is said to be loaded, and current flows through the secondary and the load. The magnitude and phase of secondary current I_2 with respect to secondary terminal voltage V_2 will depend upon the characteristic of load *i.e.* current I_2 will be in phase, lag behind and lead the terminal voltage V_2 respectively when the load is non-inductive, inductive and capacitive.

When the transformer is on no load, as shown in Fig. 10.5, it draws no-load current I_0 from the supply mains. The no-load current I_0 sets up an mmf $N_1 I_0$ which produces flux ϕ in the core. When an impedance is connected across the secondary terminals, as shown in Fig. 10.10, current I_2 flows through the secondary winding. The secondary current I_2 sets up its own mmf and hence creates a secondary flux ϕ_2 . The secondary flux ϕ_2 opposes the main flux ϕ set up by the exciting current I_0 according to Lenz's law. The opposing secondary flux ϕ_2 weakens the main flux ϕ momentarily, so primary back emf E_1 tends to be reduced. So difference of applied voltage V_1 and back emf E_1 increases, therefore, more current is drawn from the source of supply flowing through the primary winding until the original value of flux ϕ is obtained. It again causes increase in back emf E_1 and it adjusts itself as such that there is a balance between applied voltage V_1 and back emf E_1 . Let the additional primary current be I'_1 . The current I'_1 is in phase opposition with secondary current I_2 and is called the *counter-balancing current*. The additional current I'_1 sets up an mmf $N_1 I'_1$ producing flux ϕ_1' in the same direction as that of main flux ϕ and cancels the flux ϕ_2 produced by secondary mmf $N_2 I_2$ being equal in magnitude.

$$\text{So } N_1 I'_1 = N_2 I_2$$

$$\text{or } I'_1 = \frac{N_2}{N_1} I_2$$

The total primary current I_1 is, therefore, phasor sum of primary counter-balancing current I'_1 and no-load current I_0 , which will be approximately equal to I'_1 as I_0 is usually very small in comparison to I'_1 .

$$\therefore I_1 = I'_1 = \frac{N_2}{N_1} I_2$$

$$\text{or } \frac{I_1}{I_2} = \frac{N_2}{N_1} = K \text{ (transformation ratio)}$$

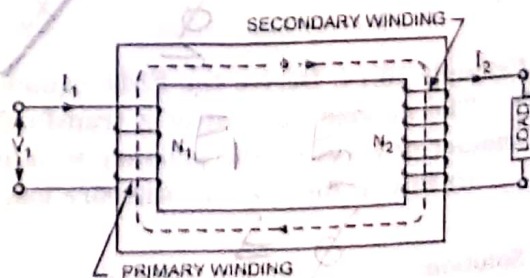


Fig. 10.10 An Ideal Transformer On Load

Hence primary and secondary currents are inversely proportional to their respective turns.

Since the secondary flux ϕ_2 produced by secondary mmf $N_2 I_2$ is neutralized by the flux ϕ_1 produced by mmf $N_1 I_1'$ set up by counter-balancing primary current I_1' , so the flux in the transformer core remains almost constant from no-load to full load.

The phasor diagrams for transformer on non-inductive, inductive and capacitive loads are shown in Figs. 10.11 (a), (b) and (c) respectively.

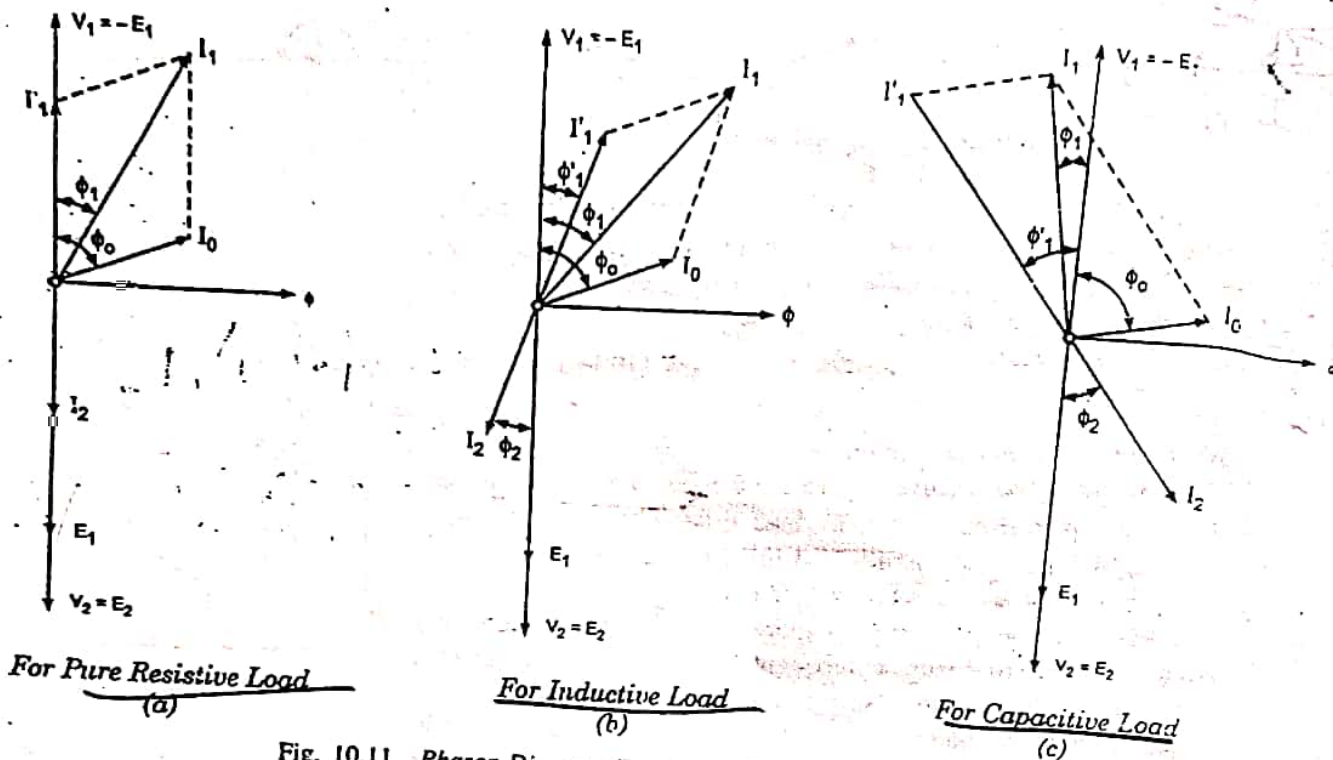


Fig. 10.11 Phasor Diagram For An Ideal Transformer On Load

Since the voltage drops in both of the windings of the transformer are assumed to be negligible, therefore

$$V_2 = E_2 \text{ and } V_1 = -E_1$$

The secondary current I_2 is in phase, lags behind and leads the secondary terminal voltage V_2 by an angle ϕ_2 for pure resistive, inductive and capacitive loads respectively.

The induced primary current I_1' , also known as counterbalancing current, is always in opposition to secondary current I_2 and since no-load current I_0 is very small, the total primary current I_1 is almost opposite in phase to I_2 and K times the secondary current I_2 , where K is transformation ratio.

Note. In phasor diagrams shown in Figs. 10.11 (a), 10.11 (b) and 10.11 (c) no-load current has been drawn on exaggerated scale for sake of clarity.

Example 10.15. Draw and explain phasor diagram of transformer under loaded condition. A single phase 440/110 V transformer takes a no-load current of 4 A at 0.2 power factor. If the secondary supplies a current of 100 A at a power factor of 0.8 lagging determine (i) the current taken by the primary winding (ii) the magnetising reactance and resistance representing core losses.

[U.P. Technical Univ. Electrical Engineering February 2002]

Solution :

Primary emf, $E_1 = 440$ V

Secondary emf, $E_2 = 110$ V

No-load current, $I_0 = 4$ A

No-Load power factor, $\cos \phi_0 = 0.2$ (lag)

Secondary current, $I_2 = 100 \text{ A}$

Load power factor, $\cos \phi_2 = 0.8 \text{ (lag)}$

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{110}{440} = \frac{1}{4}$$

$$\text{Primary balancing current, } I_1 = K I_2 = \frac{1}{4} \times 100 = 25 \text{ A}$$

lagging behind the supply voltage V_1 by an angle ϕ'_1

$$\text{where } \phi'_1 = \phi_2 = \cos^{-1} 0.8 = 36.87^\circ$$

$$\text{No-load current, } I_0 = 4 \text{ A lagging behind the supply voltage } V_1 \text{ by an angle } \phi_0 = \cos^{-1} 0.2 = 78.46^\circ$$

$$\begin{aligned} \text{(i) Primary current, } I_1 &= I'_1 + I_0 = (I'_1 \cos \phi'_1 + I_0 \cos \phi_0) + j(I'_1 \sin \phi'_1 + I_0 \sin \phi_0) \\ &= (25 \times 0.8 + 4 \times 0.2) + j(25 \times 0.6 + 4 \times 0.98) \\ &= (20.8 + j 18.92) \text{ A} = 28.12 \angle 42.29^\circ \text{ A Ans.} \end{aligned}$$

$$\text{Energy component of no-load current, } I_e = I_0 \cos \phi_0 = 4 \times 0.2 = 0.8 \text{ A}$$

$$\text{Magnetising component of no-load current, } I_m = I_0 \sin \phi_0 = 4 \times 0.98 = 3.92 \text{ A}$$

$$\text{(ii) Resistance representing core losses, } R_0 = \frac{V_1}{I_e} = \frac{440}{0.8} = 550 \Omega \text{ Ans.}$$

$$\text{Magnetising reactance, } X_0 = \frac{V_1}{I_m} = \frac{440}{3.92} = 112.25 \Omega \text{ Ans.}$$

10.11. RESISTANCE AND LEAKAGE REACTANCE

In preceding discussions we considered an ideal transformer, which according to our assumptions, has got no resistance in the windings and no leakage flux, but in actual practice it is impossible to obtain such an ideal transformer.

In actual transformer both the windings, primary and secondary windings have finite resistances R_1 and R_2 which cause copper losses and voltage drop in them. The result is that:

(i) The secondary terminal voltage V_2 is less than the secondary induced emf E_2 and is equal to phasor difference of secondary induced emf E_2 and voltage drop in the secondary winding $I_2 R_2$, if magnetic leakage is negligible, i.e.,

$$V_2 = E_2 - I_2 R_2$$

where I_2 is the secondary current and R_2 is the secondary winding resistance.

(ii) Similarly the counter emf of primary $-E_1$ is equal to phasor difference of voltage applied to the primary winding V_1 and voltage drop in the primary winding $I_1 R_1$, provided magnetic leakage is negligible, i.e.,

$$-E_1 = V_1 - I_1 R_1$$

where I_1 is the primary current and R_1 is the primary resistance.

It was previously assumed that the entire flux ϕ , developed by the primary winding, links with and cut every turn of both the primary and secondary windings. In practice, it is impossible to realize this condition.

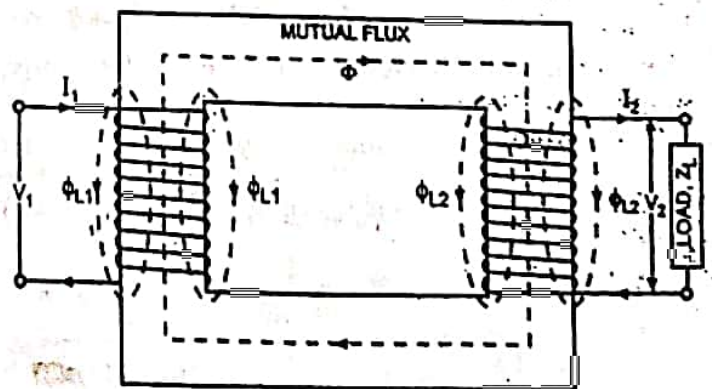


Fig. 10.12 Magnetic Fluxes in a Transformer

However, part of the flux set up by the primary winding links only the primary turns, as illustrated in Fig. 10.12 by flux ϕ_{L1} . Also, some of the flux set up by the secondary winding links only the secondary turns, as illustrated in Fig. 10.12 by ϕ_{L2} . These two fluxes ϕ_{L1} and ϕ_{L2} are known as leakage flux i.e. that flux which leaks out of the core and does not link both windings. The flux which does pass completely through the core and links both the windings is known as the mutual flux and is illustrated as ϕ .

The primary leakage flux ϕ_{L1} linking with the primary winding is produced by primary ampere-turns only, therefore, it is proportional to primary current, number of primary turns being fixed. On no load primary current is so small that leakage flux ϕ_{L1} produced by it can be neglected but on load primary current increases resulting in increase in ampere-turns and hence leakage flux ϕ_{L1} increases. The primary leakage flux ϕ_{L1} is in phase with I_1 and produces self-induced emf E_{L1} given by $E_{L1} = 2\pi f L_1 I_1$ in primary winding, where L_1 is the self inductance of the primary winding produced by primary leakage flux ϕ_{L1} .

The self induced emf E_{L1} , due to primary leakage flux, in the primary winding must lag leakage flux ϕ_{L1} and primary current I_1 by 90° . The emf necessary to balance this counter emf is opposite and equal to it and, therefore, leads the primary current I_1 by 90° . As this emf, induced by the primary leakage flux, is proportional to the current and lags it by 90° , it is nothing more than a reactance voltage and is denoted by $-I_1 X_1$. The component of line voltage that balances this emf is $+I_1 X_1$. The effect of the primary leakage flux, therefore, is to induce an emf that opposes the flow of current to the transformer. The reactance of the primary winding, X_1 can be obtained by dividing self induced emf E_{L1} by the primary current I_1 i.e.

$$X_1 = \frac{E_{L1}}{I_1} = \frac{2\pi f L_1 I_1}{I_1} = 2\pi f L_1 \quad E_{L1} = 2\pi f L_1 I_1$$

Similarly secondary leakage flux ϕ_{L2} is set up by secondary ampere-turns and is proportional to secondary current I_2 . On no load there is no current in secondary winding and, therefore, no leakage flux exists across the secondary winding on no load. On load leakage flux ϕ_{L2} in phase with secondary current I_2 and produces self-induced emf $E_{L2} = 2\pi f L_2 I_2$ in the secondary winding where L_2 is self-inductance of secondary winding due to leakage flux ϕ_{L2} . This is also a reactance voltage, and the component that balances it leads the secondary current by 90° . The secondary reactance X_2 opposes the current flowing out of the transformer and can be obtained by dividing self induced emf in secondary winding, E_{L2} by the secondary current I_2 i.e.

$$X_2 = \frac{E_{L2}}{I_2} = \frac{2\pi f L_2 I_2}{I_2} = 2\pi f L_2$$

The effect of magnetic leakage is, thus to produce in their respective windings emfs of self-inductance which are proportional to the current, and are, therefore, equivalent in effect to the addition of an inductive coil in series with each winding. the reactance of which is called the leakage reactance.

A transformer with magnetic leakage and winding resistance is equivalent to an ideal transformer (having no resistance and leakage reactance) having inductive and resistive coils connected in series with each winding as shown in Fig. 10.13.

Few important points to be kept in mind are given below:

1. The leakage flux links one or the other winding but not both, hence it in no way contributes to the transfer of energy from the primary winding to the secondary winding.
2. The applied voltage to the primary winding, V_1 will have to meet the reactive drop $I_1 X_1$ in addition to $I_1 R_1$. Similarly induced emf in the secondary winding E_2 will have to meet the resistive and reactive drops $I_2 R_2$ and $I_2 X_2$ respectively.
3. Transformation ratio is reduced* due to resistance and magnetic leakage, since these reduce the secondary terminal voltage V_2 on load for a given primary applied voltage V_1 .

4. The main useful flux ϕ decreases slightly with the increase in load but the leakage fluxes are practically proportional to the currents in the respective winding.

5. In actual transformers, the primary and secondary windings are not placed on separate legs, as shown in Fig. 10.1, as due to their being widely separate, large primary and secondary leakage fluxes would result. The leakage fluxes are reduced to a minimum by sectionalizing and interleaving the primary and secondary windings.

10.12. PHASOR DIAGRAM OF ACTUAL TRANSFORMER ON LOAD

Consider a transformer shown in Fig. 10.13 having primary and secondary windings of resistances R_1 and R_2 and reactances X_1 and X_2 respectively. The impedance of primary winding is given by $Z_1 = R_1 + jX_1$ and impedance of secondary winding is given by $Z_2 = R_2 + jX_2$.

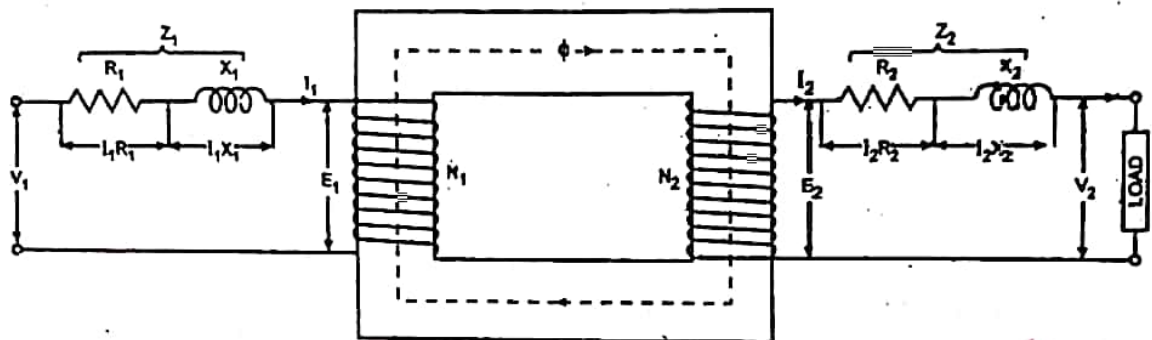


Fig. 10.13 An Equivalent Diagram of Actual Transformer

The phasor diagrams of above transformer on (i) pure resistive, (ii) resistive-inductive, and (iii) resistive-capacitive loads are shown in Fig. 10.14 (a), (b) and (c) respectively.

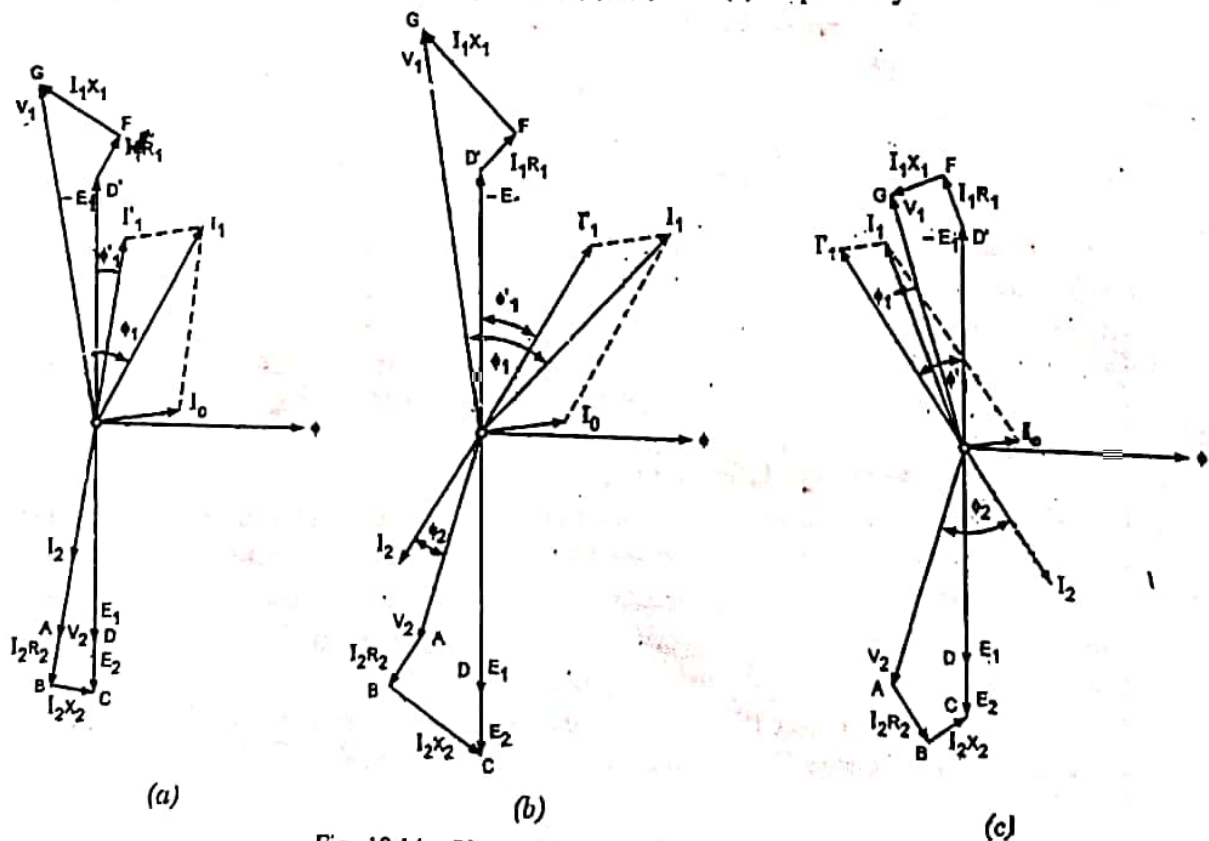


Fig. 10.14 Phasor Diagram of Actual Transformer

Draw OA representing secondary terminal voltage V_2 and OI_2 representing secondary current I_2 in phase as well as magnitude. Since voltage drops due to secondary winding resistance and reactance are $I_2 R_2$ in phase with current I_2 and $I_2 X_2$ leading current I_2 by $\frac{\pi}{2}$ respectively, so draw AB parallel to OI_2 and equal to $I_2 R_2$ in magnitude representing resistive drop in secondary winding and draw BC perpendicular to AB and equal to $I_2 X_2$ in magnitude representing reactive drop of secondary winding. Since phasor sum of terminal voltage V_2 , secondary resistive drop $I_2 R_2$ and secondary reactive drop $I_2 X_2$ is equal to induced emf E_2 in secondary winding so phasor OC represents secondary induced emf E_2 . Hence we have

$$E_2 = V_2 + I_2 (R_2 + j X_2) = V_2 + I_2 Z_2 \quad (10.15)$$

The induced emf E_1 in primary winding is in phase with E_2 and equal to $\frac{N_1}{N_2} E_2$ in magnitude. so

take $CD = \frac{N_1}{N_2} OC$ representing E_1 . Produce DO to D' taking $OD' = OD$ hence representing $(-E_1)$.

The induced primary current I_1 is equal to $-I_2 \frac{N_2}{N_1}$ so draw OI_1' equal to $OI_2 \times \frac{N_2}{N_1}$ by producing line $I_2 O$. Draw line OI_0 representing no-load current in magnitude as well as in phase. The phasor sum of induced primary current I_1' and no-load current I_0 gives primary current represented by phasor OI_1 in Fig. 10.14.

Since voltage drop due to primary winding resistance and reactance are $I_1 R_1$ in phase with primary current I_1 and $I_1 X_1$ leading current I_1 by $\frac{\pi}{2}$ respectively, so draw $D'F$ parallel to OI_1 and equal to $I_1 R_1$ in magnitude representing resistive drop in primary winding and draw FG perpendicular to $D'F$ equal to $I_1 X_1$ in magnitude representing reactive drop in primary winding. As the phasor sum of $(-E_1)$, primary resistive drop and primary reactive drop gives the applied voltage V_1 to primary winding, hence phasor OG represents the applied voltage V_1 in magnitude as well as in phase.

$$\text{i.e. } V_1 = -E_1 + I_1 (R_1 + j X_1) = -E_1 + I_1 Z_1 \quad (10.16)$$

The phase angle ϕ_1 between V_1 and I_1 gives the power factor angle of the transformer.

Since no-load current I_0 , resistive drops $I_1 R_1$ and $I_2 R_2$ and reactive drops, $I_1 X_1$ and $I_2 X_2$ are very small, so neglecting these we have $\phi_2 = \phi_1' = \phi_1 = \phi$, the phase angle of the load. In Fig. no-load current, resistive drops and reactive drops are shown, for clarity, on exaggerated scales.

From phasor diagrams we have

(a) For pure resistive load [phasor diagram 10.14 (a)]

$$E_2 = \sqrt{(V_2 + I_2 R_2)^2 + (I_2 X_2)^2} \approx V_2 + I_2 R_2 \quad (10.17)$$

$$E_1 = \frac{E_2}{K}$$

$$\text{and } V_1 = \sqrt{(E_1 + I_1 R_1)^2 + (I_1 X_1)^2} \approx E_1 + I_1 R_1 \quad (10.18)$$

(b) For resistive-inductive load [phasor diagram 10.14 (b)]

$$E_2 = \sqrt{(V_2 + I_2 R_2 \cos \phi + I_2 X_2 \sin \phi)^2 + (I_2 X_2 \cos \phi - I_2 R_2 \sin \phi)^2} \\ \approx V_2 + I_2 R_2 \cos \phi + I_2 X_2 \sin \phi \quad (10.19)$$

$$E_1 = \frac{E_2}{K}$$

$$\text{and } V_1 = \sqrt{(E_1 + I_1 R_1 \cos \phi + I_1 X_1 \sin \phi)^2 + (I_1 X_1 \cos \phi - I_1 R_1 \sin \phi)^2}$$

$$\approx E_1 + I_1 R_1 \cos \phi + I_1 X_1 \sin \phi \quad \dots(10.20)$$

(c) For resistive-capacitive load [phasor diagram 10.14 (c)]

$$E_2 = \sqrt{(V_2 + I_2 R_2 \cos \phi - I_2 X_2 \sin \phi)^2 + (I_2 X_2 \cos \phi + I_2 R_2 \sin \phi)^2}$$

$$\approx V_2 + I_2 R_2 \cos \phi - I_2 X_2 \sin \phi \quad \dots(10.21)$$

$$E_1 = \frac{E_2}{K}$$

$$\text{and } V_1 = \sqrt{(E_1 + I_1 R_1 \cos \phi - I_1 X_1 \sin \phi)^2 + (I_1 X_1 \cos \phi + I_1 R_1 \sin \phi)^2}$$

$$= E_1 + I_1 R_1 \cos \phi - I_1 X_1 \sin \phi \quad \dots(10.22)$$

Example 10.16. A 230/460 V transformer has a primary resistance of 0.2Ω and a reactance of 0.5Ω and the corresponding values for the secondary are 0.75Ω and 1.8Ω respectively. Find the secondary terminal voltage when supplying (a) 10 A at 0.8 pf lagging (b) 10 A at pf 0.8 leading.

[Allahabad Univ. Elec. Machines-I, 1993]

Solution: Transformation ratio, $K = \frac{V_2}{V_1} = \frac{460}{230} = 2$

(i) When Load is 10 A at 0.8 pf lagging i.e. when

$$I_2 = 10 \text{ A and } \cos \phi = 0.8 \text{ lagging.}$$

$$\text{Primary current } I_1 = K I_2 = 2 \times 10 = 20 \text{ A}$$

From Eq. (10.20)

$$\text{Primary induced emf, } E_1 = V_1 - (I_1 R_1 \cos \phi + I_1 X_1 \sin \phi)$$

$$= 230 - (20 \times 0.2 \times 0.8 + 20 \times 0.5 \times 0.6) = 220.8 \text{ V}$$

$$\text{Secondary induced emf, } E_2 = K E_1 = 2 \times 220.8 = 441.6 \text{ V}$$

From Eq. (10.19)

$$\text{Secondary terminal voltage, } V_2 = E_2 - (I_2 R_2 \cos \phi + I_2 X_2 \sin \phi)$$

$$= 441.6 - (10 \times 0.75 \times 0.8 + 10 \times 1.8 \times 0.6)$$

$$= 441.6 - (6 + 10.8) = 424.8 \text{ V Ans}$$

(ii) When load is 10 A at 0.8 pf leading

From Eq. (10.22)

$$\text{Primary induced emf, } E_1 = V_1 - (I_1 R_1 \cos \phi - I_1 X_1 \sin \phi)$$

$$= 230 - (20 \times 0.2 \times 0.8 - 20 \times 0.5 \times 0.6) = 232.8 \text{ V}$$

and from Eq. (10.21)

$$\text{Secondary terminal voltage, } V_2 = K E_1 - (I_2 R_2 \cos \phi - I_2 X_2 \sin \phi)$$

$$= 2 \times 232.8 - (10 \times 0.75 \times 0.8 - 10 \times 1.8 \times 0.6)$$

$$= 465.6 - (6 - 10.8) = 470.4 \text{ V Ans}$$

10.13. EQUIVALENT RESISTANCE AND REACTANCE

The two independent circuits of a transformer can be resolved into an equivalent circuit to make the calculations simple.

Let resistances and reactances of primary and secondary windings be R_1 and R_2 and X_1 and X_2 ohms respectively and let transformation ratio be K .

Resistive drop in secondary winding = $I_2 R_2$

Reactive drop in secondary winding = $I_2 X_2$

Resistive drop in primary winding = $I_1 R_1$

Reactive drop in primary winding = $I_1 X_1$

Referred To Secondary Side. Since transformation ratio is K , so primary resistive and reactive drops as referred to secondary will be K times, i.e. $K I_1 R_1$ and $K I_1 X_1$ respectively. If I_1 is substituted equal to $K I_2$ then we have primary resistive and reactive drops referred to secondary equal to $K^2 I_2 R_1$ and $K^2 I_2 X_1$ respectively.

Total resistive drop in a transformer = $K^2 I_2 R_1 + I_2 R_2 = I_2 (K^2 R_1 + R_2) = I_2 R_{02}$

Total reactive drop in a transformer = $K^2 I_2 X_1 + I_2 X_2 = I_2 (K^2 X_1 + X_2) = I_2 X_{02}$

The term $(K^2 R_1 + R_2)$ and $(K^2 X_1 + X_2)$ represent the equivalent resistance and reactance respectively of the transformer referred to secondary and let these be represented by R_{02} and X_{02} respectively. Equivalent circuit referred to secondary has been shown in Fig. 10.15 (a).

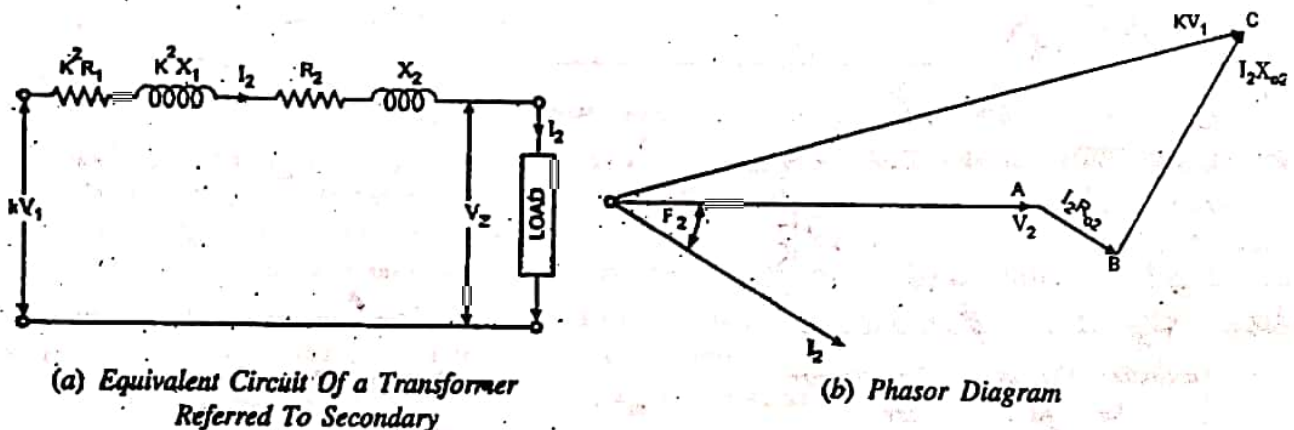


Fig. 10.15

From phasor diagram [Fig. 10.15 (b)]

$$KV_1 = \sqrt{(V_2 + I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi)^2 + (I_2 X_{02} \cos \phi - I_2 R_{02} \sin \phi)^2}$$

where V_2 is secondary terminal voltage, I_2 is secondary current lagging behind the terminal voltage V_2 by ϕ .

Since term $(I_2 X_{02} \cos \phi - I_2 R_{02} \sin \phi)$ is very small as compared to the term $(V_2 + I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi)$, so neglecting the former we have

$$KV_1 = V_2 + I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi$$

$$\text{or } V_2 = KV_1 - I_2 R_{02} \cos \phi - I_2 X_{02} \sin \phi$$

... (10.23)

where V_1 is applied voltage to primary winding.

If load is pure resistive, $\phi = 0$ and $V_2 = KV_1 - I_2 R_{02}$

... (10.24)

If load is capacitive then ϕ should be taken as -ve hence we have

$$V_2 = KV_1 - I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi$$

... (10.25)

Referred To Primary Side. Secondary resistive drop referred to primary

$$= \frac{I_2 R_2}{K} = \frac{I_1 R_2}{K^2} \text{ since } I_2 = \frac{I_1}{K}$$

Secondary reactive drop referred to primary

$$= \frac{I_2 X_2}{K} = \frac{I_1 X_2}{K^2}$$

Total resistive drop in the transformer referred to primary

$$= I_1 R_1 + \frac{I_1 R_2}{K^2} = I_1 \left(R_1 + \frac{R_2}{K^2} \right) = I_1 R_{01}$$

Total reactive drop in the transformer referred to primary

$$= I_1 X_1 + \frac{I_1 X_2}{K^2} = I_1 \left(X_1 + \frac{X_2}{K^2} \right) = I_1 X_{01}$$

The terms $\left(R_1 + \frac{R_2}{K^2} \right)$ and $\left(X_1 + \frac{X_2}{K^2} \right)$ represent the total resistance and reactance of the transformer referred to primary respectively. Let these be represented by R_{01} and X_{01} respectively.

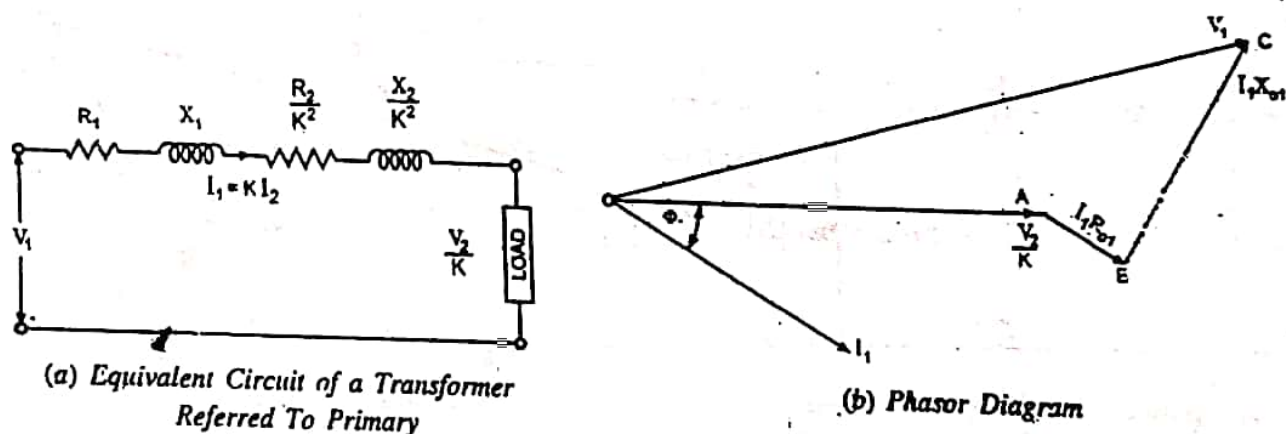


Fig. 10.16

Equivalent circuit referred to primary is shown in Fig. 10.16 (a)

From phasor diagram shown in Fig. 10.16 (b) we have

$$V_1 = \sqrt{\left(\frac{V_2}{K} + I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi \right)^2 + \left(I_1 X_{01} \cos \phi - I_1 R_{01} \sin \phi \right)^2}$$

Since term $(I_1 X_{01} \cos \phi - I_1 R_{01} \sin \phi)$ is very small as compared to the term

$\left(\frac{V_2}{K} + I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi \right)$ so neglecting the former we have

$$V_1 = \frac{V_2}{K} + I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi \quad \dots (10.26)$$

If load is pure resistive i.e. $\phi = 0$ then

$$V_1 = \frac{V_2}{K} + I_1 R_{01} \quad \dots (10.27)$$

If load is capacitive then ϕ should be taken as -ve so we have

$$V_1 = \frac{V_2}{K} + I_1 R_{01} \cos \phi - I_1 X_{01} \sin \phi \quad (10.2)$$

Example 10.17. A 30 kVA, 2000/200 V, single phase, 50 Hz transformer has a primary resistance of 3.5Ω and reactance of 4.5Ω . The secondary resistance and reactance are 0.015Ω and 0.02Ω respectively. Find (i) equivalent resistance, reactance and impedance referred to the primary side (ii) total copper loss in the transformer. [U.P. Technical Univ. Electrical Engineering First Semester 2005-06]

Solution :

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{200}{2000} = 0.1$$

$$\text{Full-load primary current, } I_1 = \frac{30 \times 1000}{2,000} = 15 \text{ A}$$

$$\text{Equivalent resistance referred to primary, } R_{01} = R_1 + \frac{R_2}{K^2} = 3.5 + \frac{0.015}{(0.1)^2} = 5 \Omega \text{ Ans.}$$

$$\text{Equivalent reactance referred to primary, } X_{01} = X_1 + \frac{X_2}{K^2} = 4.5 + \frac{0.02}{(0.1)^2} = 6.5 \Omega \text{ Ans.}$$

$$\text{Equivalent impedance referred to primary, } Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2} = \sqrt{5^2 + 6.5^2} = 8.2 \Omega \text{ Ans.}$$

$$\text{Total copper loss in transformer} = I_1^2 R_{01} = 15^2 \times 5 = 1,125 \text{ Watts Ans.}$$

Example 10.18. A 5 kVA, 240/120 V, single phase transformer supplies rated current to a load at 120 V. Determine the magnitude of load impedance as seen from input terminals of transformer. The load impedance is $(5 + j15) \Omega$. [Pb. Technical Univ. Basic Electrical Engineering December 2002]

Solution:

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{120}{240} = 1/2$$

$$\text{Load impedance referred to secondary, } Z_L = (5 + j15) \Omega$$

$$\text{Load impedance referred to primary, } Z_L = \frac{Z_L}{K^2} = \frac{5+j15}{(1/2)^2} = (20 + j60) \Omega \text{ Ans.}$$

Example 10.19. A 10 kVA, 2,500/250 V single phase transformer has resistances and leakage reactances as follows:

$$\text{HV winding : } R_1 = 4.8 \Omega, X_1 = 11.2 \Omega$$

$$\text{LV winding : } R_2 = 0.048 \Omega, X_2 = 0.112 \Omega$$

With primary voltage held constant at 2,500 V, calculate the secondary terminal voltage when the transformer delivers its rated current at 0.8 pf lagging on the lv side.

[RGPV Basic Electrical Engineering June 2012]

$$\text{Solution : Transformation ratio, } K = \frac{2500}{2,500} = 0.1$$

Equivalent resistance referred to lv side,

$$R_{02} = K^2 R_1 + R_2 = (0.1)^2 \times 4.8 + 0.048 = 0.096 \Omega$$

Equivalent reactance referred to lv side,

$$X_{02} = K^2 X_1 + X_2 = (0.1)^2 \times 11.2 + 0.112 = 0.224 \Omega$$

$$\text{Secondary current, } I_2 = \frac{kVA \times 1,000}{V_2} = \frac{10 \times 1,000}{250} = 40 \text{ A}$$

$$\text{Power factor, } \cos \phi_2 = 0.8 \text{ lagging}$$

$$\sin \phi_2 = \sin \cos^{-1} (0.8) = 0.6$$

$$\begin{aligned} \text{Secondary terminal voltage, } V_2 &= KV_1 - I_2 R_{02} \cos \phi_2 - I_2 X_{02} \sin \phi_2 \\ &= 0.1 \times 2,500 - 40 \times 0.096 \times 0.8 - 40 \times 0.224 \times 0.6 = 241.55 \text{ V Ans.} \end{aligned}$$

10.14. EQUIVALENT CIRCUIT OF A TRANSFORMER (linear dot model)

The equivalent circuit of any device can be quite helpful in predetermination of the behaviour of the device under various conditions of operation and it can be drawn if the equations describing its behaviour are known. If any electrical device is to be analysed and investigated further for suitable modifications, an appropriate equivalent circuit is necessary. The equivalent circuit for electromagnetic devices consists of a combination of resistances, inductances, capacitances, voltage etc. Such an equivalent circuit (or circuit model) can, therefore, be analysed and studied easily by the direct application of electric circuit theory.

As stated above, equivalent circuit is simply a circuit representation of the equations describing the performance of the device. Eqs. (10.15) and (10.16) describe the behaviour of the transformer under load and are helpful in arriving at the transformer equivalent circuit.

Equivalent circuit of a transformer having transformation ratio $K = \frac{E_2}{E_1}$ is shown in Fig. 10.17.

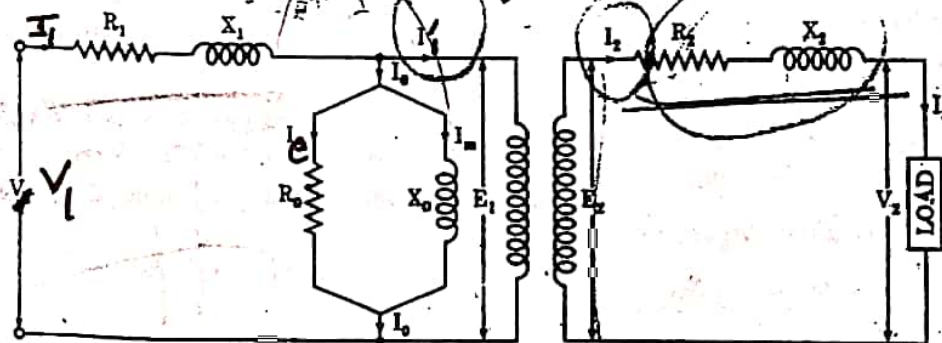


Fig. 10.17 Equivalent Circuit of a Transformer

The induced emf in primary winding E_1 is primary applied voltage V_1 less primary voltage drop. This voltage causes iron loss current I_0 and magnetising current I_m and we can, therefore, represent these two components of no-load current by the current drawn by a non-inductive resistance R_0 and pure reactance X_0 having the voltage E_1 or $(V_1 - \text{primary voltage drop})$ applied across them, as shown in Fig. 10.17.

$$\text{Secondary current, } I_2 = \frac{I_1'}{K} = \frac{I_1 - I_0}{K}$$

Terminal voltage V_2 across load is induced emf E_2 in secondary winding less voltage drop in secondary winding.

The equivalent circuit can be simplified by transferring the voltage, current and impedance to the primary side. After transferring the secondary voltage, current and impedance to primary side equivalent circuit is reduced to that shown in Fig. 10.18.

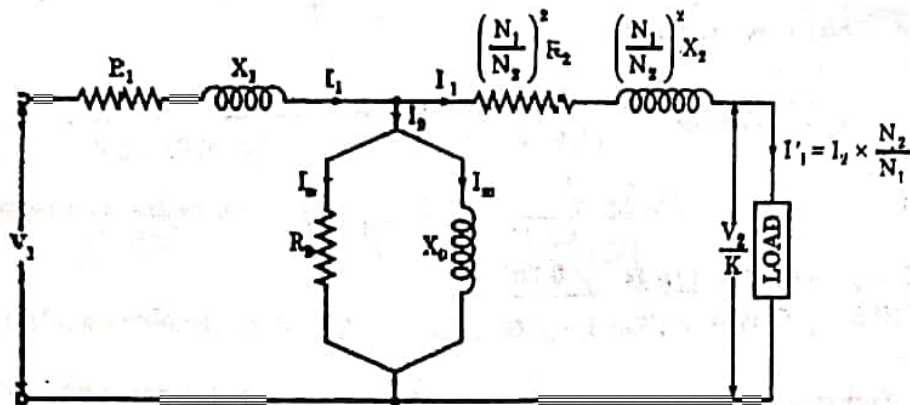


Fig. 10.18 Equivalent Circuit of a Transformer With All Secondary Impedances Transferred To Primary Side

The equivalent circuit diagram can further be simplified by transferring the resistance R_0 and reactance X_0 towards left end, as shown in Fig. 10.19. The error introduced by doing so is very small and can be neglected.

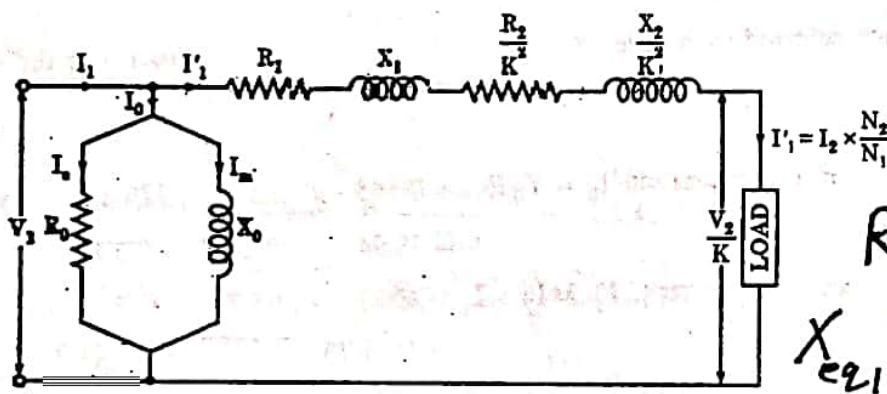


Fig. 10.19 Approximate Equivalent Circuit of a Transformer

No-load current I_0 is hardly 3 to 5 percent of the full-load rated current, the parallel branch consisting of resistance R_0 and reactance X_0 can be omitted without introducing any appreciable error in the behaviour of the transformer under loaded condition. Such a circuit is shown in Fig. 10.16 (a). The equivalent circuit referred to secondary side (neglecting no-load current I_0) is illustrated in Fig. 10.15 (a).

Example 10.20. Derive emf equation for a single phase transformer. A 25 kVA, 2200/220 V, 50 Hz single phase transformer has the following parameters.

$$r_1 = 1.2 \, \Omega ; x_1 = 6.0 \, \Omega ; r_2 = 0.03 \, \Omega ; x_2 = 0.06 \, \Omega \text{ hv side}$$

$$r_2 = 0.03 \, \Omega ; x_2 = 0.06 \, \Omega \text{ (lv side)}$$

Draw the equivalent circuit referred to lv side for a 0.8 pf lagging load being supplied at 220 V.

[U.P. Technical Univ. Electrical Engineering June 2001]

Solution: Rated secondary current, $I_2 = \frac{\text{Rated kVA} \times 1,000}{\text{Rated secondary voltage}} = \frac{25 \times 1,000}{220} = 113.64 \text{ A}$

Secondary resistance, $r_2 = 0.03 \, \Omega$

Secondary leakage reactance, $x_2 = 0.06 \, \Omega$

Power factor $\cos \phi = 0.8$; $\sin \phi = 0.6$

Secondary induced emf,

$$E_2 = \sqrt{(V_2 + I_2 R_2 \cos \phi + I_2 X_2 \sin \phi)^2 + (I_2 X_2 \cos \phi - I_2 R_2 \sin \phi)^2}$$

$$R_{eq2} = R_2 + R_1 \left(\frac{N_2}{N_1} \right)^2$$

↓
equivalent resistance referred to secondary side

$$= \sqrt{(220 + 113.64 \times 0.03 \times 0.8 + 113.64 \times 0.06 \times 0.6)^2 + (113.64 \times 0.06 \times 0.8 - 113.64 \times 0.03 \times 0.6)^2}$$

$$= \sqrt{(226.82)^2 + (3.41)^2} = 226.85 \text{ V}$$

$$E_2 = (226.82 + j 3.41) \text{ V} = 226.85 \angle 0.86^\circ \text{ V}$$

$$I_2 = 113.64 (0.8 - j 0.6) \text{ A} = (90.91 - j 68.2) \text{ A}$$

$$\text{Transformation ratio, } K = \frac{E_2}{E_1} = \frac{220}{2,200} = 0.1$$

Shunt branch resistance referred to *lv* side

$$r'_c = r_c \times K^2 = 1,000 \times (0.1)^2 = 10 \Omega$$

Shunt branch reactance referred to *lv* side

$$x'_m = x_m \times K^2 = 300 \times (0.1)^2 = 3 \Omega$$

$$\text{Shunt admittance referred to } lv \text{ side, } Y_0 = \frac{1}{r'_c} - \frac{j}{x'_m} = \frac{1}{10} - \frac{j}{3} = (0.1 - j 0.333) \text{ S}$$

$$= 0.348 \angle -73.3^\circ \text{ S}$$

$$\text{Exciting current, } I_0 = Y_0 E_2 = 0.348 \angle -73.3^\circ \times 226.85 \angle 0.86^\circ$$

$$= 78.94 \angle -72.44^\circ = (23.82 - j 75.26) \text{ A}$$

$$\text{Equivalent secondary current, } I'_2 = I_2 + I_0 = (90.91 - j 68.2) + (23.82 - j 75.26)$$

$$= (114.73 - j 143.46) = 183.7 \angle -51.35^\circ \text{ A}$$

$$KV_1 = E_2 + K^2 I'_2 (R_1 + X_1)$$

$$= (226.82 + j 3.41) + (114.73 - j 143.46) (0.012 + j 0.06) = 237 \text{ V}$$

Equivalent circuit is shown in Fig. 10.20.

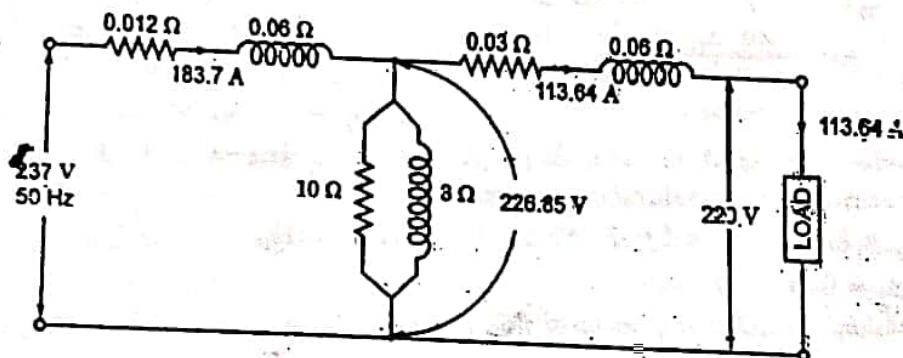


Fig. 10.20

10.15. VOLTAGE REGULATION

The way in which the secondary terminal voltage varies with the load depends upon load current, the internal impedance and the load power factor. The change in secondary terminal voltage from no-load to full-load, with primary voltage and frequency held constant, is termed the *inherent regulation*.

The figure of merit which determines the voltage characteristic of a transformer is termed as *voltage regulation*. It is usually expressed as a percentage of the rated terminal (secondary) voltage. It is defined as the change in magnitude of terminal (secondary) voltage when full-load (rated load) of specified power factor supplied at rated voltage is thrown off (reduced to no-load) with primary voltage and frequency held constant, as percentage of the rated terminal voltage.

If V_2 is the secondary (terminal) voltage at full-load at specified power factor and E_2 is the secondary (terminal) voltage at no load, then percentage regulation is given as

$$\text{Voltage regulation} = \frac{E_2 - V_2}{\text{Secondary rated voltage}} \times 100$$

As per IS, the secondary rated voltage of a transformer is equal to the secondary voltage at no load i.e. E_2 .

$$\begin{aligned} \text{So percentage regulation} &= \frac{E_2 - V_2}{E_2} \times 100 = \frac{\text{Voltage drop in transformer at full load}}{\text{No-load rated secondary voltage } E_2} \times 100 \\ &= \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{E_2} \times 100 \quad (10.29) \end{aligned}$$

When the power factor is leading, the percentage regulation is given as

$$\text{Percentage regulation} = \frac{I_2 R_{02} \cos \phi - I_2 X_{02} \sin \phi}{E_2} \times 100 \quad (10.30)$$

The percentage regulation can be given as

$$\% \text{ regulation} = \frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{E_2} \times 100$$

+ve sign for lagging power factor and
-ve sign for leading power factor

$$\begin{aligned} &= \left(\frac{I_2 R_{02}}{E_2} \cos \phi \pm \frac{I_2 X_{02}}{E_2} \sin \phi \right) \times 100 \\ &= (\text{PU resistance} \times \cos \phi \pm \text{PU reactance} \sin \phi) \times 100 \quad (10.31) \end{aligned}$$

Voltage regulation of a transformer, on an average, is about 4 percent.

From the consumer's view point, voltage regulation due to variations in load is undesirable and should be kept smallest possible.

10.15.1 Condition For Zero Regulation.

$$\text{Regulation} = \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{E_2}$$

Regulation will be zero if the numerator will be equal to zero.

$$\text{or } I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi = 0$$

$$\text{or } \tan \phi = \frac{-R_{02}}{X_{02}} \quad \dots (10.32)$$

The -ve sign indicates that zero regulation occurs at a leading power factor.

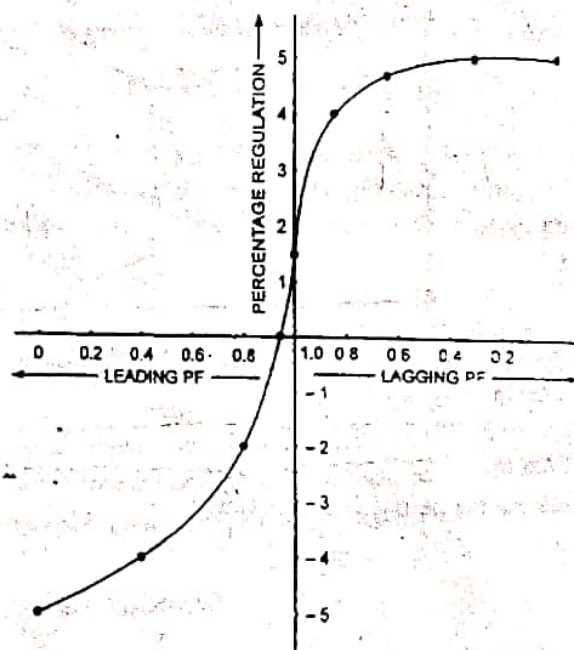
Example 10.21. A 100 kVA, single phase transformer has $N_1 = 400$, $N_2 = 80$, $R_1 = 0.3 \text{ ohm}$, $X_1 = 1.1 \text{ ohm}$, $R_2 = 0.01 \text{ ohm}$, $X_2 = 0.035 \text{ ohm}$. The applied voltage is 2200 V. Find :

(i) Total impedance as referred to primary.

(ii) Voltage regulation and output voltage at 0.8 power factor lagging at full load.

[U.P. Technical Univ; Electrical Machine 2005-06]

Solution: Transformation ratio, $K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{80}{400} = 0.2$



Percentage Regulation Vs Power Factor
($R = 0.02 \text{ PU}$, $X = 0.06 \text{ PU}$)

Fig 10.21

Primary voltage, $V_1 = 2,200$ V

Secondary voltage on no load, $E_2 = K E_1 = K V_1 = 0.2 \times 2,200 = 440$ V

Equivalent resistance as referred to primary,

$$R_{01} = R_1 + \frac{R_2}{K^2} = 0.3 + \frac{0.01}{(0.2)^2} = 0.55 \Omega$$

Equivalent reactance as referred to primary,

$$X_{01} = X_1 + \frac{X_2}{K^2} = 1.1 + \frac{0.035}{(0.2)^2} = 1.975 \Omega$$

(i) Equivalent impedance as referred to primary,

$$Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2} = (0.55 + j 1.975) \Omega \text{ or } 2.05 \angle 74.44^\circ \Omega \text{ Ans.}$$

$$\text{Primary current, } I_1 = \frac{\text{kVA} \times 1,000}{V_1} = \frac{100 \times 1,000}{2,200} = 45.45 \text{ A}$$

Power factor, $\cos \phi = 0.8$ (lagging)

$$\sin \phi = \sqrt{1 - \cos^2 \phi} = \sqrt{1 - 0.8^2} = 0.6$$

Voltage drop in transformer windings at full load

$$\begin{aligned} &= I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi \\ &= 45.45 \times 0.55 \times 0.8 + 45.45 \times 1.975 \times 0.6 = 73.86 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Output voltage, } V_2 &= [V_1 - (I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi)] \times K \\ &= (2,200 - 73.86) \times 0.2 = 425.228 \text{ V Ans.} \end{aligned}$$

$$\text{Voltage regulation} = \frac{I_1 R_{01} \cos \phi + I_1 X_{01} \sin \phi}{V_1} \times 100 = \frac{73.86}{2,200} \times 100 = 3.357\% \text{ Ans.}$$

Example 10.22. The full load copper loss on the hv side of 100 kVA, 11000/317 V, 1-phase transformer is 0.62 kW and on the lv side is 0.48 kW. (i) Calculate R_1 , R_2 and R'_2 in ohms (ii) the total reactance is 4 percent, find X_1 , X_2 and X'_2 in ohms if the reactance is divided in the same proportion as resistance.

Solution: HV side full-load current, $I_1 = \frac{\text{Rated KVA} \times 1,000}{V_1} = \frac{100 \times 1,000}{11,000} = 9.09 \text{ A}$

$$\text{LV side full-load current, } I_2 = \frac{100 \times 1,000}{317} = 315.46 \text{ A}$$

$$\text{Primary winding resistance, } R_1 = \frac{\text{Full-load copper loss in primary winding}}{I_1^2}$$

$$= \frac{0.62 \times 1,000}{(9.09)^2} = 7.5 \Omega \text{ Ans.}$$

$$\text{Secondary winding resistance, } R_2 = \frac{\text{Full-load copper loss in secondary winding}}{I_2^2}$$

$$= \frac{0.48 \times 1,000}{(315.46)^2} = 0.00482 \Omega$$

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{317}{11,000} = 0.02882$$

$$\text{Secondary resistance referred to primary, } R'_2 = \frac{R_2}{K^2} = \frac{0.00482}{(0.02882)^2} = 5.8 \, \Omega \text{ Ans.}$$

$$\text{Percentage reactance referred to hv side} = \frac{I_1 X_{01}}{V_1} \times 100 = 4$$

or transformer total reactance referred to primary,

$$X_{01} = \frac{4 \times V_1}{100 I_1} = \frac{4 \times 11,000}{100 \times 9.09} = 48.4 \, \Omega$$

$$\text{Given that } \frac{R_1}{R'_2} = \frac{X_1}{X'_2}$$

$$\text{or } \frac{R_1 + R'_2}{R'_2} = \frac{X_1 + X'_2}{X'_2}$$

$$\text{or } \frac{7.5 + 5.8}{5.8} = \frac{48.4}{X'_2}$$

$$\text{or secondary reactance referred to primary } X'_2 = \frac{48.4 \times 5.8}{13.3} = 21.1 \, \Omega \text{ Ans.}$$

$$\text{Primary winding reactance, } X_1 = X_{01} - X'_2 = 48.4 - 21.1 = 27.3 \, \Omega \text{ Ans.}$$

$$\text{Secondary winding reactance, } X_2 = K^2 X'_2 = \left(\frac{317}{11,000} \right)^2 \times 21.1 = 0.0175 \, \Omega \text{ Ans.}$$

Example 10.23. Calculate the regulation of a transformer in which ohmic drop is 1% and reactance drop 5% of the voltage at the following power factors (i) 0.8 lagging (ii) 0.8 leading.

[U.P. Technical Univ. Elec. Engineering Second Semester 2005-06]

Solution :

$$\text{Per unit resistance} = \text{Per unit resistance drop} = \frac{1}{100} = 0.01$$

$$\text{Per unit reactance} = \text{Per unit reactance drop} = \frac{5}{100} = 0.05$$

(i) When power factor, $\cos \phi = 0.8$ lagging; $\sin \phi = 0.6$

$$\begin{aligned} \text{Percentage voltage regulation} &= (\text{PU resistance} \times \cos \phi + \text{PU reactance} \times \sin \phi) \times 100 \\ &= (0.01 \times 0.8 + 0.05 \times 0.6) \times 100 = 3.8\% \text{ Ans.} \end{aligned}$$

(ii) When power factor, $\cos \phi = 0.8$ (leading); $\sin \phi = -0.6$

$$\text{Percentage voltage regulation} = [0.01 \times 0.8 + 0.05 \times (-0.6)] \times 100 = -2.2\% \text{ Ans.}$$

Example 10.24. Define voltage regulation of a transformer. A transformer has copper loss of 1.5% and reactance-drop of 3.5% when tested at full-load. Calculate its full-load regulation at :

(i) 0.8 pf lagging, (ii) 0.8 pf leading.

[U.P. Technical Univ. Electrical Machines 2003-04]

Solution :

$$\text{Per unit resistance} = \frac{\text{Percentage copper loss}}{100} = \frac{1.5}{100} = 0.015$$

$$\text{Per unit reactance} = \text{Per unit reactance drop} = \frac{3.5}{100} = 0.035$$

(i) When power factor, $\cos \phi = 0.8$ lagging; $\sin \phi = 0.6$

Percentage voltage regulation = $(0.015 \times 0.8 + 0.035 \times 0.6) \times 100 = 3.3\%$ Ans.

(ii) When power factor, $\cos \phi = 0.8$ leading; $\sin \phi = -0.6$

and Percentage voltage regulation = $[0.015 \times 0.8 + 0.035 (-0.6)] \times 100 = -0.9\%$ Ans.

Example 10.25. Define voltage regulation of transformer. A 100 kVA, 2,200/220 V transformer has leakage reactance drop of 8% and resistance drop of 2%. Find its voltage regulation at full load and 0.8 pf lagging. Find also the power factor at which the regulation will be zero.

[U.P. Technical Univ. Electro-mechanical Energy Conversion-I, 2002-03]

Solution: Regulation = PU resistance $\times$ 0.8 + PU reactance $\times$ 0.6
 $= 0.02 \times 0.8 + 0.08 \times 0.6 = 0.064$ or 6.4% Ans.

Regulation will be zero when power factor

$$\cos \phi = \cos \tan^{-1} \frac{-R_{02}}{X_{02}} = \cos \tan^{-1} - \frac{I_2 R_{02} / E_2}{I_2 X_{02} / E_2} = \cos \tan^{-1} - \frac{0.02}{0.08} = 0.97 \text{ (lead) Ans.}$$

10.16. TRANSFORMER LOSSES

As already mentioned in Art 10.1, the transformer is a static machine and, therefore, there are no friction or windage losses. The various power losses occurring in a transformer are enumerated below:

1. **Iron or Core Losses.** Iron loss is caused by the alternating flux in the core and consists of hysteresis and eddy current losses.

(a) **Hysteresis Loss.** The core of a transformer is subjected to an alternating magnetizing force and for each cycle of emf a hysteresis loop is traced out. The hysteresis loss per second is given by the equation

$$\text{Hysteresis loss, } P_h = \eta' (B_{\max})^x f V \text{ joules per second or watts} \quad \dots(10.33)$$

where f is the supply frequency in Hz, V is the volume of core in cubic metres, η' is the hysteresis coefficient, $B_{\max}$ is peak value of flux density in the core and x lies between 1.5 and 2.5 depending upon the material and is often taken as 1.6.

(b) **Eddy Current Loss.** If the magnetic circuit is made up of iron and if the flux in the circuit is variable, currents will be induced by induction in the iron circuit itself. All such currents are known as eddy currents.

Eddy currents result in a loss of power, with consequent heating of the material.

The eddy current loss is given by

$$P_e = K_e (B_{\max})^2 f^2 t^2 V \text{ watts or joules per second} \quad \dots(10.34)$$

From the above expression for eddy current loss in a thin sheet it is obvious that eddy current loss varies (i) as the square of maximum flux density (ii) as the square of the frequency and (iii) as the square of thickness of laminations.

The hysteresis and eddy current losses depend upon the maximum flux density in the core and supply frequency. Since it has been determined that the mutual flux varies somewhat with the load (its variation being 1 to 3 % from no-load to full-load), the core losses will vary somewhat with the load and its power factor. It may be emphasized here that *core-losses are assumed to remain constant from no-load to full load*, the variations in losses from no-load to full-load being very small and negligible.

These losses are determined from the open-circuit test (refer art 10.17.1)

The input to the transformer with rated voltage applied to the primary and secondary open-circuited is equal to the core loss.

These losses are minimized by using steel of high silicon content for the core and by using very thin laminations (0.3 mm to 0.5 mm) insulated from each other either by insulating varnish or by layer of papers.

$$\begin{aligned}\text{Full-load Cu loss} &= I_1^2 R_1 + I_2^2 R_2 \\ &= I_1^2 R_{eq1} \\ &= I_2^2 R_{eq2}\end{aligned}$$

Transformers

2. Copper or Ohmic Losses. These losses occur due to ohmic resistance of the transformer windings. I_1 and I_2 are the primary and secondary currents respectively and R_1 and R_2 are the respective resistances of primary and secondary windings then copper losses occurring in primary and secondary windings will be $I_1^2 R_1$ and $I_2^2 R_2$ respectively. So total copper losses will be $(I_1^2 R_1 + I_2^2 R_2)$. These losses vary as the square of the load current or kVA. For example if the copper losses at full load are P_c then copper losses

at one-half or one-third of full load will be $\left(\frac{1}{2}\right)^2 P_c$ or $\left(\frac{1}{3}\right)^2 P_c$ i.e. $\frac{P_c}{4}$ or $\frac{P_c}{9}$ respectively.

Copper losses are determined on the basis of constant equivalent resistance R_{eq} determined from the short-circuit test (refer Art 10.17.3) and then corrected to 75°C (since the standard operating temperature of electrical machines is taken 75°C).

Example 10.26. The flux in a magnetic core is alternating sinusoidally at a frequency of 600 Hz. The maximum flux density is 2 T and the eddy current loss is 15 W. Find the eddy current loss if the frequency is raised to 800 Hz and the maximum flux density is reduced to 1.5 T.

[Pb. Univ. Elements of Electrical Engineering, Nov. 1994]

Solution: Since from Eq. (10.34)

The eddy current loss, $P_e \propto B_{\max}^2 f^2$

$$\therefore \frac{P_{e2}}{P_{e1}} = \left(\frac{B_{2\max}}{B_{1\max}} \right)^2 \left(\frac{f_2}{f_1} \right)^2$$

$$\text{or } P_{e2} = P_{e1} \times \left(\frac{B_{2\max}}{B_{1\max}} \right)^2 \left(\frac{f_2}{f_1} \right)^2 = 15 \times \left(\frac{1.5}{2} \right)^2 \times \left(\frac{800}{600} \right)^2 = 15 \text{ watts Ans.}$$

10.17. TESTING OF TRANSFORMERS

The main difficulties encountered in testing of large power transformers by direct loading are (i) wastage of large amount of energy and (ii) a stupendous (impossible for large transformers) task of arranging a load large enough for direct loading. The performance characteristics of a transformer can be conveniently computed from the knowledge of its equivalent circuit parameters which, in turn, may be determined by conducting simple tests called the open-circuit or no-load test and short-circuit or impedance test involving very little power consumption (power needed to supply the losses incurred).

10.17.1. Open-Circuit Test (or No-Load Test). The purpose of this test is to determine the core (or iron or excitation) loss, P_i and no-load current I_0 and thereby the shunt branch parameters R_0 and X_0 of the equivalent circuit.

In this test, one of the windings (usually high voltage winding) is kept open circuited and the rated voltage at rated frequency is applied to the other winding, as shown Fig. 10.22. No doubt, the core loss will be the same whether the measurements are made on lv winding or hv winding so long as the rated voltage of that winding is applied to it but in case the measurements are made on hv winding, the voltage required to be applied would often be inconveniently large while the current I_0 would be inconveniently small.

(Either an auto-transformer or a voltage divider (VD) is used for varying the voltage applied to the low-voltage winding. Ammeter A and wattmeter W are connected to measure no-load current I_0 and input power P_0 . Voltmeter V is connected to measure the applied voltage.)

Since no current flows in the open-circuited secondary, the current in the primary will be merely that necessary to magnetize the core at normal voltage. Moreover, this magnetising current is a very small fraction of the full load current (usually 3 to 10 % of full load current) and may be neglected as far as the copper loss is concerned consequently, the test gives core loss alone practically.

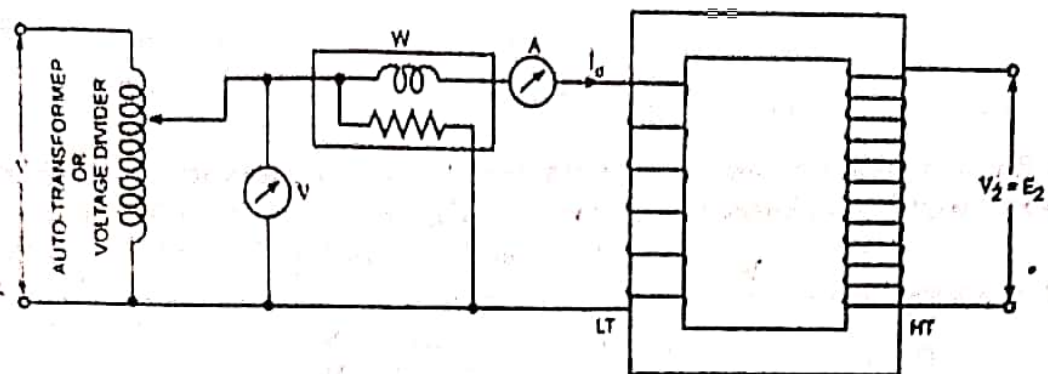


Fig. 10.22 Circuit Diagram For Open-Circuit Test

With normal voltage applied to the primary, normal flux will be set up in the core and, therefore, normal iron (or core) loss will occur which are recorded by a wattmeter W. ✓

The open-circuit test gives enough data to compute the equivalent circuit constants R_0 , X_0 , no-load power factor $\cos \phi_0$, no-load current I_0 and no load power loss (iron loss) of a transformer.

✓ Iron loss, $P_i = \text{Input power on no load} = P_0 \text{ watts (say)}$ ✓

No-load current = I_0 amperes

Applied voltage to primary = V_1 volts

$$P_0 = V_1 I_0 \cos \phi_0 \quad \cos \phi_0 = \frac{P_0}{V_1 I_0} \quad \dots(10.35)$$

$$\text{No-load current energy component, } I_e = I_0 \cos \phi_0 = \frac{P_0}{V_1} \quad \dots(10.36)$$

$$\text{No-load current magnetizing component, } I_m = \sqrt{I_0^2 - I_e^2} \quad \cos \phi_0 = \frac{P_0}{V_1 I_0} \quad \dots(10.37)$$

$$\text{Equivalent circuit parameter, } R_0 = \frac{V_1}{I_e} = \frac{V_1^2}{P_0} \quad \dots(10.38)$$

$$\text{Equivalent circuit parameter, } X_0 = \frac{V_1}{I_m} = \frac{V_1}{\sqrt{I_0^2 - I_e^2}} \quad \dots(10.39)$$

Note 1. Since no-load current I_0 is very small, therefore, pressure coils of wattmeter and the voltmeter should be connected such that the currents drawn by them do not flow through the current coils of the wattmeter and ammeter.

2. Since power factor at no-load is quite low (in the range of 0.1–0.2 lag) a low power factor wattmeter should be used to ensure accurate measurements.

3. The error due to power loss in ammeter can be eliminated by short-circuiting the ammeter while reading wattmeter.

4. Sometimes a high resistance voltmeter is connected across the secondary to indicate the emf induced in the secondary (hv winding). This helps in determination of transformation ratio K .

5. It must, however, be remembered that in making this test, hv side is hot and, therefore, its terminals must be properly insulated.

10.17.2. Separation of Hysteresis and Eddy Current Losses. In Art 10.16 it has been mentioned that

$$\text{hysteresis loss, } P_h \propto (B_{\max})^{1.6} f$$

$$\text{and eddy current loss, } P_e \propto (B_{\max})^2 f^2$$

It shows that for constant peak flux density the hysteresis loss varies as the frequency of supply while eddy current loss varies as the square of supply frequency.

For any transformer $\frac{V}{f} \propto B_{\max}$ and for a particular value of $\frac{V}{f}$ or of peak flux density core loss per cycle are given as

$$\frac{P_i}{f} = A - Bf \quad \dots(10.40)$$

where A is a constant and is equal to $K_h (B_{\max})^{1.6}$ and B is another constant having value of $K_e (B_{\max})^2$.

The values of constants A and B in Eq. (10.40) can be determined by performing open-circuit test on the transformer. The connections are the same as for open-circuit test (Fig. 10.22) except that the primary of the transformer is connected to variable voltage variable-frequency source (a dc motor driven alternator). During this test, the applied voltage V and frequency f are varied together (by adjusting the excitation

and speed of the alternator feeding the transformer under test) so as to keep $\frac{V}{f}$ (and, therefore, $B_{\max}$) constant. The wattmeter connected in the circuit indicates the iron loss P_i .

For a series of frequencies, with corresponding changes in applied voltage, the measured input power divided by frequency $\left(\frac{P_i}{f}\right)$ and plotted against frequency will give a straight line, as shown in Fig. 10.23.

The intercept on the vertical axis will give the value of constant A while the slope of the line will give the value of constant B. Knowing the values of constants A and B the hysteresis and eddy current loss components for any given frequency can be determined.

10.17.3. Short-Circuit Test (or Impedance Test). The purpose of this test is to determine full-load copper loss and equivalent resistance and equivalent reactance referred to metering side.

In this test, the terminals of secondary winding (usually of low-voltage winding*) are short-circuited by a thick wire or strip or through an ammeter (which may serve the additional purpose of indicating secondary rated load current) and variable low voltage is applied to the primary through an auto-transformer or potential divider, as shown in Fig. 10.24. The transformer now becomes equivalent to a coil having an impedance equal to impedance of both the windings.

* Voltage required for the short-circuit test is about 5 per cent of the rated value. For a 200 kVA, 2200/220 V transformer, test on high voltage side would need the voltage (to be applied) of $2,200 \times \frac{5}{100}$ i.e. 110 V (which is standard voltage for instrument coils) and a current of $\frac{200 \times 1,000}{2,200} = 91 \text{ A}$.

If the test is conducted on low voltage side of the above transformer, the voltage needed would be $220 \times \frac{5}{100} = 11 \text{ V}$ and the current would be $\frac{200 \times 1,000}{220} = 910 \text{ A}$ (very high). At this low voltage, high precision would not be readily obtainable with ordinary instruments.

Thus we see that if the measurements were made on low voltage side, the voltage needed would be inconveniently low, while the current would often be inconveniently large.

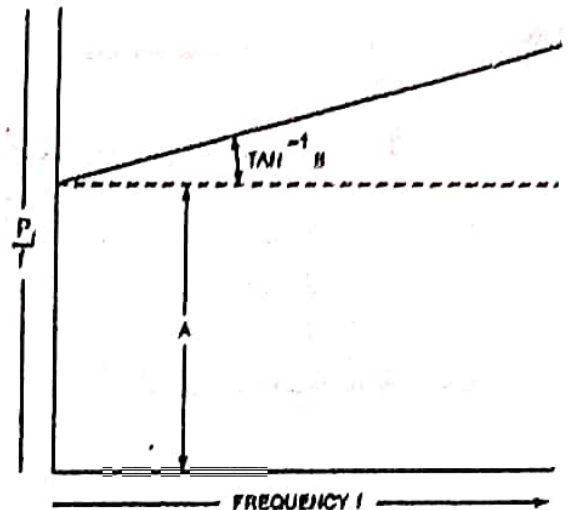


Fig. 10.23

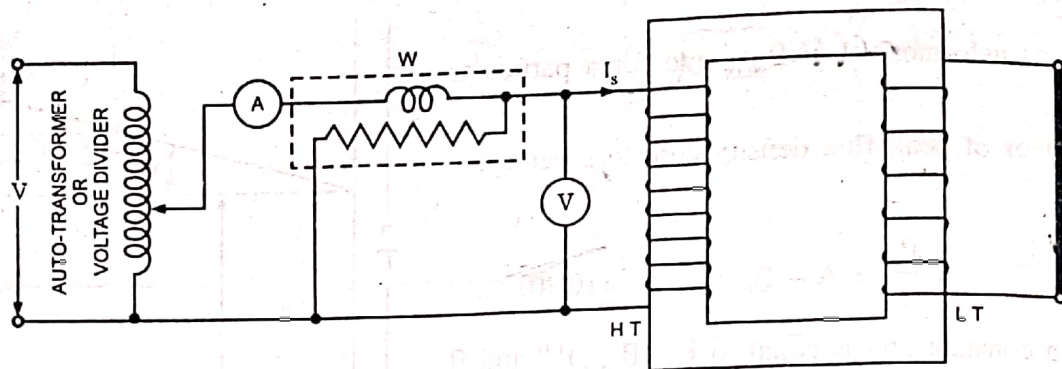


Fig. 10.24 Circuit Diagram For Short-Circuit Test

The applied voltage, V_s to the primary is gradually increased till the ammeter A indicates the full-load (rated) current of the metering side. Since applied voltage is very low (5–8% of the rated voltage) so flux linking with the core is very small and, therefore, iron losses are so small that these can be neglected. Thus the power input (reading of wattmeter W) gives total copper loss at rated load, output being nil. Let the readings of voltmeter, ammeter and wattmeter be V_s , I_s and W_s respectively.

Full-load copper loss, $P_c = I_s^2 R_{eq} = W_s$ ✓ *out put voltage* ... (10.41)

Equivalent resistance, $R_{eq} = \frac{W_s}{I_s^2}$ ✓ *V_s is low becoz of Sx, whole of primary voltage is used in supplying voltage* ... (10.42)

Equivalent impedance, $Z_{eq} = \frac{V_s}{I_s}$ ✓ *total impedance referred to primary* ... (10.43)

Equivalent reactance, $X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$ ✓ ... (10.44)

The above values are referred to the metering side (high voltage side in above case). If desired, the values could be easily determined referred to the other side, as discussed in Art 10.13.

Example 10.27. In a test for the determination of losses of a 440 V, 50 Hz transformer the total iron losses were found to be 2,500 W at normal voltage and frequency. When the applied voltage and frequency were 220 V, 25 Hz the iron losses were found to be 850 watts. Calculate the eddy current and hysteresis losses at normal voltage and frequency.

[U.P. Technical Univ, Tutorial Question Bank]

Solution: Since in second case voltage is also reduced in the proportion of frequency, flux density remains the same in both the cases.

At the same flux density

$$\text{Hysteresis loss, } P_h \propto f = A f \text{ (say)}$$

$$\text{Eddy current loss, } P_e \propto f^2 = B f^2 \text{ (say)}$$

$$\text{Total losses, } P_{h+e} = P_h + P_e = A f + B f^2$$

Substituting $P_{h+e} = 2,500$ watts when $f = 50$ Hz and $P_{h+e} = 850$ watts when $f = 25$ Hz we get

$$2,500 = A \times 50 + B \times (50)^2 \quad \dots (i)$$

$$850 = A \times 25 + B \times (25)^2 \quad \dots (ii)$$

Multiplying expression (ii) by 2 and subtracting it from expression (i) we get

$$800 = 1250 B$$

$$\text{or } B = \frac{800}{1,250} = 0.64$$

Eddy current loss at normal voltage and frequency (50 Hz) = $B f^2 = 0.64 \times (50)^2 = 1,600$ watts Ans.

Hysteresis loss at normal voltage and frequency = $2,500 - 1,600$ watts = 900 watts Ans.

Example 10.28. If in a transformer core, the hysteresis and eddy current losses are 80 W and 50 W at normal voltage and frequency, Calculate the losses when voltage and frequency are increased by 20%.

[U.P. Technical Univ. Elec. Engineering First Semester 2004-05]

Solution: When voltage and frequency are increased in the same ratio, flux density remains unchanged. Thus

$$\text{Hysteresis loss, } P_h' = P_h \times \frac{f_2}{f_1} = 80 \times 1.2 = 96 \text{ W Ans.}$$

$$\text{Eddy current loss, } P_e' = P_e \times \left(\frac{f_2}{f_1}\right)^2 = 50 \times (1.2)^2 = 72 \text{ W Ans.}$$

Example 10.29. Single phase transformer gave following test results: Rating 100 kVA, 11 kV/220 V, 50 Hz

OC test : 220 V, 45 A, 2 kW

SC test : 500 V, 9.09 A, 3 kW

Determine equivalent circuit parameters referred to lv side.

[U.P. Technical Univ. Electrical Engineering Second Semester 2005-06]

Solution: Assuming low-voltage winding as primary

$$\text{Energy component of no-load current, } I_e = \frac{P_0}{V} = \frac{2 \times 1,000}{220} = 9.091 \text{ A}$$

$$\text{Magnetising component of no-load current, } I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{45^2 - 9.091^2} = 44.07 \text{ A}$$

$$\text{Equivalent circuit parameter } R_0 = \frac{V_1}{I_e} = \frac{220}{9.091} = 24.2 \Omega \text{ Ans.}$$

$$\text{Equivalent circuit parameter, } X_0 = \frac{V_1}{I_m} = \frac{220}{44.07} = 4.992 \Omega \text{ Ans.}$$

$$\text{Equivalent impedance referred to hv side, } Z_{02} = \frac{V_s}{I_s} = \frac{500}{9.09} = 55 \Omega$$

$$\text{Equivalent resistance referred to hv side, } R_{02} = \frac{W_s}{I_s^2} = \frac{3 \times 1,000}{(9.09)^2} = 36.3 \Omega$$

$$\text{Equivalent reactance referred to hv side, } X_{02} = \sqrt{(Z_{02})^2 - (R_{02})^2} = \sqrt{55^2 - 36.3^2} = 41.31 \Omega$$

$$\text{Transformation ratio, } K = \frac{11,000}{220} = 50$$

$$\text{Equivalent resistance referred to lv side} = \frac{R_{02}}{K^2} = \frac{36.3}{50^2} = 0.0145 \Omega$$

$$\text{Equivalent reactance referred to lv side} = \frac{X_{02}}{K^2} = \frac{41.31}{50^2} = 0.0165 \Omega \text{ Ans.}$$

Example 10.30. A 200/100 V, 50 Hz transformer has an impedance of $(0.3 + j 0.8)$ ohm in the 200 V winding and impedance of $(0.1 + j 0.25)$ ohm in the 100 V winding. What are the currents on the high and low voltage sides, if a short-circuit occurs on the 100 V side with 200 V applied to the hv side.

[U.P. Technical Univ. Electrical Engineering September 2001]

$$\text{Solution: Transformation ratio, } K = \frac{V_2}{V_1} = \frac{100}{200} = 0.5$$

Equivalent impedance referred to lv side,

$$Z_{02} = K^2 Z_1 + Z_2 = 0.5^2(0.3 + j0.8) + (0.1 + j0.25) \\ = (0.175 + j0.45) = 0.483 \angle 68.8^\circ$$

Short-circuit current on 100 V side, $I_2 = \frac{V_2}{Z_{02}} = \frac{100}{0.483} = 207 \text{ A Ans.}$

Short-circuit current on 200 V side, $I_1 = K I_2 = 0.5 \times 207 = 103.5 \text{ A Ans.}$

10.18. DETERMINATION OF REGULATION OF TRANSFORMER FROM OPEN-CIRCUIT AND SHORT-CIRCUIT TESTS

Percentage regulation is given as

$$\% \text{ age regulation} = \frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{\text{No-load rated secondary voltage}} \times 100$$

...refer Eqs. (10.29) and (10.30)

Equivalent resistance R_{02} and reactance X_{02} referred to secondary can be determined from short-circuit test, as explained in Art 10.17.3. I_2 and $\cos \phi$ are the load current and power factor (lagging or leading) the load, so known. No-load secondary terminal voltage is equal to emf induced in the secondary E_2 .

$$\text{So percentage regulation} = \frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{E_2} \times 100$$

Note: Open-circuit test data are not needed for determination of voltage regulation of transformer.

Example 10.31. Define voltage regulation of a transformer. The following test data were taken on 25 kVA, 2,400/240 V, 50 Hz transformer with low-voltage winding shorted and measurement made on hv side :

$$V = 41.5 \text{ V}, I = 21.7 \text{ A}, P = 350 \text{ W.}$$

Calculate the voltage regulation of this transformer when it supplies full load at 0.8 pf lagging. Neglect the magnetising current.

[U.P. Technical Univ. Electrical Machines 2002-03]

Solution : Equivalent impedance referred to hv (or primary) side, $Z_{01} = \frac{V_s}{I_s} = \frac{41.5}{21.7} = 1.9124 \Omega$

$\therefore$ lv (secondary) winding is short-circuited and measurements are carried out on hv side.

Equivalent resistance referred to primary, $R_{01} = \frac{P_s}{I_s^2} = \frac{350}{(21.7)^2} = 0.743 \Omega$

Equivalent reactance referred to primary, $X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2} = \sqrt{(1.9124)^2 - (0.743)^2} \\ = 1.762 \Omega$

Turn-ratio, $K = \frac{V_2}{V_1} = \frac{240}{2,400} = 0.1$

Equivalent resistance referred to secondary, $R_{02} = K^2 R_{01} = (0.1)^2 \times 0.743 = 0.00743 \Omega$

Equivalent reactance referred to secondary, $X_{02} = (0.1)^2 \times 1.762 = 0.01762 \Omega$

Full-load secondary current, $I_2 = \frac{\text{Rated kVA} \times 1,000}{V_2} \\ = \frac{25 \times 1,000}{240} = 104.167 \text{ A}$

Power factor $\cos \phi_2 = 0.8$ lagging; $\sin \phi_2 = \sqrt{1 - 0.8^2} = 0.6$

$$\begin{aligned} \text{Voltage regulation} &= \frac{I_2 (R_{02} \cos \phi + X_{02} \sin \phi_2)}{V_2} \times 100 \\ &= \frac{104.167 (0.00743 \times 0.8 + 0.01762 \times 0.6)}{240} \times 100 \\ &= 0.72\% \text{ Ans.} \end{aligned}$$

Example 10.32. Consider a 20 kVA, 2,200/230 V, 50 Hz transformer. The OC/SC test results are as follows:

OC Test: 220 V, 4.2 A, 148 W (lv side)

SC Test: 86 V, 10.5 A, 360 W (hv side)

Determine the regulation at 0.8 pf lagging and at full load. What is the pf on short-circuit?

[Nagpur Univ. Elec. Machines, 1993]

Solution : Equivalent impedance referred to primary, $Z_{01} = \frac{V_s}{I_s} = \frac{86}{10.5} = 8.19 \Omega$

∵ Short-circuit test has been conducted on the hv (primary) side

Equivalent resistance referred to primary, $R_{01} = \frac{P_s}{I_s^2} = \frac{360}{10.5^2} = 3.27 \Omega$

Equivalent reactance referred to primary, $X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2} = \sqrt{(8.19)^2 - (3.27)^2} = 7.51 \Omega$

Full load primary current, $I_1 = \frac{\text{KVA (rated)} \times 1,000}{V_1} = \frac{20 \times 1,000}{2,200} = 9.09 \text{ A}$

$$\begin{aligned} \text{Percentage regulation} &= \frac{I_1 (R_{01} \cos \phi + X_{01} \sin \phi)}{V_1} \times 100 \\ &= \frac{9.09 (3.27 \times 0.8 + 7.51 \times 0.6)}{2,200} \times 100 = 2.94\% \end{aligned}$$

Power factor on short-circuit, $\cos \phi_s = \frac{R_{01}}{Z_{01}} = \frac{3.27}{8.19} = 0.399 \text{ (lag) Ans.}$

10.19. TRANSFORMER EFFICIENCY

The rated capacity of a transformer is defined as the product of rated voltage and full-load (rated) current on the output side. The power output depends upon the power factor of the load.

The efficiency (η) of a transformer, like that of any other apparatus, is defined as the ratio of useful power output to the input power, the two being measured in same units (either in watts or kilowatts).

$$\begin{aligned} \text{i.e. Transformer efficiency, } \eta &= \frac{\text{Output}}{\text{Input}} = \frac{\text{Output}}{\text{Output} + \text{losses}} = \frac{\text{Output}}{\text{Output} + \text{iron loss} + \text{copper loss}} \\ &= 1 - \frac{\text{iron loss} + \text{copper loss}}{\text{Output} + \text{iron loss} + \text{copper loss}} \end{aligned}$$

(Now power output $= V_2 I_2 \cos \phi$)

where V_2 is the secondary terminal voltage on load, I_2 is the secondary current at load and $\cos \phi$ is the power factor of the load.

Iron loss, P_i = Hysteresis loss + eddy current loss

Copper loss = $I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{01} = I_2^2 R_{02}$

10.19.1. Determination of Transformer efficiency. The ordinary transformer has a very high efficiency (in the range of 96–99%). Hence the transformer efficiency cannot be determined with high precision by direct measurement of output and input, since the losses are of the order of only 1–4 %. The difference between the readings of output and input instruments is then so small that an instrument error as low as 0.5 % would cause an error of the order of 15 % in the losses. Further, it is inconvenient and costly to have the necessary loading devices of the correct current and voltage ratings and power factor to load the transformer. There is also a wastage of large amount of power (equal to that of power output + losses) and no information is available from such a test about the proportion of copper and iron losses.

The best and accurate method of determining of efficiency of a transformer would be to compute losses from open-circuit and short-circuit tests and determine efficiency as follows :

Iron loss, $P_i = W_0$ or P_0 , determined from open-circuit test

Copper loss at full load, $P_c = W_s$ or P_s , determined from short-circuit test

Copper loss at a load x times full load = $I_2^2 R_{02} = x^2 P_c$

where x is the ratio of load current I_2 to secondary full-load secondary current.

and transformer efficiency, $\eta = \frac{V_2 I_2 \cos \phi}{V_2 I_2 \cos \phi + P_i + x^2 P_c} \quad \text{--- (10.45)}$

In Eq. (10.45) the effect of instrument readings is confined to losses only so that overall efficiency obtained from it is far more accurate than that obtained by direct loading. Another great advantage of this method is that it is not necessary for the transformer to be loaded to its full-load rating during testing, and the kW rating of the test plant need be equal to the value of the individual transformer losses.

10.19.2. Efficiency Versus Load. It has been pointed out that with constant voltage the mutual flux of the transformer is practically constant from no-load to full load (maximum variation is from 1 to 3 %). The core or iron loss is, therefore, considered constant regardless of load. Copper loss varies as the square of the load current or kVA output, as mentioned in Art 10.16. The variations in copper loss with the increase in load current (or kVA) is shown in Fig. 10.25. The efficiency vs load curve as deduced from

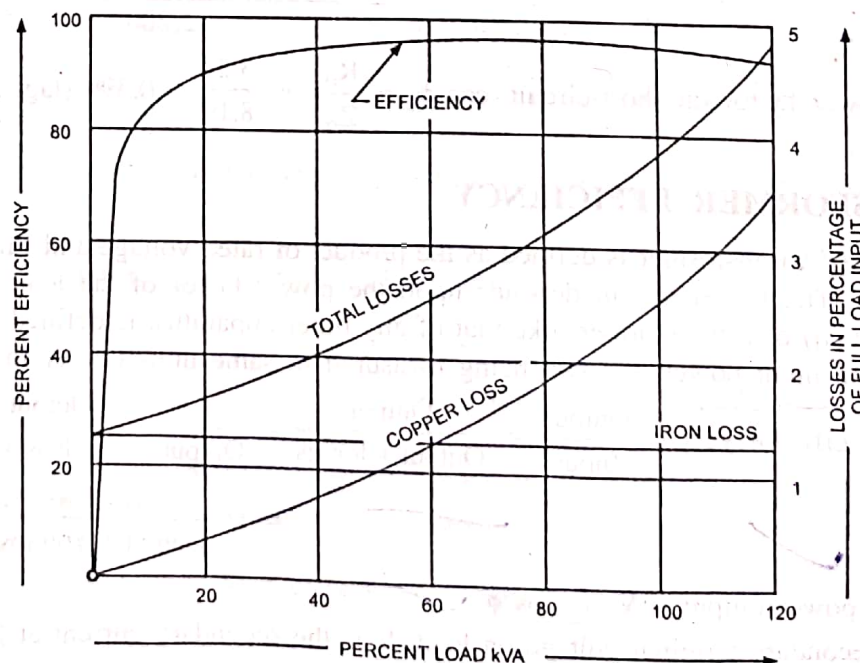


Fig. 10.25

$$\eta = \frac{x \times (\text{VArating}) \times \cos \phi_2}{x \times (\text{VArating}) \cos \phi_2 + P_i + x^2 P_{cu} FL} \times 100$$

these is also shown in the Fig. From the efficiency-load curve shown in Fig. 10.25 it is obvious that the efficiency is very high even at light load, as low as 10 % of rated load. The efficiency is practically constant from about 20% rated load to about 20% overload. At light loads the efficiency is poor because of constant iron loss whereas at high loads the efficiency falls off due to increase in copper loss as the square of load. From Fig. 10.25 it is also obvious that the transformer efficiency is maximum at the point of intersection of copper loss and iron loss curves i.e. when copper loss equals iron loss.

10.19.3. Condition For Maximum Efficiency.

From Eq. (10.45)

$$\text{Transformer efficiency, } \eta = \frac{V_2 I_2 \cos \phi}{V_2 I_2 \cos \phi + P_i + x^2 P_c} = \frac{x P}{x P + P_i + x^2 P_c} \quad \dots(10.46)$$

where P is equal to the full load (rated output in volt-amperes or kVA) $\times$ load power factor ($\cos \phi$), P_i is the total iron loss, P_c is the full-load copper loss and x is the fraction of full load-kVA at which efficiency is maximum.

Differentiating both sides of Eq. (10.46) we get

$$\frac{d\eta}{dx} = \frac{(x P + x^2 P_c + P_i) P - x P (P + 2x P_c)}{(x P + P_i + x^2 P_c)^2} = \frac{P(P_i - x^2 P_c)}{(x P + P_i + x^2 P_c)^2}$$

Efficiency η will be maximum if $\frac{d\eta}{dx} = 0$

$$\text{or } P_i - x^2 P_c = 0$$

$$\text{or } x^2 P_c = P_i$$

...(10.47)

Hence the efficiency will be maximum when variable loss (copper loss) is equal to constant loss (iron loss).

$$\text{Output kVA corresponding to maximum efficiency} = x \times \text{Rated kVA} = \text{Rated kVA} \times \sqrt{\frac{P_i}{P_c}} \quad \dots(10.48)$$

$$\text{Since from Eq. (10.47) } x = \sqrt{\frac{P_i}{P_c}}$$

Output current I_2 corresponding to maximum efficiency is determined as below:

Copper loss at given load at which efficiency is maximum $= I_2^2 R_{02}$

and since for maximum efficiency to occur it is necessary that copper loss equals P_i

$$\text{So } I_2^2 R_{02} = P_i$$

$$\text{So current corresponding to maximum efficiency, } I_2 = \sqrt{\frac{P_i}{R_{02}}} \quad \dots(10.49)$$

Power transformers employed for bulk power transmission are operated continuously near about full load and are, therefore, designed to have maximum efficiency at full load. On the other hand, the distribution transformers, which supply load varying over the day through a wide range, are designed to have maximum efficiency at about three-fourths the full load.

It is important to appreciate that, since copper loss depends on current and the iron loss depends on voltage, the total loss in the transformer depends on the volt-ampere product, and not on the phase angle between voltage and current, i.e. is independent of the load power factor. The transformers are, therefore, rated in kilo-volt amperes (KVA) and not in kilowatts.

$$\text{At maximum efficiency } \eta_{\max} = \frac{x (\text{rated VA}) \cos \phi_2}{x (\text{rated VA}) \cos \phi_2 + 2P_i}$$

10.19.4. Efficiency Versus Power Factor. Transformer efficiency is given as

$$\begin{aligned}
 \eta &= \frac{\text{Output}}{\text{Output} + \text{losses}} \\
 &= 1 - \frac{\text{losses}}{\text{output} + \text{losses}} \\
 &= 1 - \frac{\text{losses}}{V_2 I_2 \cos \phi + \text{losses}} \\
 &= 1 - \frac{\text{losses}/V_2 I_2}{\cos \phi + \text{losses}/V_2 I_2}
 \end{aligned}$$

Substituting

$$\frac{\text{losses}}{V_2 I_2} = x \text{ in above equation we get}$$

$$\eta = 1 - \frac{x}{\cos \phi + x} = 1 - \frac{x/\cos \phi}{1 + x/\cos \phi} \quad \dots(10.50)$$

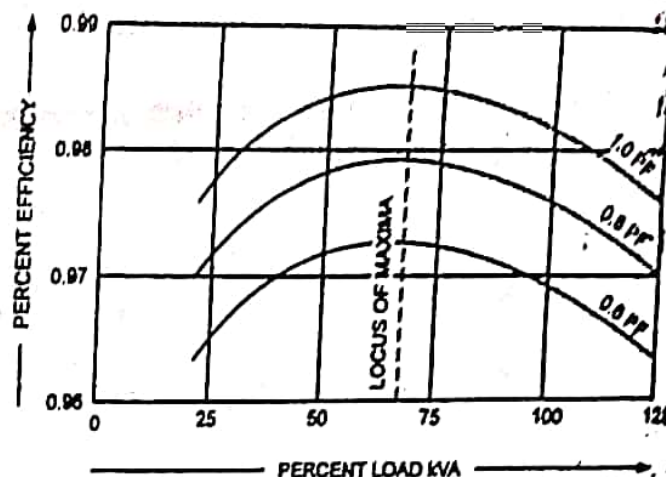


Fig. 10.26

The variations of efficiency with power factor at different loadings for a typical transformer are illustrated in Fig. 10.26.

Example 10.33. In a 50 kVA, 11 kV/400 V transformer, the iron and copper losses are 500 W and 600 W respectively under rated conditions. Calculate the efficiency on unity power factor at full load. Find the load for maximum efficiency and the iron and copper losses corresponding to this load.

[GATE 1998]

Solution: Full-load copper loss, $P_c = 600$ watts or 0.6 kW

Iron loss, $P_i = 500$ watts or 0.5 kW

At full load and at unity power factor

$$\text{Output, } P = \text{Rated kVA} \times \text{power factor} = 50 \times 1 = 50 \text{ kW}$$

$$\text{Efficiency, } \eta = \frac{P}{P + P_i + P_c} \times 100 = \frac{50}{50 + 0.5 + 0.6} \times 100 = 97.85 \% \text{ Ans.}$$

Let the transformer maximum efficiency occur at x times full load

Since maximum efficiency occurs when

$$\text{Copper loss, } x^2 P_c = \text{Iron loss, } P_i$$

$$\text{so } x = \sqrt{\frac{P_i}{P_c}} = \sqrt{\frac{500}{600}} = 0.9129$$

Load in kVA corresponding to maximum efficiency

$$= x \times \text{rated kVA} = 0.9129 \times 50 = 45.645 \text{ kVA Ans.}$$

Copper and iron losses will be equal at a load corresponding to maximum efficiency so each of copper and iron loss will be equal to 500 W Ans.

Example 10.34. In a 50 kVA, 1-phase 3300/230 V, transformer, the iron losses and full-load copper losses were found to be as 500 W and 650 W respectively. Calculate efficiency at half load and 0.6 power factor. Determine maximum efficiency and corresponding load.

[U.P. Technical Univ. Electrical Engineering First Semester 2004-05]

Solution: Iron loss, $P_i = 500$ W

Copper loss at full load, $P_c = 650$ W

$$\text{Output at half full load} = \frac{1}{2} \times \text{rated kVA} \times \text{pf} = \frac{1}{2} \times 50 \times 0.6 = 15 \text{ kW} = 15,000 \text{ W}$$

$$\text{Input at half full load} = P_{\text{out}} + P_i + \left(\frac{1}{2}\right)^2 P_c = 15,000 + 500 + \frac{650}{4} = 15,662.5 \text{ W}$$

$$\text{Efficiency at half full load} = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100 = \frac{15,000}{15,662.5} \times 100 = 95.77\% \text{ Ans.}$$

$$\text{Load corresponding to maximum efficiency} = \text{Rated kVA} \times \sqrt{\frac{P_i}{P_c}} = 50 \times \sqrt{\frac{500}{650}} = 43.853 \text{ kVA}$$

$$\text{Maximum efficiency, } \eta_{\text{max}} = \frac{43.853 \times 1.0}{43.853 + 500 + 500} \times 100 \text{ assuming unity pf} = 97.77\% \text{ Ans.}$$

Example 10.35. A transformer rated 200 V/50 V, 10 kVA has a core loss of 100 watts. What is the maximum efficiency of the transformer at 0.8 lagging power factor? Assume full-load copper loss as 200 watts. At what load this maximum efficiency is obtained? How and why would your answer be different if the power factor were unity? [R.G.P.V. Bhopal Basic Electrical Engineering December 2004]

Solution:

$$\text{Full-load copper loss, } P_c = 200 \text{ W} = 0.2 \text{ kW}$$

$$\text{Iron loss, } P_i = 100 \text{ W} = 0.1 \text{ kW}$$

$$\text{The load corresponding to maximum efficiency} = \text{Rated kVA} \times \sqrt{\frac{P_i}{P_c}} = 10 \times \sqrt{\frac{0.1}{0.2}} = 7.07 \text{ kVA} \text{ Ans.}$$

$$\text{Maximum efficiency at 0.8 lagging power factor} = \frac{7.07 \times 0.8}{7.07 \times 0.8 + 0.1 + 0.1} \times 100 = 96.585\% \text{ Ans.}$$

$$\text{Maximum efficiency when pf is improved to unity} = \frac{7.07 \times 1.0}{7.07 \times 1.0 + 0.1 + 0.1} \times 100 = 97.249\% \text{ Ans.}$$

Example 10.36. A 4 kVA, 220/440 V, 50 Hz, single phase transformer gave the following test figures.

No-load test performed on 220 V side keeping 440 V side open: 220 V, 0.7 A, 60 W.

Short-circuit test performed short-circuiting the 440 V side through an ammeter: 9 V, 6 A, 21.6 W.

Calculate (a) the magnetising current and the component corresponding to iron loss at normal voltage and frequency (b) The efficiency on full load at unity power factor and the corresponding secondary terminal voltage (c) Draw the phasor diagram corresponding to full-load operation.

[Pb. Technical Univ. Electrical Engineering January 2000]

Solution:

$$\text{Primary (lv) voltage, } V_1 = 220 \text{ V}$$

$$\text{No-load power factor, } \cos \phi_0 = \frac{P_0}{V_1 I_0} = \frac{60}{220 \times 0.7} = 0.3896 \text{ (lagging)}$$

(a) Component of current corresponding to iron loss at normal voltage and frequency.

$$I_w = I_0 \cos \phi_0 = 0.7 \times 0.3896 = 0.2727 \text{ A} \text{ Ans.}$$

$$\text{Magnetising current, } I_m = \sqrt{I_0^2 - I_w^2} = \sqrt{(0.7)^2 - (0.2727)^2} = 0.6447 \text{ A} \text{ Ans.}$$

$$(b) \text{ Full-load secondary current, } I_2 = \frac{S \text{ in kVA} \times 1,000}{V_2} = \frac{4 \times 1,000}{440} = 9.091 \text{ A}$$

$$\text{Full-load primary current, } I_1 = \frac{4 \times 1,000}{440} = 18.182 \text{ A}$$

$$\text{Full-load copper loss, } P_c = \left(\frac{I_1}{I_r} \right)^2 \times P_r = \left(\frac{18.182}{6} \right)^2 \times 21.6 = 198.35 \text{ W}$$

$$\begin{aligned} \text{Full-load efficiency, } \eta &= \frac{\text{Output in kVA} \times \cos \phi}{\text{Output in kVA} \times \cos \phi + P_c + P_i} \times 100 \\ &= \frac{4 \times 1,000 \times 1}{4 \times 1,000 \times 1 + 198.35 + 60} \times 100 \\ &= \frac{4,000}{4,000 + 258.35} \times 100 = 93.93\% \text{ Ans.} \end{aligned}$$

$$\text{Equivalent impedance referred to primary, } Z_{01} = \frac{V_1}{I_1} = \frac{9}{6} = 1.5 \Omega$$

$$\text{Equivalent resistance referred to primary, } R_{01} = \frac{P_1}{I_1^2} = \frac{21.6}{6^2} = 0.6 \Omega$$

$$\text{Equivalent impedance referred to secondary, } Z_{02} = Z_{01} \times \left(\frac{V_2}{V_1} \right)^2 = 1.5 \times \left(\frac{440}{220} \right)^2 = 6 \Omega$$

$$\text{Equivalent resistance referred to secondary, } R_{02} = R_{01} \times \left(\frac{V_2}{V_1} \right)^2 = 0.6 \times \left(\frac{440}{220} \right)^2 = 2.4 \Omega$$

$$\text{Equivalent reactance referred to secondary, } X_{02} = \sqrt{(Z_{02})^2 - (R_{02})^2} = \sqrt{6^2 - 2.4^2} = 5.5 \Omega$$

$$\text{Since power factor, } \cos \phi_2 = 1.0, \sin \phi_2 = 0$$

$$\text{and } E_2^2 = (V_2 + I_2 R_{02})^2 + (I_2 X_{02})^2$$

$$\begin{aligned} \text{So secondary voltage, } V_2 &= \sqrt{E_2^2 - (I_2 X_{02})^2} - I_2 R_{02} \\ &= \sqrt{440^2 - (9.091 \times 5.5)^2} - 9.091 \times 2.4 \\ &= 415.9 \text{ V Ans.} \end{aligned}$$

Phasor diagram is shown in Fig. 10.27.



Fig. 10.27

Example 10.37. A 10 kVA, 500/250 V, 50 Hz single phase transformer gave the following test results:
SC test (HV side): 60 V, 20 A, 150 W.

The maximum efficiency occurs at unity power factor and at 1.2 full-load current. Determine the full-load efficiency at 0.8 power factor lagging. Also calculate the maximum efficiency.

[R.G.P.V. Bhopal Basic Electrical Engineering June 2005]

$$\text{Solution: Full-load current on hv side, } I_2 = \frac{10 \times 1,000}{500} = 20 \text{ A}$$

Since short-circuit test is conducted on hv side, and copper loss with a short-circuit current is 150 W, full-load copper loss = 150 W

Iron loss, P_i = Copper loss corresponding to load at which maximum efficiency occurs
 $= (1.2)^2 \times 150 = 216 \text{ W}$

Full-load output at 0.8 pf lagging, $P_{out} = 10 \times 0.8 = 8 \text{ kW}$ or 8,000 W

$$\text{Full-load efficiency at 0.8 pf lagging, } \eta = \frac{P_{out}}{P_{out} + P_r + P_i} \times 100$$

$$= \frac{8,000}{8,000 + 150 + 216} \times 100 = 95.625\% \quad \text{Ans.}$$

Load corresponding to maximum efficiency = $1.2 \times 10 = 12 \text{ kVA}$

$$\text{Maximum efficiency assuming power factor unity} = \frac{12 \times 1,000 \times 1.0}{12 \times 1,000 \times 1.0 + 216 + 216} \times 100 = 96.525\% \quad \text{Ans.}$$

Example 10.38. A 100 kVA, 50 Hz, 440/11,000 V single phase transformer has an efficiency of 98.5 % when supplying full-load currents at 0.8 pf lagging and an efficiency of 99 % when supplying half-full load current at unity power factor. Find the core loss and copper losses corresponding to full-load current. At what value of load current the maximum efficiency be attained ?

[R.G.P.V. Bhopal Basic Electrical Engineering December 2001]

Solution: Let the iron loss and full-load copper loss P_i and P_r respectively.

At full-load and 0.8 power factor lagging

Output power, $P = 100 \times 0.8 = 80 \text{ kW}$

Transformer efficiency, $\eta = 98.5\%$ or 0.985

$$\text{Power input} = \frac{P}{\eta} = \frac{80}{0.985} = 81.218 \text{ kW}$$

$$\text{Total losses, } (P_i + P_r) = \text{Input} - \text{output} = 81.218 - 80 = 1.218 \text{ kW}$$

At half full-load and unity power factor

$$\text{Output power} = \frac{1}{2} \times 100 \times 1.0 = 50 \text{ kW}$$

Transformer efficiency, $\eta = 99\% = 0.99$

$$\text{Power input} = \frac{50}{0.99} = 50.505 \text{ kW}$$

$$\text{Total losses} = P_i + \left(\frac{1}{2}\right)^2 P_r = 50.505 - 50 = 0.505 \text{ kW}$$

Solving Eqs. (i) and (ii) we have

Iron loss, $P_i = 0.2674 \text{ kW}$ Ans

Full-load copper loss, $P_r = 0.9506 \text{ kW}$ Ans

$$\text{Load corresponding to maximum efficiency} = \text{Rated kVA} \times \sqrt{\frac{P_i}{P_r}} = 100 \times \sqrt{\frac{0.2674}{0.9506}} = 53.04$$

$$\text{Current corresponding to maximum efficiency} = \frac{\text{kVA corresponding to maximum efficiency} \times 1,000}{\text{Secondary rated voltage}}$$

$$= \frac{53.04 \times 1,000}{11,000} = 4.82 \text{ A Ans.}$$

10.20. ALL-DAY EFFICIENCY (OR ENERGY EFFICIENCY)

The transformer efficiency discussed so far is the ordinary, also called *commercial efficiency*, which is defined as the ratio of power output to power input.

There are certain types of transformers whose performance cannot be judged by ordinary or commercial efficiency. For instance, distribution transformers are energized for 24 hours, but they deliver very low loads for major portion of the day. Thus iron or core loss occur for the whole day but the copper loss occurs only when the transformer is loaded. The performance of such a transformer must be judged by *all-day efficiency*, also called the *energy efficiency* or *operational efficiency* which is computed on the basis of energy consumed during the whole day (24 hours).

The all-day efficiency is defined as the ratio of energy (kWh) output over 24 hours to the energy input in the same period.

$$\text{i.e. All-day efficiency} = \frac{\text{Output in kWh}}{\text{Input in kWh}} \quad \dots(10.51)$$

Since the distribution transformer does not supply the rated load for the whole day so the all-day efficiency of such a transformer will be less than ordinary or commercial efficiency.

For determination of all-day efficiency of a transformer, it is necessary, of course, to know how the load varies from hour to hour during the day.

Higher energy efficiencies are achieved by designing distribution transformers to yield maximum commercial efficiency at less than full load (usually 50–75 percent of full load). This is achieved by restricting the core flux density to the lower values by using a relatively larger x-section. Thus the ratio of iron loss to copper loss is reduced.

Example 10.39. A 100 kVA transformer has its maximum efficiency of 98% at full load and unity pf. During the day it is loaded as follows:

12 hrs — 20 kW at a pf of 0.5 lag

6 hrs — 50 kW at a pf of 0.9 lag

6 hrs — 75 kW at a pf of 0.8 lag

Calculate the “all day efficiency” of the transformer. [U.P.S.C. I.E.S. Electrical Engineering-II, 2003]

Solution: For maximum efficiency

$$\text{Output} = 100 \times 1.0 = 100 \text{ kW}$$

$$\text{Input} = \frac{\text{Output}}{\eta} = \frac{100}{0.98} = 102.04 \text{ kW}$$

$$\text{Total losses} = \text{Input} - \text{output} = 102.04 - 100 = 2.04 \text{ kW}$$

$$\text{Full-load copper loss, } P_c = \text{Iron loss, } P_i = \frac{\text{Total losses}}{2} = \frac{2.04}{2} = 1.02 \text{ kW}$$

$$\text{All day output} = 20 \times 12 + 50 \times 6 + 75 \times 6 = 990 \text{ kWh}$$

$$\text{Iron loss for 24 hours} = 24 \times 1.02 = 24.48 \text{ kWh}$$

$$\begin{aligned} \text{Copper loss for 24 hours} &= \left(\frac{20/0.5}{100}\right)^2 \times 1.02 \times 12 + \left(\frac{50/0.9}{100}\right)^2 \times 1.02 \times 6 + \left(\frac{75/0.8}{100}\right)^2 \times 1.02 \times 6 \\ &= 1.9584 + 1.8889 + 5.3789 = 9.226 \text{ kWh} \end{aligned}$$

$$\text{All-day efficiency, } \eta = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{990}{990 + 24.48 + 9.226} \times 100 = 96.7\% \quad \text{Ans.}$$

Example 10.40. Two-100 kW transformers each has a maximum efficiency of 98 % but in one the maximum efficiency occurs at full load while in the other it occurs at half load. Each transformer is on full load for 4 hours, on half-load for 6 hours and one-tenth load for 14 hours per day. Determine the efficiency of each transformer.

[Vikram Univ. Elec. Machines-I, 1988]

Solution: Transformer 1

$$\text{Total losses on full load, } P_c + P_i = \frac{\text{Output}}{\eta} - \text{output} = \frac{100}{0.98} - 100 = 2.04 \text{ kW}$$

$$\text{Full-load copper loss, } P_c = P_i = \frac{P_c + P_i}{2} = \frac{2.04}{2} = 1.02 \text{ kW}$$

Since maximum efficiency occurs at full load and for maximum efficiency $P_c = P_i$

$$\text{All-day output} = 100 \times 4 + \frac{100}{2} \times 6 + \frac{100}{10} \times 14 = 840 \text{ kWh}$$

$$\text{Iron loss for 24 hours} = 1.02 \times 24 = 24.48 \text{ kWh}$$

$$\text{Copper loss for 24 hours} = 1.02 \times 4 + \left(\frac{1}{2}\right)^2 \times 1.02 \times 6 + \left(\frac{1}{10}\right)^2 \times 1.02 \times 14 = 5.753 \text{ kWh}$$

$$\text{All-day efficiency, } \eta_1 = \frac{840}{840 + 24.48 + 5.753} \times 100 = 96.53 \% \text{ Ans.}$$

Transformer 2

$$\text{Half-load output} = \frac{1}{2} \times 100 = 50 \text{ kW}$$

$$\text{Half-load input} = \frac{50}{0.98} = 51.02 \text{ kW}$$

$$\text{Total losses at half-full load} = 51.02 - 50 = 1.02 \text{ kW}$$

$$\text{Copper loss at half-full load} = \left(\frac{1}{2}\right)^2 \times P_c = \frac{P_c}{4} \quad \text{where } P_c \text{ is full load copper loss}$$

$$\text{Half-load copper loss, } \frac{P_c}{4} = P_i = \frac{\text{Total loss}}{2} = \frac{1.02}{2} = 0.51 \text{ kW}$$

$$\text{or } P_c = 0.51 \times 4 = 2.04 \text{ kW}$$

$$\text{All-day output} = 100 \times 4 + \frac{100}{2} \times 6 + \frac{100}{10} \times 14 = 840 \text{ kWh as above}$$

$$\text{Iron loss for 24 hours} = 24 \times 0.51 = 12.24 \text{ kWh}$$

$$\text{Copper loss for 24 hours} = 2.04 \times 4 + \left(\frac{1}{2}\right)^2 \times 2.04 \times 6 + \left(\frac{1}{10}\right)^2 \times 2.04 \times 14 = 11.506 \text{ kWh}$$

$$\text{All-day efficiency, } \eta_2 = \frac{840}{840 + 12.24 + 11.506} \times 100 = 97.25 \% \text{ Ans.}$$

10.21. TRANSFORMER CONSTRUCTION

The transformer is very simple in construction and consists of magnetic circuit linking with two windings known as primary and secondary windings. Besides magnetic circuit and windings it consists of a suitable container for the assembled core and windings, such as a tank, a suitable medium for insulating the core and windings from its container such as transformer oil, suitable bushings (either of porcelain, oil-filled or condenser type) for insulating and bringing out terminals of the windings from the container, temperature gauge for measurement of temperature of hot oil or hottest spot temperature, and oil gauge to indicate the oil level inside the tank. Some transformers are also provided with conservator tank in order to slowdown deterioration of oil and keep the main tank full of oil, emergency vent to relieve the pressure inside the tank in case the pressure inside the transformer rises to a danger point and gas operated relay (Buchholz relay) in order to give alarm to indicate the presence of gas in case of some minor fault and take out the transformer out of circuit in case of serious fault.

Magnetic circuit consists of an iron core. Since core is magnetic link between the two systems in a transformer and it itself contains a lot of energy, it is not by any means the passive component it would first appear.

Ideally the steel, of which the transformer core is made, is simple in chemical composition, the hot rolled steel has 4-5% silicon and the cold rolled has 3%. Now-a-days cold rolled steel is used almost exclusively in spite of its high price, because flux density 30% higher can be used. This is because of high permeability and lower losses at given flux density. Since flux is an alternating one, in order to reduce eddy current losses the core is laminated. The thickness of laminations or stampings varies from 0.35 mm to 0.5 mm. The laminations are insulated from each other by coating them with a thin coat of varnish. The iron core is rectangular in cross-section for small size transformers and circular for large size ones to reduce the quantity of copper required.

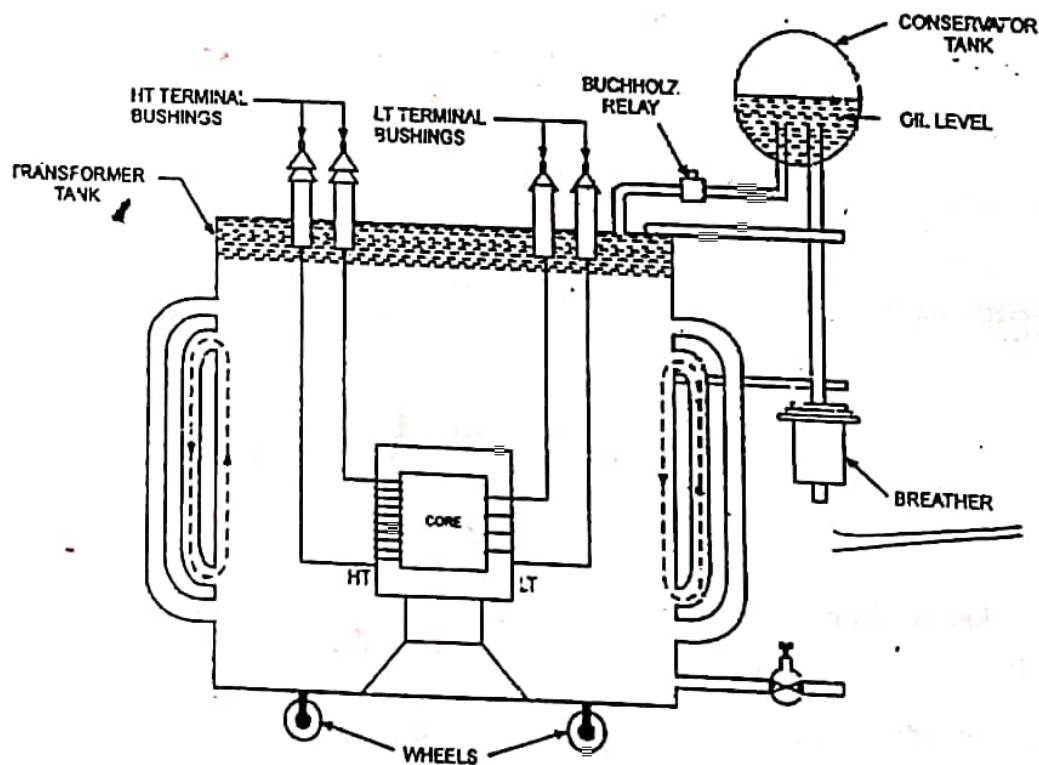


Fig. 10.28

As already mentioned, the electric circuit consists essentially of two windings, each being split up into two equal number of coils and the coils of two windings are arranged on each limb of iron core to reduce the magnetic leakage. In order to minimise the amount of high voltage insulation, low voltage

10.22. TYPES OF TRANSFORMERS

Depending upon the type of construction used, the transformers are classified into two categories viz (i) *core type* and (ii) *shell type*.

Depending upon the type of service, in the field of power system, they are classified as (i) *power transformers* (ii) *distribution transformers*.

10.22.1. Core Type and Shell Type Transformers.

In *core type construction*, as shown in Fig. 10.29, the coils are wound around the two limbs of a rectangular magnetic core. Each limb carries one half of the primary winding and one half of the secondary winding so as to reduce the leakage reactance to the minimum possible. The lv winding is wound on the inside nearer to the core while the hv winding is wound over the lv winding away from the core in order to reduce the amount of insulation materials required.

Small transformers may have cores of rectangular or square x-section with rectangular or circular coils but it is wasteful in case of large capacity transformers. In case of large sized transformers stepped cruciform core with circular cylindrical coils is employed, as illustrated in Fig. 10.30. Such a core employs laminations of different sizes. Though the cost of manufacturing of such a cruciform core is much greater, but the circular coils that are used are easier to wind and provide more mechanical strength, especially when short-circuit occurs. Other advantages of using cruciform core are high space factor and reduced mean length of turns resulting in reduced copper loss.

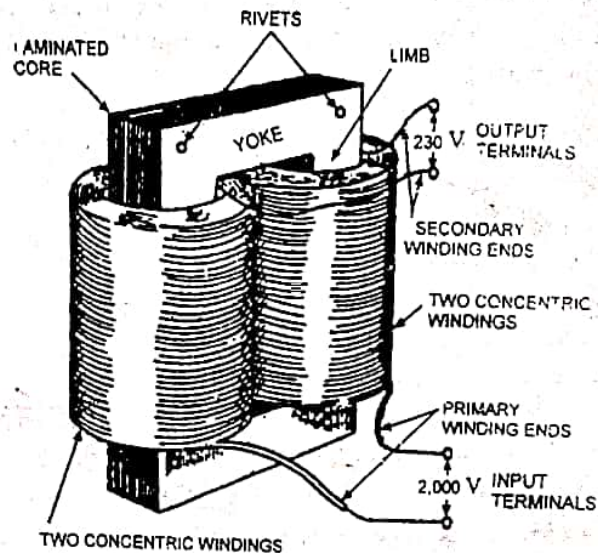


Fig. 10.29 Core Type Transformer

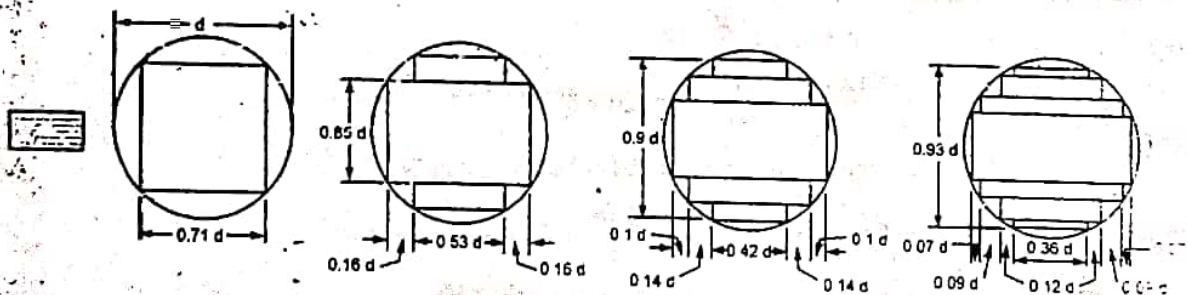


Fig. 10.30

In *shell type construction*, as shown in Fig. 10.31, the coils are wound on the central limb of a three limb core. The entire flux passes through the central limb and divides into two parts going to side limbs. Consequently, the cross-sectional area (and hence width) of the central limb is twice of that of each of the side limbs. Sandwich type winding is used in such a construction.

If the core type and shell type transformers are compared, then we conclude that the core type transformer has a longer mean length of iron core and a shorter mean length of coil turn. The core type transformer has smaller x-section of iron and, therefore, a greater number of turns. Core type construction provides more space for insulation making it best suited for EHV requirements.

In shell type transformers advantage is gained through the core being used to protect the windings from mechanical damage. The shell type construction gives better support against electro-magnetic forces between current carrying conductors. These forces are of considerable magnitude under short-circuit conditions. The shell type construction is commonly used for small transformers where a square or

rectangular core cross-section is suitable for economic consideration. The shell type construction needs more specialized fabrication facilities than core type, while the latter offers an additional advantage of permitting visual inspection of coils in the case of a fault and ease of repair at substation site. For these reasons, the present practice is to use core type transformers in large high voltage installations.

10.22.2. Power Transformers. The term is used to include all transformers of large size (250 kVA and above) used in generating stations and substations for transforming the voltage at each end of a power transmission line. They may be single phase or 3-phase, 3-wire delta/delta or delta/star connected; and voltage rating 220/11 kV or in high voltage range.

They are put in operation during load hours and disconnected during light load hours i.e. they are usually operated on approximately full load. This is possible because they are arranged in banks and can be thrown in parallel with other units or disconnected at will. So power transformers are designed to have maximum efficiency at or near full load (i.e. with iron loss to full load copper loss ratio of 1 : 1). Power transformers are designed to have considerable greater leakage reactance than is permissible in distribution transformers because in power transformers inherent voltage regulation is not as much important as current limiting effect of the higher leakage reactance. Power transformers usually make use of flux density of 1.5 to 1.7 T, have percentage impedance ranging from 6–18 % and regulation of 6–10 percent. They may be self oil cooled, forced air cooled or forced water cooled.

Distribution Transformers. Transformers of rating upto 200 kVA, used to step down the distribution voltage to a standard service voltage are known as *distribution transformers*. They are kept in operation all the 24 hours a day whether they are carrying any load or not. In such transformers iron loss occurs for all the time where copper loss occurs only when they are loaded. Therefore, distribution transformers should be designed with iron loss smaller in comparison to full-load copper loss (say with iron loss to full load copper loss ratio of 1 : 3). In other words, they are designed for good 'all-day efficiency' and for highest efficiency on full load. They are of the self cooling type and are almost invariably oil-immersed. The distribution transformers are usually 3-phase, 4-wire 11 kV/415 V delta/star connected. Such transformers are designed to have an inherently good voltage regulation which is possible by arranging the coils in such a way as to have minimum leakage reactance. Such transformers have percentage impedance of 4–5 % and voltage regulation of 4–8 %. They make use of cold rolled steel with flux density of 1.7 T.

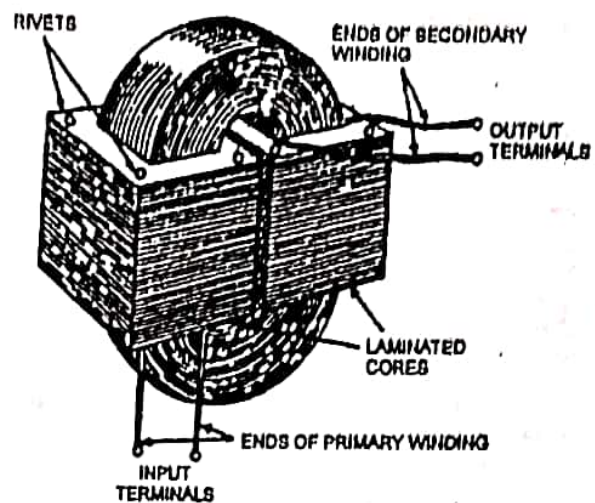


Fig. 10.31 Shell Type Transformer

HIGHLIGHTS

1. Transformer is an ac machine that (i) transfers electrical energy from one electric circuit to another (ii) does so without a change of frequency (iii) does so by the principle of electro-magnetic induction and (iv) has electric circuits that are linked by a common magnetic circuit. The energy transfer usually takes place with a change of voltage, although this is not always necessary. It is a static machine and has very high efficiency (in the range of 96–99%).
A transformer never operates on dc. If dc supply is connected to the transformer primary or secondary, it may burn out since there is no self induced emf and heavy current will flow through it.
2. In an *ideal transformer*, winding resistance, magnetic leakage, and iron losses are neglected and core is assumed to have infinite permeability and zero reluctance.
3. The emfs induced in the transformer windings are given as

$$E_1 = 4.44 f N_1 \Phi_{\max} \text{ volts and } E_2 = 4.44 f N_2 \Phi_{\max} \text{ volts}$$

4. Voltage transformation ratio, $K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$
5. If the transformer is operated with the frequency and voltage changed in the same proportion, the flux density will remain unchanged and so the no-load current.
6. The current drawn by transformer on no load is called the no-load current or exciting current. It has two components (i) in-phase, active or energy component I_e ($I_e \cos \Phi_0$) used to meet the iron loss and small amount of copper loss occurring in primary winding and (ii) quadrature or wattless component, called the magnetizing component,

I_m ($I_0 \sin \Phi_0$) used to create alternating flux in the core. No-load phase angle, $\Phi_0 = \tan^{-1} \frac{I_m}{I_e}$.

7. When a transformer is loaded, the secondary ampere-turns set up a magnetic field that opposes the main field. To neutralize this effect, transformer draws extra primary current from the mains. Flux remains almost constant from no-load to full-load.

The primary current, $I_0 = I'_1 + I_0 = K I_2 + I_0$

8. When a practical transformer is loaded

Induced emf in secondary winding $E_2 = V_2 + I_2 (R_2 + j X_2) = V_2 + I_2 Z_2$

Induced emf in primary $E_1 = E_2/K$

and applied voltage to primary winding, $V_1 = -E_1 - I_1 (R_1 + j X_1) = -E_1 + I_1 Z_1$

9. The total resistance and reactance of the transformer referred to primary are given as

$$R_{01} = R_1 + \frac{R_2}{K^2}, \quad X_{01} = X_1 + \frac{X_2}{K^2} \quad \text{and} \quad Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

10. The total resistance and reactance of the transformer referred to secondary are given as

$$R_{02} = (K^2 R_1 + R_2), \quad X_{02} = (K^2 X_1 + X_2) \quad \text{and} \quad Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

11. Approximate voltage drop in the transformer referred to secondary = $I_2 (R_{02} \cos \Phi \pm X_{02} \sin \Phi)$
+ ve sign for lagging power factor and - ve sign for leading power factor

$$12. \text{Percentage voltage regulation} = \frac{E_2 - V_2}{E_2} \times 100 = \frac{I_2 (R_{02} \cos \Phi \pm X_{02} \sin \Phi)}{\text{No-load secondary voltage, } E_2} \times 100$$

13. The condition for zero regulation is

$$\text{Load phase angle, } \phi = \tan^{-1} \left(\frac{-R_{02}}{X_{02}} \right)$$

-ve sign indicates that zero regulation occurs at a leading power factor.

14. Transformer losses are of two types namely iron loss and copper loss. Iron losses are almost constant from no-load to full-load while copper losses vary as the square of load in kVA.

15. Transformer efficiency is given as

$$\eta = \frac{V_2 I_2 \cos \Phi}{V_2 I_2 \cos \Phi + P_i + x^2 P_c}$$

Transformer efficiency will be maximum when copper losses equal iron or core losses.

$$\text{Output kVA corresponding to maximum efficiency} = \text{Rated kVA} \sqrt{\frac{\text{Iron losses}}{\text{full-load copper losses}}}$$

16. Open-circuit test is performed to determine the core or iron losses, P_i and no-load current I_0 and thereby the shunt branch parameters R_0 and X_0 of the equivalent circuit.

17. Short-circuit test is performed to determine the copper losses at rated load and equivalent resistance and reactance referred to metering side.

18. The performance of distribution transformers is judged by its all-day efficiency, also called the energy efficiency or operational efficiency which is completed on the basis of energy consumed during the whole day (24 hours) and is given as

$$\text{Transformer all-day efficiency, } \eta = \frac{\text{kWh output in 24 hours}}{\text{kWh input in 24 hours}}$$

EXERCISES

1. Why is efficiency of a transformer maximum among all electrical machines? Explain.
[Pb. Technical Univ. Basic Electrical Engineering and Electronics First Semester, 2004-05]
2. Explain basic principle of operation of a single phase transformer. Where are they used ?
[U.P. Technical Univ. Elec. Engineering Second Semester 2005-06]
3. Give principle, construction and working of transformer. [M.D. Univ. Rohtak Electrical Technology May 2003]
4. What is an ideal transformer? Draw the no-load phasor diagram for such transformer.
[R.G.P.V. Bhopal Basic Electrical Engineering June 2002]
5. Explain the working of a transformer. Derive its emf equation. [M.D. Univ. Rohtak Electrical Technology May 2001]
6. Derive the emf equation of a single phase transformer. [G.G.S.I.P. Delhi Univ. Electrical Science, May 2005]
7. Explain the working of an ideal transformer under (i) no-load and (ii) loaded conditions and derive expressions for voltage and current ratio; relating to transformer turns ratio.
[U.P. Technical Univ. Elec. Engineering First Semester 2004-05]
8. Why does the primary current in a transformer increase when a load is connected across the secondary? Explain the mechanism through which primary draws more current on loading the secondary. Is the ratio of primary current to secondary current exactly equal to the ratio of secondary turns to primary turns? Explain.
[R.G.P.V. Bhopal Basic Electrical Engineering December 2004]
9. Discuss working of a two-winding transformer. Develop its phasor diagram with an inductive load connected to secondary winding. [Pb. Technical Univ. Basic Electrical Engineering and Electronics First Semester, 2004-05]
10. Draw and explain phasor diagram of single phase transformer on load with lagging power factor.
[U.P. Technical Univ. Elec. Engineering First Semester 2005-06]
11. Explain the working of a transformer at no load and loaded conditions. [U.P.S.C. I.E.S. Elec. Engineering II, 1993]
12. Draw and explain the phasor diagram of a transformer on load. How it affects the power factor of the loaded transformer ?
[Pb. Univ. Elec. Machines-I, June 1992]
13. Draw the phasor diagram of single phase transformer at lagging power factor load. Give step by step development of equivalent circuit of transformer.
[RGPV Basic Electrical Engineering June 2004]
14. Draw an equivalent circuit of a transformer referred to primary. [G.G.S.I.P. Delhi Univ. Electrical Science, May 2005]
15. Draw the equivalent circuit of a single phase transformer and derive expressions for all its parameters.
[Pb. Technical Univ. Electrical Engineering June 2000; December 2000; May 2001]
16. Develop the equivalent circuit of a single phase transformer. Mention the physical significance of all its parameters.
[U.P. Technical Univ. Electrical Engineering September 2001]
17. (i) What is meant by leakage flux and leakage reactance in reference to a transformer?
(ii) Define voltage regulation of a transformer and deduce expression for lagging power factor load.
[U.P. Technical Univ. Electrical Engineering Second Semester 2003-04]
18. Describe the various losses occurring in a transformer and also explain how they vary with load current.
[R.G.P.V. Bhopal Basic Electrical Engineering February 2005]
19. Describe the power losses that take place in a transformer. On what factors and how do each of these losses depend ?
[U.P. Technical Univ. Electrical Engineering First Semester 2003-04]
20. With necessary circuit diagrams explain the open-circuit test and short-circuit test conducted on single phase transformer. Justify that the copper loss in the open-circuit test condition and the iron loss in the short-circuit test condition are normally neglected.
[U.P.S.C. I.E.S. Electrical Engineering-II, 1999]
21. Give circuit diagrams and explain the procedures of open-circuit and short-circuit tests on a two winding transformer. How can parameters of equivalent circuit be determined from these tests?

22. How short-circuit test is performed on transformer ? Draw its set-up and discuss the results obtained from the test
[Pb. Technical Univ. Basic Electrical Engineering December 2002]
23. What is the purpose of performing open-circuit and short-circuit tests on transformers ? Explain why open-circuit and short-circuit tests should be performed on lv and hv sides respectively.
[R.G.P.V. Bhopal Basic Electrical Engineering June 2004]
24. Draw the circuit diagrams for conducting open-circuit and short-circuit tests on a single phase transformer. Also explain how the efficiency and regulation can be estimated by these tests.
[R.G.P.V. Bhopal Basic Electrical Engineering December 2003]
25. Derive the condition for maximum efficiency of a transformer at a given power factor.
[R.G.P.V. Bhopal Basic Electrical Engineering December 2003, Jan/Feb. 2006]
26. What is all day efficiency and what is its importance? For what type of transformer it is used.
[U.P. Technical Univ. Electromechanical Energy Conversion-I. 2005-06]
27. Explain the working principle and types of transformers. [M.D. Univ. Rohtak Electrical Technology, May, 2003]
28. Describe the constructional features and principle of operation of a single phase transformer. Classify the transformers on the basis of service rendered and discuss.
[R.G.P.V. Bhopal Basic Electrical Engineering 2001]
29. Explain core and shell type of transformers with their applications. What is one to one transformer ? What is its application?
[R.G.P.V. Bhopal Basic Electrical Engineering June 2002]
30. Write short notes on the following:
- (i) Equivalent circuit of a transformer.
 - (ii) Regulation of transformer.
 - (iii) Open-circuit test on transformer.
 - (iv) Short-circuit test on transformer.
 - (v) Losses and efficiency of a single phase transformer.
 - (vi) Transformer construction.
 - (vii) Types of transformers.

SHORT ANSWER TYPE QUESTIONS AND ANSWERS

- Q. 1. What is a function of transformer ?
Ans. The basic function of a transformer is to transform energy from one voltage level to another (may be from low voltage to high voltage or vice-versa).
- Q. 2. What do you understand by step-up and step-down transformers ?
Ans. When transformer raises the voltage i.e. when the output voltage is higher than its input voltage, it is called the step-up transformer.
When the transformer lowers the voltage, it is called the step-down transformer.
- Q. 3. Why does a transformer has iron core ? [PTU December 2000]
Ans. The use of iron or steel for transformer core ensures a high permeability of the magnetic circuit and because of high permeability the magnitude of exciting current necessary to create the required flux in the core is small. The presence of steel core also causes 100% of magnetic flux created by primary to be linked with secondary.
- Q. 4. Why is the efficiency of a transformer maximum among electrical equipments ? Explain. [PTU 1999]
Ans. Owing to lack of rotating parts in a transformer, there are no friction and windage losses. The other losses such as iron losses are also comparatively small because of better magnetic material and less handling of the material. So the efficiency of transformer is maximum among electrical equipments.
- Q. 5. What is normally the efficiency of a transformer ?
Ans. Typical transformer efficiencies at full load lie between 96 % and 97 % and with extremely large transformers the may be as high as 99 %.
- Q. 6. What name is given to the coils through which current flows from the source ?
Ans. Primary winding.
- Q. 7. What name is given to the coils across which load is connected ?
Ans. Secondary winding.
- Q. 8. Why the transformer is also called the static transformer ?
Ans. Since basic construction of transformer does not require any moving part, it is also called the static transformer.
- Q. 9. What is one to one transformer ? What is its application ? [RGPV June 2002]
Ans. The transformer having equal turns in primary and secondary windings is called the one to one transformer. Such transformer is used for isolating the load from supply.

- Q. 10. Explain with reasons what happens when a power transformer is connected to dc supply of the same voltage rating ?
 Ans. If a rated dc voltage is applied to the primary of the transformer, the flux produced in the transformer core will not vary but remain constant in magnitude and, therefore, there will be no self induced emf in the primary winding (which is only possible with varying flux linkage) to oppose the applied voltage and since the resistance of the primary winding is quite low, a heavy current will flow through the primary winding that may cause burning out of the primary winding.
- Q. 11. What are the properties of an ideal transformer?
 Ans. The properties of an ideal transformer are (i) no winding resistance (ii) no magnetic leakage (iii) no iron loss and (iv) zero magnetizing current i.e. an ideal transformer is assumed to consist of two purely inductive coils wound on a loss-free core. [G.G.S.I.P. Delhi Univ., May, 2005]
- Q. 12. What is the difference between an ideal and practical transformers ?
 Ans. An *ideal transformer* is an imaginary transformer that has no winding resistance, no magnetic leakage, no iron loss and zero reluctance while a *practical transformer* is an actual transformer which operates at a particular frequency and has winding resistance, magnetic leakage, iron loss, copper loss, and magnetic reluctance. [PTU January 2000]
- Q. 13. The primary of a single phase transformer having a turn-ratio 1:2 is connected to a 10 V dc source while its secondary is connected to a 10 Ω resistor. What is the value of the steady state current in the resistor ?
 Ans. Zero
- Q. 14. Is there a definite relationship between the number of turns and voltages in transformers ?
 Ans. Yes; the voltage varies in exact proportion to the number of turns connected in series in each winding.
- Q. 15. What is transformer and transformation ratio?
 Ans. The transformer is an ac machine that (i) transfers electrical energy from one electric circuit to another (ii) does so without a change of frequency (iii) does so by the principle of electromagnetic induction and (iv) has electric circuits that are linked by a common magnetic circuit. Voltage transformation ratio is defined as the ratio of secondary induced emf to primary induced emf. [M.D. Univ., July, 2001]
- Q. 16. Does the flux linking both the windings of a transformer remain same ?
 Ans. Yes; the flux linking the primary and secondary windings is the same.
- Q. 17. Does the transformer draw any current when its secondary is open ?
 Ans. Yes; no-load primary current.
- Q. 18. What is meant by magnetising current ?
 Ans. Magnetising current is quadrature or wattless component of exciting current of transformer and is used to create alternating flux in the core.
- Q. 19. What do you know about the no-load current of a transformer ?
 Ans. The current drawn by primary winding of a transformer on no load is called the no-load or exciting current. It is very small in magnitude in comparison to the full-load primary current (2 to 5 per cent of full-load current). It has two components—energy component (comparatively very small) to meet the iron loss in addition to small amount of copper loss occurring in the primary winding and magnetising component relatively large to create the alternating flux in the core.
- Q. 20. Does the magnetising current of a transformer lie in phase with the applied voltage ? [A.M.I.E. Sec. B. Summer 1992]
 Ans. No; the magnetising current of a transformer lags behind the applied voltage by 90° .
- Q. 21. Ordinarily, what is the phase relationship between the primary and secondary voltages of a transformer ?
 Ans. 180° out of phase.
- Q. 22. Explain how in a transformer, as the secondary current increases, primary current also increases. [PTU May 2002]
 Ans. When the load secondary current of a transformer increases, the secondary ampere turns ($N_2 I_2$) creates more secondary flux. This increased secondary flux opposes the main flux ϕ set up by the exciting current I_0 . Thus main flux ϕ is weakened momentarily by the opposing increased secondary flux ϕ_2 , and, therefore, difference of applied voltage V_1 and back emf E_1 increases, consequently, more current is drawn from the source of supply flowing through the primary winding until the original value of flux ϕ restores. The extra current drawn from the supply source increases the primary ampere-turns ($N_1 I_1$) and compensates the extra secondary ampere-turns i.e. $N_1 I_1' = N_2 I_2$.
 Thus in a transformer when the secondary current increases, primary current also increases.
- Q. 23. Why the main flux in a transformer remains practically constant from no-load to full-load ?

Ans. When the transformer is on no load, it draws no-load or exciting current I_0 from the supply mains. The no-load current I_0 sets up mmf $N_1 I_0$ which produces flux ϕ in the core. When an impedance is connected across the secondary terminals, current I_2 flows through the secondary. The secondary current I_2 sets up its own mmf and hence creates a secondary flux ϕ_2 opposing the main flux ϕ . In other words, the secondary ampere-turns weaken the main flux in the core. Momentarily, self induced emf E_2 decreases due to which primary draws extra current from the supply source. This extra current drawn from the supply source increases the primary ampere-turns ($N_1 I_1'$) and compensates the secondary ampere-turns $N_2 I_2$.

Since the secondary flux ϕ_2 produced by secondary mmf $N_2 I_2$ is neutralized by the flux ϕ_1' produced by mmf $N_1 I_1'$ set up by counter-balancing primary current I_1' , the flux in the transformer core remains constant from no-load to full-load.

Q. 24. How does the leakage flux occur in a transformer ?

[PTU December 2000]

Ans. In actual practice the total flux created does not link the primary and secondary windings but is divided into three parts namely the main flux ϕ linking both the primary and secondary windings, primary leakage flux ϕ_{L1} linking with primary winding only and secondary leakage flux ϕ_{L2} linking with secondary winding only.

The flux set up by the primary or secondary which is set up in the core linking with its own turns and not linking with the other is known as leakage flux.

Q. 25. What is meant by leakage flux and leakage reactance in reference to a transformer ?

[UP Technical Univ. Second Semester 2003-04]

Ans. The flux set up by the primary or secondary which is set up in the core linking with its own turns and not linking with the other is known as leakage flux.

The reactances produced by leakage fluxes are called the leakage reactances.

Q. 26. What is leakage reactance of transformers ?

[PTU December 2003]

Ans. Leakage fluxes of primary and secondary produce self inductances L_1 given as $\frac{N_1 \phi_{L1}}{I_1}$ and L_2 given as $\frac{N_2 \phi_{L2}}{I_2}$ in

the respective windings i.e. primary and secondary. These inductances, in turn, produce inductive reactances X_1 given as $2\pi f L_1$ and X_2 given as $2\pi f L_2$ which cause voltage drop in respective windings lagging behind the currents I_1 and I_2 respectively. The reactances produced by leakage fluxes are called the leakage reactances.

Q. 27. What is the effect of leakage flux in a transformer ? What is meant by magnetising current ? [PTU June 2000]

Ans. The effect of leakage flux is to develop into their respective windings emfs of self inductance which are proportional to the current, and are, therefore, equivalent in effect to the addition of an inductive coil in series with each winding. The component of no-load current used to create the flux in the transformer core is known as magnetising current.

Q. 28. Explain with reasons what happens when the primary terminals of a single phase transformer are connected to the rated ac supply while secondary terminals are short-circuited.

Ans. The transformer with secondary short-circuited becomes equivalent to a coil having an impedance equal to impedance of both windings, which is quite small. When the primary terminals of the transformer are connected to the rated ac supply, heavy current will flow through primary and secondary windings that may cause burning out of these windings.

Q. 29. Define voltage regulation of a transformer.

[UP Technical Univ. Second Semester 2003-04; G.G.S.I.P. Univ. Delhi May 2005]

Ans. Voltage regulation of a transformer is defined as the change in magnitude of terminal or secondary voltage, when full load (rated load) of specified power factor supplied at rated voltage is thrown off (reduced to no load) with primary voltage and frequency held constant, as percentage of the rated terminal voltage.

Q. 30. Is voltage regulation better or worse at 0.6 p.f. lagging than at 0.8 p.f. lagging for the same kVA output explain.

[PTU May 2001]

Ans. The voltage regulation will be the maximum and worst when power factor becomes equal to $\cos \tan^{-1} \frac{X_{02}}{R_{02}}$ (lagging).

Since equivalent reactance of transformer is usually more than double its equivalent resistance, the voltage regulation will increase (i.e. become more and more worse) till power factor falls even below $\cos \tan^{-1} 2$ i.e. 0.45 (lagging). the voltage regulation will worsen at 0.6 p.f. lagging than at 0.8 p.f. lagging.

Q. 31. What is meant by regulation of a transformer and how is it affected by the power factor of the load supplied by it ?

Ans. The change in secondary terminal from no-load to full-load is termed the regulation. It is usually expressed as a percentage or a fraction of the rated no-load terminal voltage.

The voltage regulation increases with the fall in power factor of the load, if the load is inductive, and becomes maximum when the power factor of the inductive load becomes equal to $\cos \left(\tan^{-1} \frac{X_{02}}{R_{02}} \right)$.

The voltage regulation drops with the fall in power factor of the load, if the load is capacitive, and becomes zero, when the power factor of the capacitive load becomes equal to $\cos \left(\tan^{-1} \frac{-R_{02}}{X_{02}} \right)$, and then increases with the fall in power factor of load in the negative direction.

Q. 32. Is the regulation at rated load of a transformer same at 0.8 pf lag and 0.8 pf lead ? [A.M.I.E. Sec B. Summer 1992]

Ans. No; the regulation obtained is different at leading and lagging power factors.

Q. 33. What type of load should be connected to a transformer for getting negative voltage regulation ?

Ans. Highly capacitive load.

Q. 34. Is the percentage impedance of a transformer different on the primary or on the secondary side ?

Ans. No; the percentage impedance of a transformer remains the same on both sides.

Q. 35. Will the voltage regulation of a transformer based on its primary referred equivalent circuit be the same as that when the equivalent circuit is referred to the secondary ?

Ans. Yes; voltage regulation will be the same if magnetising component of current is neglected.

Q. 36. What factors constitute the major losses produced in a transformer ?

Ans. Eddy currents and hysteresis cause iron loss in the magnetic core and yoke and resistance of primary and secondary windings cause copper loss when the transformer is loaded.

Q. 37. On what factors does the hysteresis loss depend ? Why can this loss be assumed to be constant ?

[PTU December 2000]

Ans. Hysteresis loss depends upon the type of material, volume of the material, frequency of reversal of magnetic field and maximum value of flux density in the core of the material i.e.

$$P_h = \eta (B_{\max})^{1.6} f V \text{ watts}$$

Since from no-load to full-load the flux linking with the core and frequency remains constant, hysteresis loss is assumed to remain constant.

Q. 38. On what factors does the eddy current loss depend ? Why this loss can be assumed to be constant ?

[PTU June 2000]

Ans. The eddy current loss depends on the maximum flux density in the core and supply frequency for a given core (i.e. nature of magnetic material, volume of the core, and thickness of laminations remaining unchanged). Since from no-load to full-load the flux linking with the core and frequency remain constant, the eddy current loss remains constant.

Q. 39. List out various losses that take place in a transformer. Which loss is independent of load ? [PTU June 2003]

Ans. Iron or core losses consisting of hysteresis and eddy current losses. These losses occur in the core and yoke due to alternating flux.

Copper losses occurring in primary and secondary windings due to their resistances when the transformer is loaded. Iron losses are independent of load, and therefore, are called the constant losses.

Q. 40. Does eddy current loss of a transformer depend on the applied voltage ?

Ans. Yes, eddy current loss depends upon the applied voltage. If the transformer is operated with increased voltage but at the same frequency the eddy current loss increases due to increase in value of peak value of flux density and if applied voltage is increased with increase in frequency, the proportion of increase being the same in both, the eddy current loss increases due to increase in supply frequency.

Q. 41. If the supply frequency is doubled, which loss component in transformer will double ?

Ans. Due to doubling of supply frequency, the hysteresis loss component will be double.

Q. 42. Under what conditions, open-circuit test is performed on transformer ? What are the results obtained ?

[PTU June 2003]

Ans. Open-circuit test is performed to determine the core or iron or excitation loss, P_0 and no-load current I_0 and thereby the shunt branch parameters R_0 and X_0 of the equivalent circuit.

In this test, one of the windings (usually h.v winding) is kept open-circuited and the rated voltage at rated frequency is applied to the other winding, as shown in Fig. 10.22. Either an auto-transformer or voltage divider (VD) is used for applying the voltage applied to the h.v winding. Ammeter A and wattmeter W are connected to measure no-load

Q. 43. What is the advantage of predetermining the performance of electrical machines through open-circuit and short-circuit test data ?

[PTU January 2000]

Ans. The advantage of predetermining the performance of electrical machines through open-circuit and short-circuit test data is that we can know the behaviour and characteristics of the machine that will show during actual operating conditions and on this basis we can select the machine meeting our requirements.

Q. 44. How can eddy current loss be minimised ?

Ans. By using laminated iron core.

Q. 45. What is meant by equivalent circuit of a transformer ?

Ans. An electrical circuit that can represent behaviour of the transformer to an approximation is called its equivalent circuit.

Q. 46. How may the iron loss be reduced to a minimum in a transformer ?

Ans. The iron loss may be minimised by using steel of high silicon content for the transformer core and by using very thin laminations (0.35 to 0.5 mm thick) insulated from each other by insulating varnish or by layers of papers.

Q. 47. How can iron-loss be measured in a transformer ?

Ans. Iron loss can be measured from open-circuit test. Iron loss equals the power supplied on no-load with normal (rated) voltage applied to the primary winding of the transformer.

Q. 48. How can copper loss be measured in a transformer ?

Ans. Copper loss can be determined from short-circuit test. The power supplied during short-circuit is equal to copper loss corresponding to short-circuit current.

Q. 49. What are the various tests that give the complete parameters of the equivalent circuit of the transformer ?

Ans. Open-circuit and short-circuit tests.

Q. 50. What test provides the copper loss of a transformer ?

Ans. Short-circuit test.

Q. 51. What are typical informations that can be had by performing open- and short-circuit tests on transformer ?

Ans. Iron and copper losses and the parameters of the equivalent circuit of the transformer can be determined from the open-circuit and short-circuit tests and so the voltage regulation and efficiency of the transformer can be determined.

Q. 52. Why core loss is neglected in short-circuit test and copper loss in open-circuit test ?

Ans. Since the voltage applied to the primary of transformer is very low (5 to 8% of rated voltage), the flux produced in the core is very small and, therefore, iron losses are very small and can be neglected.

Since no current flows in the open-circuited secondary and the current in the primary will also be very small (usually 3 to 10% of full-load current), the copper losses in the open-circuit test will be so small that these can be neglected.

Q. 53. Why are iron or core losses assumed to remain constant from no load to full-load in a power transformer ?

Ans. Since iron or core losses depend on the flux density and supply frequency for a given transformer core and flux density and supply frequency remain the same from no-load to full-load, iron or core losses are assumed to remain constant from no-load to full-load.

Q. 54. Why the lv winding is normally short-circuited during short-circuit test ?

Ans. Voltage required for the short-circuit test is very low (about 5 percent of the rated value) and if the measurements are made, by short-circuiting lv winding, on lv side, the voltage required would be inconveniently low, while the current would often be inconveniently large.

Q. 55. Why are the iron loss in a transformer substantially independent of the load current ? Explain.

[U.P. Technical Univ. Electrical Machines 2001-02]

Ans. Iron or core loss is caused by the alternating flux in the core and consists of hysteresis and eddy current losses which depend upon the maximum flux density in the core and supply frequency. Since it has been determined that mutual flux varies somewhat with the load (its variation being 1 to 3 percent from no-load to full-load), the core losses will vary somewhat with the load and its power factor. It may be emphasized here that core loss is assumed to remain constant from no-load to full-load, the variations in losses from no-load to full-load being very small and negligible.

Q. 56. Explain how a short-circuit test is performed on a single phase transformer and why it gives the copper losses and not the iron losses ?

[U.P. Technical Univ. Electrical Machines 2001-02]

Ans. Short-circuit test is performed by short-circuiting the secondary (low-voltage) winding by a thick wire or strap and applying variable low voltage to the primary (lv winding) through an auto-transformer or potential divider as illustrated in Fig. 10.24. The applied voltage to the primary is gradually increased till the ammeter indicates the full-load current of metering side.

Since applied voltage is very low (5–8% of the rated voltage) so flux linking with the core is very small and therefore iron losses are so small that these can be neglected and thus the power input gives the copper losses only and not the

conditions. The shell type construction is commonly

What is the condition for maximum efficiency in transformer ?

[PTU December 2003]

Ans. Efficiency of a transformer will be maximum on such a load at which variable loss (copper loss) becomes equal to constant loss (iron loss).

Q. 58. Is the efficiency of a transformer same at the same load at 0.8 pf lag and 0.8 pf lead ?

Ans. Yes, the efficiency of a transformer remains the same for a given load at a given power factor irrespective of the fact whether the pf is lagging or leading.

Q. 59. What is meant by all-day efficiency of a power transformer and why is it lower than commercial efficiency ?

Ans. The all-day efficiency, also known as energy efficiency or operational efficiency, is defined as the ratio of energy (kWh) output over 24 hours to the energy input over the same period

$$\text{i.e. All-day efficiency} = \frac{\text{Output in kWh}}{\text{Input in kWh}}$$

There are certain types of transformers, such as distribution transformers, which remain energized for 24 hours but they supply very light loads for major portion of the day. Thus iron or core loss occurs for the whole day but the copper loss occurs only when the transformer is loaded and so the all-day efficiency of transformer is lower than its commercial efficiency.

Q. 60. Why is an iron-silicon alloy core used in a transformer?

[PTU December, 2005]

Ans. Iron-silicon alloy core is used in a transformer to ensure a high permeability of the magnetic circuit and keep the iron loss the minimum possible.

PROBLEMS

- The maximum flux density in the core of 250/3,000 V, 50 Hz single phase transformer is 1.2 Wb/m². If the emf per turn is 1 V, determine (i) primary and secondary turns (ii) area of the core. [Nagpur Univ. Elec. Engineering-I, 1991]
[Ans (i) 32; 375 (ii) 0.03 m²]
- A single phase transformer has 400 primary and 1,000 secondary turns. The net cross-sectional area of the core is 60 cm². If the primary winding be connected to a 50-Hz supply at 520 V, calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding. [Pune Univ. Elec. Engineering-I, 1989]
[Ans 0.976 T; 1,300 V]
- A single phase transformer has 500 turns in the primary and 1,200 turns in the secondary. The cross-sectional area of the core is 80 cm². If the primary winding is connected to a 50 Hz supply at 500 V, calculate (peak flux density, and (ii) voltage induced in the secondary). [Bharathiyar Univ., November 1997]
[Ans. (i) 0.563 T (ii) 1200 V]
- A single phase 50 Hz transformer has 80 turns on the primary winding and 280 turns on the secondary winding. The voltage across the primary winding is 240 V at 50 Hz. Calculate (i) the maximum flux density in the core and (ii) induced emf in secondary. The net cross-sectional area of the core be taken 200 cm². [Ans. (i) 0.6757 T (ii) 840 V]
- A 200 kVA, 6600/400 V, 50 Hz single phase transformer has 80 turns on the secondary. Calculate (i) the approximate values of the primary and secondary currents (ii) the approximate number of primary turns and (iii) the maximum value of flux. [Pb. Univ. Elec. Engineering-I, 1994]
[Ans. (i) 30.3 A, 500 A; (ii) 1,320 V (iii) 22.5 mWb]
- A single phase transformer with a ratio of 440/110 V takes a no-load current of 5 A at 0.2 power factor lagging. If the secondary supplies a current of 120 A at a pf of 0.8 lagging, estimate the current taken by the primary. [Elec. Machines-I, Bangalore Univ. 1992, S. C. E. & T.E.W.B. 1996 and Elec. Engg. Pb. Univ, 1991]
[Ans. (i) 33.9 A, 42.49° (lag)]
- A single phase transformer has 1,000 turns on the primary and 200 turns on the secondary. The no-load current is 3 A at a pf 0.2 lagging. Calculate the primary current and power factor when the secondary current is 280 A at a pf of 0.8 lagging. [Nagpur Univ., Nov. 1997]
[Ans. 58.28 A; 38.83° (lagging)]
- A single phase transformer takes 10 A on no load at a pf of 0.2 lagging. The turns ratio is 4 : 1 (step down). If the load on the secondary is 200 A at a pf of 0.85 lagging, find the primary current and power factor. Neglect the voltage drop in the winding. [Nagpur Univ., November 1999]
[Ans. 57.33 A; 39.1° (lagging)]
- A 230/115 V, single phase transformer is supplying a load of 5 A, at pf 0.866 lagging. The no-load current is 0.2 A at pf of 0.208 lagging. Calculate the primary current and primary power factor. [Nagpur Univ. Summer 2000]
[Ans. 2.64 A; 33.26° (lagging)]

10. A 10 kVA, 1-phase transformer with 2,000/400 V, at no load, has resistance and leakage reactance of primary winding 5.5Ω and 12Ω respectively, the corresponding values of secondary winding being 0.5Ω and 0.45Ω . Determine the value of secondary voltage at full load, 0.8 pf lagging, when the primary applied voltage is 2,000 V.
[U.P. Technical Univ. Electrical Engineering July 2002]
[Ans. 377.65 V]
11. Calculate the percentage voltage drop for a transformer with a percentage resistance of 2.5 % and a percentage reactance of 5 % of rating 500 kVA when it is delivering 400 kVA at 0.8 pf lagging. [Indore Univ. Elec. Machinery-I, 1987]
[Ans. 4%]
12. Calculate the regulation of a transformer in which the ohmic loss is 1% of the output and reactance drop of 5% of the voltage, when the power factor is (i) 0.8 lag (ii) unity (iii) 0.8 leading.
[Madras Univ., 1997]
[Ans. (i) 3.8% (ii) 1% (iii) - 2.2%]
13. A transformer has copper loss of 1.5% and reactance drop of 3.5% when tested at full load. Calculate its full-load regulation at (i) 0.8 pf (ii) 0.8 pf lagging and (iii) 0.8 pf leading.
[Bharathidhasan Univ., April 1997]
[Ans. (i) 1.5%, (ii) 3.33%, (iii) - 0.9%]
14. If in a transformer core, the hysteresis and eddy current losses are 80 W and 50 W at normal voltage and frequency, Calculate the losses when voltage and frequency are increased by 20%.
[U.P. Technical Univ. Elec. Engineering First Semester 2004-05]
[Ans. 96 W, 72 W]
15. A transformer is connected to 2200 V, 40 V Hz supply. The core loss is 800 watts out of which 600 watts are due to hysteresis and remaining eddy current losses. Determine the core loss if the supply voltage and frequency are 3300 V and 60 Hz respectively.
[Bharathiyar Univ. Nov., 1997]
[Ans. 1,350 W]
16. The following data were obtained on a 50 kVA, 2400/120 V transformer.
Open-circuit test, instruments on lv side:
Wattmeter reading = 396 W; Ammeter reading = 9.65 A; Voltmeter reading = 120 V
Short-circuit test, instruments on hv side:
Wattmeter reading = 810 W; Ammeter reading = 20.8 A; Voltmeter reading = 92 V
Find the equivalent circuit parameters referred to high voltage side.
[Pb Technical Univ. Electrical Engineering May 2002]
[Ans. $14.5 \text{ k}\Omega$, $5.3 \text{ k}\Omega$, 1.872Ω , 4Ω]
17. A short-circuit test when performed on the hv side of a 10 kVA 2,000/400 V single phase transformer, gave the following data: 60 V, 4 A, 100 W.
If the lv side is delivering full load current at 0.8 pf lag and at 400 V, find the applied voltage to hv side.
[Nagpur Univ. Elec. Machines-I, 1993]
[Ans. 2,065.9 V]
18. Open-circuit and short-circuit tests on a single phase transformer gave the following results:
 $V_0 = 200 \text{ V}$, $I_0 = 0.7 \text{ A}$, $W_0 = 20 \text{ W}$ — Test from primary side
 $V_s = 10 \text{ V}$, $I_s = 10 \text{ A}$, $W_s = 40 \text{ W}$ — Test from primary side
Determine the equivalent circuit referred to primary side. [RGPV Basic Electrical Engineering December 2002]

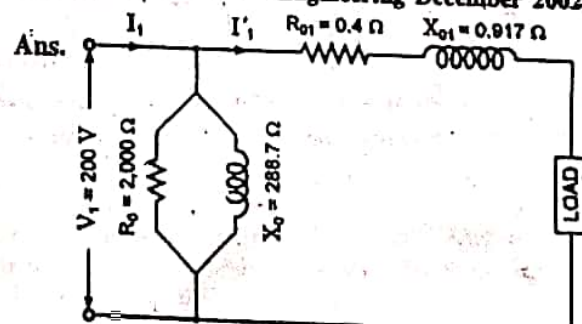


Fig. 1 Equivalent Circuit

19. The parameters of the equivalent circuit of a 100 kVA, 2000/200 V single phase transformer are as follows:
Primary resistance = 0.2Ω ;
Secondary leakage reactance = $4.5 \text{ m}\Omega$;

Secondary resistance = $2 \text{ m } \Omega$;

Core-loss resistance = $10 \text{ k } \Omega$;

Primary leakage reactance = $0.45 \text{ } \Omega$;

Magnetising reactance = $1.55 \text{ k } \Omega$.

Using the circuit referred to primary, determine the (a) voltage regulation (b) efficiency of the transformer operating at rated load with 0.8 lagging power factor. [Pb. Technical Univ. Electrical Engineering June 2000]

[Ans. (a) 2.1% (b) 98.28%]

20. The parameters of the equivalent circuit of a 200 kVA, 2500/250 V single phase transformer are as follows:

Primary resistance = $0.2 \text{ } \Omega$;

Secondary resistance = $2 \text{ m } \Omega$

Primary leakage reactance = $0.45 \text{ } \Omega$;

Secondary leakage reactance = $4.5 \text{ m } \Omega$

Core-loss resistance = $10 \text{ k } \Omega$;

Magnetizing reactance = $1.55 \text{ k } \Omega$

Using the circuit referred to primary, determine (a) voltage regulation (b) efficiency of the transformer at rated load with 0.8 lagging power factor. [Pb. Technical Univ. Electrical Engineering May 2001]

[Ans. (a) 2.686% (b) 98.048%]

21. In a 25 kVA, 2000/200 V transformer, the constant and variable losses are 350 W and 400 W respectively. Calculate the efficiency on unity power factor at (i) full load and (ii) half-full load. [Pb. Technical Univ. Electrical Engineering 1999]

[Ans. (i) 97.09% (ii) 96.53%]

22. The no-load transformation ratio in a 200 kVA, 50 Hz single phase transformer is 4,000/250 V. For a maximum core flux of 0.06 Wb, find

(i) the efficiency at half the rated kVA and at unity power factor,

(ii) the efficiency at full load, 0.8 power factor lagging and.

(iii) kVA load for maximum efficiency, given that the full-load copper losses are 1.8 kW and core losses are 1.5 kW. [U.P. Technical Univ. Electrical Machines 2001-2002]

23. A 40 kVA, 1-ph, transformer has an iron loss of 400 W. and full-load copper loss of 800 W. Find the load at which maximum efficiency is achieved at unity pf. [Amravati Univ., Winter 1999]

[Ans. 28.28 kW]

24. A 40 kVA transformer has a core loss of 400 watts and full-load copper loss of 800 watts. If the power factor of the load is 0.9 lagging, calculate:

(i) the full-load efficiency, and

(ii) percentage of the full-load at which the maximum efficiency occurs.

[U.P. Technical Univ. Electrical Engineering Second Semester 2002-03]

[Ans. (i) 96.774% (ii) 70.71%]

25. A 1,000 / 230 V, 150 kVA, 1-phase, 50 Hz transformer has core loss of 1.4 kW and full-load copper loss of 1.6 kW. Determine (i) kVA load for maximum efficiency and value of maximum efficiency at unity pf. (ii) the efficiency at half-full load 0.8 pf leading. [Nagpur Univ. Basic Elec. Machines 1993]

[Ans. 140.0511 kVA; 98.04 % (ii) 97.09 %]

26. A 250 / 500 V transformer gave the following test results:

Short-circuit test with low voltage winding short-circuited: 20 V, 12 A, 100 W

Open-circuit test : 250 V, 1 A, 80 W on low-voltage side.

Determine the circuit constants, insert these on the equivalent circuit diagram and calculate applied voltage and efficiency when the output is 10 A at 500 V and 0.8 pf lagging. [Nagpur Univ. Elec. Machines 1995]

[Ans. $R_0 = 781.25 \text{ } \Omega$, $X_0 = 263.88 \text{ } \Omega$, $R_{01} = 0.1736 \text{ } \Omega$; $X_{01} = 0.3788 \text{ } \Omega$, 257.32 V, 96.4 %]

27. The maximum efficiency of a 100 kVA, single phase transformer is 98 % and occurs at 80 % of full load. If the leakage impedance of the transformer is 5 %, find the voltage regulation at rated load of 0.8 pf lagging. [Nagpur Univ. Elec. Machines -I, 1993]

[Ans. 3.74 %]

28. A 5 kVA, 1000/200 V, 50 Hz single phase transformer has the following test results:

Open-circuit test (Low voltage side) : 200 V, 1.2 A, 90 W

Short-circuit test (High voltage side) : 50 V, 5 A, 110 W

Calculate iron losses, copper losses, resistance of two windings of transformer and efficiency at full load and 0.8 pf lagging. [Pb Technical Univ. Basic Electrical Engineering June 2003]

[Ans. 90 W, 110 W, 4.4 Ω , 95.238%]

29. A 250/500 volts, single-phase transformer gave the following test results:

Short-circuit test (LV short-circuited) : 20 V, 12 A, 100 Watts

Determine the efficiency of the transformer when the output is 12 Amp, 500 Volts at 0.85 power factor lagging.
[U.P. Technical Univ. Electrical Engineering Second Semester 2003-04]
[Ans. 96.59%]

30. A 4 kVA, 400/200 V, 50 Hz, single phase transformer gave the following results :

No-load test : 200 V, 0.7 A, 60 W (LV side)

Short-circuit test : 15 V, 10 A, 61 W (HV side)

Calculate: (i) components of no-load current. (ii) efficiency on full-load at 0.8 pf lagging.
(iii) regulation at full load at 0.8 pf lagging.

[R.G.P.V. Bhopal Basic Electrical Engineering February 2005]
[Ans. (i) 0.3 A, 0.632 A (ii) 96.36% (iii) 3.275%]

31. A 4 kVA, 200/400 V, 50 Hz single phase transformer gives the following test results:

OC Test (on primary side): 200 V, 0.8 A, 50 W.

SC Test (on secondary side): 17.5 V, 9 A, 50 W.

Calculate the full-load efficiency and secondary terminal voltage when supplying full-load secondary current at 0.8 pf lagging.

[RGPV Basic Electrical Engineering Jan/Feb: 2006]

[Ans. 96.93%, 384 V]

32. Consider a 4 kVA, 200/400 V single phase transformer supplying full load current at 0.8 lagging power factor. The OC / SC test results are as follows:

OC Test : 200 V 0.8 A 70 W (lv side)

SC Test : 20 V 10 A 60 W (hv side)

Calculate efficiency, secondary voltage and current into primary at the above load.

Calculate the load at unity power factor corresponding to maximum efficiency. [Nagpur Univ. Elec. Machines 1993]
[Ans. 96.1%, 383.75 V, 4.32 kVA]

33. The following readings were obtained from OC and SC tests on 8 kVA, 400/100 V, 50 Hz transformer.

OC Test (LV Side) : 100 V, 4 A, 60 W

SC Test (HV Side) : 10 V, 20 A, 100 W

Calculate: (i) Voltage regulation at full load and 0.8 power factor lagging (ii) Efficiency at half-load and 0.8 power factor lagging.

[U.P. Tech. Univ. Electrical Engineering First Semester 2005-06]

[Ans. 2.3%, 97.41%]

34. A 600 kVA, single-phase transformer when working at unity power factor, has an efficiency of 92 % at full-load and also at half-load. Determine its efficiency when it operates at unity pf and 60 % of full-load.

[U.P. Technical Univ. Electrical Machines 2003-04]

[Ans. 93.53%]

35. A 150 kVA single phase transformer working at unity power factor has an efficiency of 96 % both at 70 percent of full load and at full-load. Find the load at which maximum efficiency occurs and also the value of maximum efficiency.

[R.G.P.V. Bhopal Basic Electrical Engineering June 2003]

[Ans. 125 kVA, 96.044 %]

36. A transformer rated 200 V/50 V, 10 kVA has a core loss of 100 watts. What is the maximum efficiency of the transformer at 0.8 lagging power factor? Assume full-load copper loss as 200 watts. At what load this maximum efficiency is obtained? How and why would your answer be different if the power factor were unity?

[R.G.P.V. Bhopal Basic Electrical Engineering December 2004]

[Ans. 7.07 kVA, 96.585%, 97.249%]

37. The efficiency of a 400 kVA single-phase transformer is 98.77 % when delivering full-load at 0.8 pf and 99.13 % at half load and unity power factor. Calculate:

(i) Iron losses (ii) The full-load copper losses. [R.G.P.V. Bhopal Basic Electrical Engineering Jan/Feb: 2006]

[Ans. 1.012 kW, 2.973 kW]

38. Find all-day efficiency of a transformer having maximum efficiency of 98 % at 15 kVA at unity power factor and loaded as follows:

12 hours — 2 kW at 0.5 pf lag:

6 hours — 12 kW at 0.8 pf lag

6 hours — at no-load.

[Nagpur Univ. Elec. Machines 1993]

[Ans. 95.31%]

A transformer has its maximum efficiency of 0.98 at 15 kVA at unity power. During the day it is loaded as follows:
2 hours: 2 kW at 0.5 pf lagging.
6 hours: 12 kW at 0.8 pf lagging.
8 hours: No-load.
Find the all day efficiency of the transformer. [R.G.P.V. Bhopal Basic Electrical Engineering February 2005]
[Ans. 96.98%]

20. A transformer has its maximum efficiency of 0.98 at 15 kVA at unity power factor. During the day it is loaded as follows:
2 hours: 2 kW at power factor 0.5.
6 hours: 12 kW at power factor 0.8.
8 hours: 18 kW at power factor 0.9.
Find the all day (energy) efficiency. [R.G.P.V. Bhopal Basic Electrical Engineering June 2004]

21. A 500 kW transformer is required to operate for 5 hours at 1/4 full-load; for 8 hours at full load and at no load for rest of the period. Calculate their all-day efficiencies and decide which of the two transformers be chosen if the full load losses in them are as follows :

Transformer A – Iron loss 1 % and copper loss 2.5 %

Transformer B – Iron loss 1.5 % and copper loss 1.5 %

[Ans. 95.38 %; 95.02 %; Transformer A]